Functions

Version 5.0.1

16-7-2022

Contents

Introduction

Function is an important concept in mathematics. This concept is involved in every mathematical structure since this is a rigorous way to relate different objects together.

In Olympiad, one major kind of problems involving functions is functional equations. This means we are given an equation relating some variables like x, y and their images $f(x), f(y)$, etc. Usually we are asked to find all such functions. To deal with this kind of problem, one only needs to carry out various substitutions and finds as many properties as possible to seek for the function. Although it sounds a simple job, sometimes it is not obvious which substitutions could give useful results.

It is quite difficult to tell how to tackle function problems in general. In spite of this, we shall go through various problems to learn different standard tricks that are commonly used. One should bear in mind that there is no powerful method that can solve every single problem. One should be flexible and think of some useful substitutions depending on the problem.

In section 1, we provide the formal definition of a function and introduce some common terminologies related to functions. In section 2, we mainly work on functional equations. We shall introduce many tricks and substitutions via different problems in functional equations. In addition, we shall introduce the concepts of injectivity and surjectivity, and see how these important concepts can be used to solve some problems.

Next, section 3 covers some function problems related to inequalities. Such problems are normally related to analysis, especially those harder problems. Therefore, we first give a brief introduction to some basic concepts in analysis such as limits of sequences and functions, and continuity of functions. Some of these concepts and results are needed in solving some function problems. Also, functional equation problems involving functions defined on $\mathbb{R}^+$ or $\mathbb{N}$ are usually solved with the use of some simple inequalities or ideas in analysis. We will go through more problems of this kind in this section. Moreover, we shall go through some problems in functional inequalities.

Lastly, in section 4, we investigate some function problems related to number theory. In particular, there are occasionally some divisibility problems that involve functions. There are additional tricks which are needed to solve such problems apart from substitutions.

1 Basic Terminologies

To begin with, we provide the formal definition of a function using set notations.

1.1 Set Notations

Definition 1.1. A *set*(集合) is a collection of certain objects. The things belonging to a set are called the *elements*(元素/元) of the set.

In most cases, the elements of a set are simply numbers. We use $S = \{a, b, c, \ldots\}$ to represent a set S, where $a, b, c, \ldots$ are the elements of S. We write $a \in S$ to mean a belongs to S. If x does not belong to S, we write $x \notin S$.

Another common way to represent a set is to write $S = \{a : \cdots\}$. This means S is the collection of all elements a that satisfy certain properties as indicated in the $\cdots$ part. For example, $\{n : 3 \mid n\}$ is the set of all integers which are divisible by 3, while $\{x : x^2 \leqslant 4\}$ is the set of all real numbers x whose squares are at most 4.

Definition 1.2. Let A and B be any sets. We say that A and B are the same, and write $A = B$, if each element in A belongs to B, and each element in B belongs to A.

In other words, two sets are equal if they contain the same elements. Here, repeated elements are regarded as the same, and the order does not matter. For example, the sets $\{1, 1, 2, 2, 3\}$, $\{2, 3, 1, 3\}$ and $\{1, 2, 3\}$ are the same.

Definition 1.3. Let A and B be any sets. If every element in A belongs to B, then A is called a *subset*(子集) of B, and we use the notation $A \subset B$ or $A \subseteq B$. In addition, if $A \neq B$, then A is called a *proper subset*(真子集) of B, and we use the notation $A \subsetneq B$.

For example, $\{1, 2, 4\}$ is a proper subset of $\{1, 2, 3, 4, 5\}$.

The following are the notations of some commonly used sets.

- $\mathbb{R}$: the set of all real numbers
- $\mathbb{Q}$: the set of all rational numbers
- $\mathbb{Z}$: the set of all integers
- $\mathbb{N}$: the set of all positive integers

- $\mathbb{R}^+$: the set of all positive real numbers
- $\mathbb{Q}^+$: the set of all positive rational numbers
- $\mathbb{R}_0^+$: the set of all nonnegative real numbers
- $\mathbb{N}_0$: the set of all nonnegative integers
- $\varnothing$: the *empty set*(空集), which is the set containing no element

Note that the notations $\mathbb{R}_0^+$ and $\mathbb{N}_0$ are not that commonly used.

1.2 Functions

We now introduce a few concepts about functions.

Definition 1.4. A *function*(函數) or a *mapping*(映射) is a relation between two sets A and B such that every element $a \in A$ corresponds to a unique element $b \in B$.

The set A is the *domain*(定義域) of the function while B is the *codomain*(陪域). We write $f : A \to B$ to denote a function f with domain A and codomain B. For each $a \in A$, we write $f(a)$ to denote the unique element in set B to which a corresponds under the mapping f. If $f(a) = b$, then b is called the *image*(像) of a, while a is called a *preimage*(原像) of b.

The set $\{f(a) : a \in A\}$ is called the *range*(值域) of the function, which means the set of all images under f. By definition, the range is a subset of B but may not necessarily be the same as B. Indeed, for each $b \in B$, it is possible that the equation $f(x) = b$ has no solution in $x \in A$, and it is possible that it has more than one solution.

Some simple functions can be represented using a nice formula, such as $f(x) = x$ for $x \in \mathbb{R}$, or $g(x) = \sqrt{x} - 1$ for $x \in \mathbb{R}_0^+$. However, it should be emphasized that not all functions can be given explicitly by a single formula. One example is the *Dirichlet function*(狄利克雷函數)

$$h(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q}, \\ 0 & \text{if } x \notin \mathbb{Q}. \end{cases}$$

But actually this function is still simple since there are many functions which cannot be expressed in an explicit way.

Definition 1.5. Let $f : A \to B$ be a function, and let C be a subset of A. The *restriction*(限制) of f on C is the function $f|_C : C \to B$ defined by $f|_C(x) = f(x)$ for any $x \in C$.

This means we change the domain of the function from A to C. Formally, the functions f and $f|_C$ are distinct, and so we should use different notations. However, sometimes we may still use f to mean both functions if there is no confusion.

Definition 1.6. Let $f : A \to B$ and $g : B \to C$ be two functions. The *composition*(複合) of f and g is the function $g \circ f : A \to C$ defined by $(g \circ f)(x) = g(f(x))$ for any $x \in A$.

This means we first apply f and then g. The composition $g \circ f$ is only defined when the range of f is contained in the codomain of g. In general, $f \circ g$ and $g \circ f$ are different, even if both are well-defined.

In particular, for $f : A \to A$, we can define $f \circ f$. This is denoted by $f^2(x)$. Similarly, we write $f^3(x)$ to mean $f(f(f(x)))$, etc. These are also known as *nested functions*(嵌套函數). On the other hand, the notation $f^{(3)}(x)$ is more commonly used to mean the third derivative of f, etc., which will not appear in this note.

Definition 1.7. Let A be a nonempty set. The *identity function*(恆等函數) on A is the function $f : A \to A$ defined by $f(x) = x$ for any $x \in A$.

Definition 1.8. A *constant function*(常數函數) is a function $f : A \to B$ such that the image $f(x)$ is the same for all $x \in A$.

Definition 1.9. Let $f : A \to B$ be a function. The *inverse*(反函數) of f is the function $f^{-1} : B \to A$ such that $f^{-1} \circ f$ is the identity function on A, and $f \circ f^{-1}$ is the identity function on B.

Equivalently, this means $(f^{-1} \circ f)(x) = x$ for all $x \in A$, and $(f \circ f^{-1})(y) = y$ for all $y \in B$. Note that the inverse may not exist. However, it must be unique if it exists.

Definition 1.10. A function $f : \mathbb{R} \to \mathbb{R}$ is called *bounded*(有界) if there exists a constant M such that $|f(x)| \leqslant M$ for any $x \in \mathbb{R}$. Otherwise, f is said to be *unbounded*(無界).

If there is a constant M such that $f(x) \leqslant M$ for any $x \in \mathbb{R}$, we say that f is *bounded above*(有上界). Similarly, f is *bounded below*(有下界) if there is a constant m such that $f(x) \geqslant m$ for any $x \in \mathbb{R}$. Note that f is bounded if and only if(iff) it is both bounded above and bounded below.

Definition 1.11. A function $f:\mathbb{R}\to\mathbb{R}$ is called *monotonic increasing*(單調遞增) if $f(x)\leqslant f(y)$ for any $x<y$. It is called *monotonic decreasing*(單調遞減) if $f(x)\geqslant f(y)$ for any $x<y$. In either case, it is called a *monotonic function*(單調函數).

In particular, f is called *strictly increasing*(嚴格遞增) if $f(x)<f(y)$ for any $x<y$. Similarly, it is called *strictly decreasing*(嚴格遞減) if $f(x)>f(y)$ for any $x<y$.

Definition 1.12. A function $f:\mathbb{R}\to\mathbb{R}$ is called an *even function*(偶函數) if $f(x)=f(-x)$ for any $x\in\mathbb{R}$, and is called an *odd function*(奇函數) if $f(x)=-f(-x)$ for any $x\in\mathbb{R}$.

A polynomial $P(x)=a_nx^n+a_{n-1}x^{n-1}+\cdots+a_1x+a_0$ is an even function iff $a_k=0$ whenever k is odd. It is an odd function iff $a_k=0$ whenever k is even.

Definition 1.13. A function $f:\mathbb{R}\to\mathbb{R}$ is called a *periodic function*(周期函數) if there exists a constant $d>0$ such that $f(x+d)=f(x)$ for any $x\in\mathbb{R}$. In that case, d is called a *period*(周期) of f.

Usually, we use the same word 'period' to mean the smallest period of a periodic function. However, the smallest period needs not exist. A counterexample is the Dirichlet function $f:\mathbb{R}\to\mathbb{R}$ defined by

$$f(x)=\begin{cases}1 & \text{if } x\in\mathbb{Q},\\ 0 & \text{if } x\notin\mathbb{Q}.\end{cases}$$

It is periodic and any positive rational number d is a period. Indeed, if $x\in\mathbb{Q}$, then $x+d\in\mathbb{Q}$, and so $f(x)=1=f(x+d)$. If $x\notin\mathbb{Q}$, then $x+d\notin\mathbb{Q}$, and so $f(x)=0=f(x+d)$.

2 Functional Equations

A large class of Olympiad problems involving functions is related to functional equations. This is a kind of problem where one or more equations and conditions relating the variables and their images under a function are given. Usually, we are asked to find out all possible functions satisfying these conditions, to find some particular values, or to prove some facts about the function.

2.1 Substitutions

To deal with problems in functional equations, the major technique is to apply various substitutions to the given equation, so as to find out the relationships among different terms, and finally obtain the explicit form of $f(x)$. Though we have mentioned that not all functions have explicit formulae, the solutions to functional equations are usually nice enough. However, we cannot assume they are of those forms before we have proved so mathematically.

We first go through some fundamental techniques for solving functional equations.

Problem 2.1. Find all function(s) $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(x+3) = x^2 - 2x$$

for all $x \in \mathbb{R}$.

Analysis. To find f means to find $f(x)$ for any $x \in \mathbb{R}$. Since the condition only involves one f, using a simple change of variable $y = x+3$ would lead to a formula for $f(y)$. This already gives us a formula for f since x and y are just dummy variables.

Solution. Let $y = x + 3$. Then $x = y - 3$ and hence

$$f(y) = (y-3)^2 - 2(y-3) = y^2 - 8y + 15.$$

Since x can be any real number, y runs through all real numbers so that this formula is true for all $y \in \mathbb{R}$. $\square$

Remarks. In general, if $f(g(x)) = h(x)$ for some functions g and h, and g has an inverse g^{-1}, then $f(x) = h(g^{-1}(x))$ for any x.

Problem 2.2. Find all function(s) $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(x^2+1) = x^2 + 2$$

for all $x \in \mathbb{R}$.

Analysis. This problem is similar to problem 2.1. The only thing that we need to care is the range of $y = x^2 + 1$.

Solution. Let $y = x^2 + 1 \geqslant 1$. Then $x = \sqrt{y-1}$ and hence

$$f(y) = \sqrt{y-1}^2 + 2 = y + 1.$$

Note that y can only take all real values $\geqslant 1$. So $f(y) = y + 1$ for $y \geqslant 1$ and $f(y)$ can be anything for $y < 1$.

Conversely, suppose f is such a function. As $x^2 + 1 \geqslant 1$ for any $x \in \mathbb{R}$, we have $f(x^2+1) = (x^2+1)+1 = x^2+2$ so that the condition is satisfied. So any such function f is a solution. $\square$

Remarks. This problem illustrates one common mistake when solving functional equations. When one tries to replace an expression by another variable, one needs to check the range that this expression can take. It is a good habit to write down the range of the variables in each step.

When working on problems in functional equation, the step of checking solution cannot be omitted, even though it could be obvious. This is because in the intermediate steps, we usually only use part of the conditions and these steps are not reversible.

Problem 2.3. Find all function(s) $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(x) + 2f(1-x) = 12 - 3x \tag{1}$$

for all $x \in \mathbb{R}$.

Analysis. This time there are two f's in the equation. Putting $y = 1 - x$ does not yield the result directly. However, this is still useful as it creates another similar equation. We can view $f(x)$ and $f(1-x)$ as variables and solve for them.

Solution. Putting $x = 1 - y$ in (1), we get $f(1-y) + 2f(y) = 9 + 3y$, i.e.

$$f(1-x) + 2f(x) = 9 + 3x \tag{2}$$

for all $x \in \mathbb{R}$. Considering $2\times$(2)$-$(1), we obtain

$$f(x) = \frac{1}{3}\Big(2\big(f(1-x) + 2f(x)\big) - \big(f(x) + 2f(1-x)\big)\Big) = 3x + 2$$

for all $x \in \mathbb{R}$. It remains to check

$$f(x) + 2f(1-x) = (3x+2) + 2(5-3x) = 12 - 3x.$$

$\square$

General problems in functional equation are much more difficult to solve since the equations may involve more variables, more f's or nested functions. The first basic idea is to consider the simple substitutions like $x = 0$, $x = 1$, $y = x$, etc. These could help us explore more information.

Problem 2.4. Find all function(s) $f : \mathbb{R} \to \mathbb{R}$ such that

$$xf(x+y) + y^2 = f(x-y)^2 + 3xy \tag{1}$$

for all $x, y \in \mathbb{R}$.

Analysis. As mentioned, we first consider $x = y = 0$. This gives us the value $f(0)$ and we can use this to eliminate a function term.

Solution. Put $x = y = 0$ in (1). We get $f(0)^2 = 0$ so that $f(0) = 0$.

Next, put $x = y \neq 0$ in (1) so as to create a term $f(0)$. Note that

$$\begin{aligned}
& & xf(2x) + x^2 &= f(0)^2 + 3x^2 \\
&\Rightarrow & xf(2x) &= 2x^2 \\
&\Rightarrow & f(2x) &= 2x \\
&\Rightarrow & f(t) &= t
\end{aligned}$$

for any $t \neq 0$. Together with $f(0) = 0$, we have $f(x) = x$ for any $x \in \mathbb{R}$. Lastly, we check that

$$xf(x+y) + y^2 = x(x+y) + y^2 = (x-y)^2 + 3xy = f(x-y)^2 + 3xy.$$

$\square$

Remarks. By putting $y = 0$ in (1), we immediately get $xf(x) = f(x)^2$, i.e. $f(x) = 0$ or $f(x) = x$. However, this does not mean that the only possible functions are the zero function and the identity function. It could happen that $f(a) = 0$ for some $a \in \mathbb{R}$ and $f(b) = b$ for some other $b \in \mathbb{R}$. This is a common mistake due to the abuse of the notation $f(x)$ (which sometimes means the image of x under f and sometimes means the function f).

Problem 2.5. Find all function(s) $f : \mathbb{R} \to \mathbb{R}$ such that

$$f\big((x-y)^2\big) = f(x)^2 - 2xf(y) + y^2 \tag{1}$$

for all $x, y \in \mathbb{R}$.

Analysis. This kind of problem is simple since it does not contain nested functions, the equation is non-symmetric in x and y and there are some variables x, y lying outside the functions. These mean there are more variations when applying substitutions. In particular, substituting $x = 0$ or $y = 0$ (or both) always gives useful information.

Solution. Putting $x = y = 0$ in (1), we get $f(0) = f(0)^2$. This gives $f(0) = 0$ or $f(0) = 1$. Knowing the value of $f(0)$, we can substitute $x = 0$ or $y = 0$ to get a simplified relation.

Case 1. $f(0) = 0$

Putting $x = 0$ in (1), we get $f(y^2) = y^2$. This means $f(x) = x$ for any $x \geqslant 0$. Putting $x = 1$ in (1), since $(1-y)^2 \geqslant 0$, we have

$$(1-y)^2 = 1 - 2f(y) + y^2$$

for any $y \in \mathbb{R}$. After simplification, we obtain $f(y) = y$ for any $y \in \mathbb{R}$.

Case 2. $f(0) = 1$

Putting $x = 0$ in (1), we get $f(y^2) = y^2 + 1$. This means $f(x) = x + 1$ for any $x \geqslant 0$. Putting $x = 1$ in (1), we have

$$(1-y)^2 + 1 = 4 - 2f(y) + y^2$$

for any $y \in \mathbb{R}$. This implies $f(y) = y + 1$ for any $y \in \mathbb{R}$.

It is easy to check that both $f(x) = x$ and $f(x) = x + 1$ are solutions. □

Problem 2.6. Find all function(s) $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(f(x) + y) = f(x^2 - y) + 4f(x)y \tag{1}$$

for all $x, y \in \mathbb{R}$.

Analysis. Apart from setting some variables to be 0, another useful way is to set the variables to cancel two function signs. For example, we may equate $f(x) + y$ and $x^2 - y$ here. The solution is almost complete, even though the answer may not be obvious at first glance.

Solution. Put $y = \dfrac{x^2 - f(x)}{2}$ in (1) so that $f(x) + y = x^2 - y$. Then we have

$$f(f(x) + y) = f(x^2 - y)$$

and hence $4f(x)y = 0$. This yields $f(x) = 0$ or $y = 0$. Thus, for every $x \in \mathbb{R}$, we have $f(x) = 0$ or $f(x) = x^2$. Again, note that the choice is independent for each x, although it is natural to guess that we cannot have $f(a) = 0$ and $f(b) = b^2$ simultaneously for $a, b \neq 0$. The following provides a standard way to justify this.

Suppose on the contrary that $f(a) = 0$ and $f(b) = b^2$ for some $a, b \neq 0$. We put $x = a$ and $y = b$ in (1). This yields

$$b^2 = f(a^2 - b).$$

Note that the right-hand side can only be 0 or $(a^2 - b)^2$. The former case is rejected since $b \neq 0$. Thus, we have $b^2 = (a^2 - b)^2$, which implies $a^2 = 2b$. However, there are infinitely many choices of a or infinitely many choices of b. So the relation $a^2 = 2b$ cannot always hold. This is a contradiction.

Therefore, either $f(x) = 0$ for all $x \in \mathbb{R}$ or $f(x) = x^2$ for all $x \in \mathbb{R}$. The first case is obviously a solution. We check also the second case as follows:

$$f(f(x) + y) = (x^2 + y)^2 = x^4 + 2x^2y + y^2 = (x^2 - y)^2 + 4x^2y = f(x^2 - y) + 4f(x)y.$$

□

Problem 2.7. Find all function(s) $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(f(x) + y) + f(x + f(y)) = f(x) + x \tag{1}$$

for all $x, y \in \mathbb{R}$.

Analysis. For functional equations which are not symmetric in the variables, sometimes it is useful to swap the variables and compare the equation with the original one. In this simple problem, we are almost done after carrying out this step.

Solution. Swapping the variables x and y, we obtain

$$f(f(y) + x) + f(y + f(x)) = f(y) + y. \tag{2}$$

Comparing (1) and (2), we find that

$$f(x) + x = f(y) + y$$

for any $x, y \in \mathbb{R}$. In particular, by putting $y = 0$, we get $f(x) + x = f(0)$. This implies $f(x) = -x + c$ for some constant c. Now,

$$f(f(x) + y) + f(x + f(y)) = -(-x + c + y) + c - (x - y + c) + c = 0$$

and

$$f(x) + x = (-x + c) + x = c$$

are equal iff $c = 0$. This shows the only solution is $f(x) = -x$ for all $x \in \mathbb{R}$. □

Remarks. In view of the above argument, if we can transform the equation to the form $g(x) = h(y)$ where g, h are some functions (possibly $g = h$), then we can conclude that $g(x) = h(0)$ is a constant for any x. If g is a simple function in x and $f(x)$, we can easily get an expression for $f(x)$ using this.

Problem 2.8. Find all function(s) $f : \mathbb{R} \to \mathbb{R}$ such that

$$(x - y)f(x + y) - (x + y)f(x - y) = 4xy(x^2 - y^2) \tag{1}$$

for all $x, y \in \mathbb{R}$.

Analysis. As the equation mainly involves $x - y$ and $x + y$, we may try to use substitutions to replace them. Then we see that the new variables can be separated into two independent parts. So we can reduce the equation to the form $g(x) = h(y)$ and make use of this symmetry.

Solution. By putting $x = \dfrac{u+v}{2}$ and $y = \dfrac{u-v}{2}$ in (1), or equivalently $u = x + y$ and $v = x - y$, we obtain

$$vf(u) - uf(v) = (u^2 - v^2)uv. \tag{2}$$

For nonzero u, v, this simplifies to $\dfrac{f(u)}{u} - \dfrac{f(v)}{v} = u^2 - v^2$. For such an expression, we can group the terms involving u and v to different sides, i.e.

$$\frac{f(u)}{u} - u^2 = \frac{f(v)}{v} - v^2.$$

This shows $\dfrac{f(u)}{u} - u^2$ is a constant (in particular equals $f(1) - 1$ by putting $v = 1$). Hence, we have $\dfrac{f(x)}{x} - x^2 = c$ for nonzero x where c is a constant. This yields

$$f(x) = x^3 + cx \tag{3}$$

for $x \neq 0$. Now, putting $u = 1$ and $v = 0$ in (2), we have $f(0) = 0$. Therefore, (3) holds for all $x \in \mathbb{R}$. The checking is simple:

$$\begin{aligned}
&(x-y)f(x+y) - (x+y)f(x-y) \\
&= (x-y)\big((x+y)^3 + c(x+y)\big) - (x+y)\big((x-y)^3 + c(x-y)\big) \\
&= (x-y)(x+y)(x^2 + 2xy + y^2 + c - x^2 + 2xy - y^2 - c) \\
&= 4xy(x^2 - y^2).
\end{aligned}$$

□

Problem 2.9. Find all function(s) $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(xf(y)) = f(x) + xy \tag{1}$$

for all $x, y \in \mathbb{R}$.

Analysis. Many functional equations involve nested functions. It is sometimes useful to replace x by $f(x)$ or y by $f(y)$, especially when we have obtained a relation that expresses $f(f(x))$ in terms of $f(x)$ or x.

Solution. Putting $x = 1$ in (1), we get

$$f(f(y)) = f(1) + y.$$

To use this relation, we replace x by $f(x)$ in (1) to generate the term $f(f(x))$. This implies

$$f(f(x)f(y)) = f(f(x)) + f(x)y = f(1) + x + f(x)y. \tag{2}$$

Noting that the left-hand side is symmetric in x and y while the right-hand side is not, we interchange x and y to obtain

$$f(f(x)f(y)) = f(1) + y + f(y)x. \tag{3}$$

Comparing (2) and (3), we obtain

$$(f(x) - 1)y = (f(y) - 1)x. \tag{4}$$

Putting $x = 0$ in (4), we have $(f(0)-1)y = 0$ for any $y \in \mathbb{R}$. So we must have $f(0) = 1$. For $x, y \neq 0$, the equation (4) can be rewritten as

$$\frac{f(x) - 1}{x} = \frac{f(y) - 1}{y}.$$

This shows $\dfrac{f(x) - 1}{x}$ is a constant. Therefore, $f(x) = cx + 1$ for some constant c. Note that this also holds when $x = 0$. It suffices to consider functions of the form $f(x) = cx + 1$. Now,

$$\begin{aligned} & f(xf(y)) = f(x) + xy \\ \Leftrightarrow \quad & cx(cy + 1) + 1 = cx + 1 + xy \\ \Leftrightarrow \quad & (c^2 - 1)xy = 0. \end{aligned}$$

This holds for any $x, y \in \mathbb{R}$ iff $c^2 = 1$. Therefore, the solutions are $f(x) = x + 1$ for all $x \in \mathbb{R}$ and $f(x) = -x + 1$ for all $x \in \mathbb{R}$. □

Before doing substitutions, we may first guess the answers. For many problems, these would be some rather obvious and simple functions like

$$f(x) = \pm x, \quad f(x) = \pm x + c, \quad f(x) = c, \quad f(x) = \frac{1}{x}, \quad f(x) = x^2$$

(where c is a constant). This might be useful as we have a clearer goal when solving the problem.

Next, we shall go through a well-known functional equation.

Problem 2.10. (Cauchy equation(柯西方程)) Find all function(s) $f : \mathbb{Q} \to \mathbb{Q}$ such that

$$f(x + y) = f(x) + f(y) \tag{1}$$

for all $x, y \in \mathbb{Q}$.

Analysis. Firstly, it is not hard to observe that $f(x) = cx$ is a solution for any $c \in \mathbb{Q}$. So we may assign any value to $f(1)$. Since we have a proposed answer, it is natural to use induction to verify this for $x \in \mathbb{Z}$. The general case in $\mathbb{Q}$ can be handled similarly, using the results in $\mathbb{Z}$.

Solution. Put $x = y = 0$ in (1) to get $f(0) = 2f(0)$, i.e. $f(0) = 0$.

Next, we let $f(1) = c$ for some $c \in \mathbb{Q}$. We shall prove that $f(n) = cn$ for any $n \in \mathbb{Z}$. Indeed, suppose $f(k) = ck$ for some $k \in \mathbb{Z}$. By putting $x = k$, $y = 1$ in (1), we have

$$f(k+1) = f(k) + f(1) = ck + c = c(k+1).$$

Similarly, by putting $x = k-1$, $y = 1$ in (1), we have

$$f(k-1) = f(k) - f(1) = ck - c = c(k-1).$$

By induction (forward for positive integers and backward for negative integers), the claim that $f(n) = cn$ for $n \in \mathbb{Z}$ is true.

Also, note that the above argument can be modified to show that $f(kx) = kf(x)$ for any $k \in \mathbb{Z}$ and $x \in \mathbb{Q}$ (by changing $c = f(1)$ to $c = f(x)$). Now, by putting $k = n$ and $x = \frac{m}{n} \in \mathbb{Q}$ where $n \neq 0$, we obtain

$$f\left(\frac{m}{n}\right) = \frac{1}{n} f\left(n \cdot \frac{m}{n}\right) = \frac{f(m)}{n} = c \cdot \frac{m}{n}.$$

This shows $f(x) = cx$ for all $x \in \mathbb{Q}$. We check that

$$f(x+y) = c(x+y) = cx + cy = f(x) + f(y).$$

$\square$

Remarks. Induction is a useful technique if the domain of the function is $\mathbb{N}, \mathbb{Z}, \mathbb{Q}$. Even if we are working on real-valued functions, it is sometimes useful if we know the images of a class of numbers such as integers. In some simple problems, we can follow the steps below to work out the values of $f(r)$ for $r \in \mathbb{Q}$.

- Find the values of $f(0)$ and $f(1)$ by simple substitutions.
- Guess a formula for $f(x)$.
- Show that $f(n)$ satisfies the formula for all $n \in \mathbb{N}$ by induction.
- Show that $f(n)$ satisfies the formula for all $n \in \mathbb{Z}$ by backward induction or by finding a relation between $f(x)$ and $f(-x)$.
- Show that $f(r)$ satisfies the formula for all $r \in \mathbb{Q}$ by using substitutions such as $x = \frac{m}{n}$ where $m, n \in \mathbb{Z}$.

For the Cauchy equation $f : \mathbb{R} \to \mathbb{R}$ where $f(x+y) = f(x) + f(y)$ for any $x, y \in \mathbb{R}$, the functions $f(x) = cx$ are not the only possible solutions. There are strange solutions which cannot be easily described without knowledge in linear algebra. Those functions cannot be described by a simple formula.

However, if additional conditions are given, then it is possible to reduce to this case.

Proposition 2.1. Let $f : \mathbb{R} \to \mathbb{R}$ be a function satisfying the Cauchy equation

$$f(x+y) = f(x) + f(y)$$

for all $x, y \in \mathbb{R}$. If, in addition, f satisfies one of the following conditions, then $f(x) = cx$ for all $x \in \mathbb{R}$ where c is a real constant.

(i) $f(xy) = f(x)f(y)$ for all $x, y \in \mathbb{R}$

(ii) $f(x) \geqslant 0$ for all $x \geqslant 0$

(iii) f is monotonic

(iv) f is continuous

The proof of this proposition will be given in section 3.

Many functional equations have the only solution $f(x) = x$. If we are able to derive the conditions $f(x+y) = f(x) + f(y)$ and $f(xy) = f(x)f(y)$, then we are almost done. Note that in this case, we must have $c = 0, 1$. This means f is either the zero function or the identity function. On the other hand, the condition of continuity seldom appears in Olympiad problems.

Problem 2.11. Find all function(s) $f : \mathbb{Q} \to \mathbb{Q}$ such that

$$f(x) + f(y) = 2f\left(\frac{x+y}{2}\right) \tag{1}$$

for all $x, y \in \mathbb{Q}$.

Analysis. There are some variants of the Cauchy equation. Very often, if we obtain an additive relation between the terms (without products or nested functions), then we can transform it to the Cauchy equation or apply the same trick to work out $f(r)$ for $r \in \mathbb{Q}$.

In this problem, since the equation remains the same if we add a constant to f, sometimes we can use a simple trick to assume $f(0) = 0$ for convenience.

Solution. Note that the equation is the same as

$$\big(f(x) - f(0)\big) + \big(f(y) - f(0)\big) = 2\left[f\left(\frac{x+y}{2}\right) - f(0)\right].$$

By letting $g(x) = f(x) - f(0)$, we obtain the same equation $g(x) + g(y) = 2g\left(\frac{x+y}{2}\right)$ for g. But then we have an extra condition that $g(0) = f(0) - f(0) = 0$. Therefore, without loss of generality(WLOG), we may assume $f(0) = 0$.

Putting $y = 0$ in (1), we obtain $f(x) = 2f\left(\frac{x}{2}\right)$. Replacing x by $2x$, this means

$$f(2x) = 2f(x).$$

We further replace x by $\dfrac{x+y}{2}$ and combine the result with (1). This gives the Cauchy equation

$$f(x)+f(y)=f\left(2\cdot\frac{x+y}{2}\right)=f(x+y).$$

Therefore, we must have $f(x)=cx$ for all $x\in\mathbb{Q}$ where $c\in\mathbb{Q}$ is a constant. In general, we have $f(x)=cx+d$ where $c,d\in\mathbb{Q}$. It is easy to check that all linear functions are solutions. □

Problem 2.12. Find all function(s) $f:\mathbb{Q}\to\mathbb{Q}$ such that

$$f(x-y+f(y))=f(x)+f(y) \tag{1}$$

for all $x,y\in\mathbb{Q}$.

Analysis. It seems not easy to find $f(0)$ immediately. Also, knowing $f(0)=0$ does not give us much information. Indeed, our first step is to put $x=y$ so that the term on the left-hand side is simplified. This gives us a relationship between $f(f(x))$ and $f(x)$. Again, we can use the trick of replacing y by $f(y)$.

Solution. Substitute $x=y$ in (1). This gives

$$f(f(x))=2f(x). \tag{2}$$

Replace y by $f(y)$ in (1). We get $f\big(x-f(y)+f(f(y))\big)=f(x)+f(f(y))$. Using (2), we obtain

$$f(x+f(y))=f(x)+2f(y). \tag{3}$$

We compare (1) and (3) and try to eliminate the clumsy term on the left. This can be done by replacing x by $x-y$ in (3), yielding

$$f(x-y+f(y))=f(x-y)+2f(y). \tag{4}$$

Equating (1) and (4), we obtain $f(x)=f(x-y)+f(y)$. This is simply the Cauchy equation. Thus, $f(x)=cx$ for any $x\in\mathbb{Q}$ where c is a constant. Putting this in (1), we have

$$c(x-y+cy)=f(x-y+f(y))=f(x)+f(y)=cx+cy.$$

This holds for all $x,y\in\mathbb{Q}$ iff $c=0$ or $c=2$. Thus, the solutions are $f(x)=0$ for all $x\in\mathbb{Q}$ or $f(x)=2$ for all $x\in\mathbb{Q}$. □

Remarks. In view of the method we use to solve the Cauchy equation, this problem can also be solved using induction. But knowing the solutions to the Cauchy equation saves us some time.

Problem 2.13. Find all function(s) $f:\mathbb{Q}\to\mathbb{Q}$ such that

$$f(x+y)+f(x)f(y)=f(x)+f(y)+f(xy) \tag{1}$$

for all $x,y\in\mathbb{Q}$.

Analysis. There are a few solutions by checking linear f. Since f is only defined on $\mathbb{Q}$, induction is a useful tool. To use induction to find the images of all integers or rational numbers, we normally first try to look for the values of $f(0), f(1), f(2)$. If the functional equation involves both $x+y$ and xy, the substitution $x=y=2$ may be useful since $2+2=(2)(2)$. Thus, there is some hope that we can compute $f(2)$.

Solution. Put $y=0$ in (1). This gives $f(x)f(0)=2f(0)$. If $f(0)\neq 0$, then $f(x)=2$ for all $x\in\mathbb{Q}$, which is obviously a solution. Now, suppose $f(0)=0$.

Put $x=y=2$ in (1). This gives $f(2)^2=2f(2)$ so that $f(2)=0$ or $f(2)=2$. Substituting $y=1$ in (1), we obtain

$$f(x+1)+f(x)f(1)=2f(x)+f(1). \tag{2}$$

We further substitute $x=1$ in (2) to obtain $f(2)+f(1)^2=3f(1)$. Each value of $f(2)$ corresponds to 2 possible values of $f(1)$. There are a few cases.

- If $f(2)=2$ and $f(1)=2$, then (2) becomes $f(x+1)=2$ for all $x\in\mathbb{Q}$. This contradicts $f(0)=0$.

- If $f(2)=2$ and $f(1)=1$, then (2) becomes

  $$f(x+1)=f(x)+1. \tag{3}$$

  By induction, this yields $f(n)=n$ for all $n\in\mathbb{Z}$. To use this result, we assume $y=k$ is an integer in (1). This implies

  $$f(x+k)+kf(x)=f(x)+k+f(kx).$$

  Using (3) repeatedly, we have $f(x+k)=f(x)+k$ and hence this becomes $kf(x)=f(kx)$. For $x\in\mathbb{Q}$, we set k to be the denominator of x so that $kx\in\mathbb{Z}$. Then $f(x)=\dfrac{1}{k}(kx)=x$. The identity function is clearly another solution.

- If $f(2)=0$ and $f(1)=0$, then we can use similar method as the previous case. Firstly, (2) becomes $f(x+1)=2f(x)$ so that $f(n)=0$ for all $n\in\mathbb{Z}$ by induction. Also, we have $f(x+k)=2^kf(x)$ for any $x\in\mathbb{Q}$ and $k\in\mathbb{Z}$. Consider (1) and assume $y=k$ is an integer. Then

  $$f(kx)=f(x+k)-f(x)=(2^k-1)f(x).$$

  For $x\in\mathbb{Q}$, we choose k to be the denominator of x so that $f(kx)=0$. This implies $f(x)=0$ since $k\neq 0$. The zero function is clearly the third solution.

- If $f(2)=0$ and $f(1)=3$, it is natural to guess this yields a contradiction if one has tested all linear functions at the beginning. Indeed, if we apply the same method, we can show that $f(2k)=0$ and $f(2k-1)=3$ for any $k\in\mathbb{Z}$ by induction, and $f(x+1)+f(x)=3$ so that $f(x+2)=f(x)$ for any $x\in\mathbb{Q}$. Then we put $x=2$ and $y=\dfrac{1}{2}$ in (1) to obtain

  $$f\left(2+\frac{1}{2}\right)+f(2)f\left(\frac{1}{2}\right)=f(2)+f\left(\frac{1}{2}\right)+f(1).$$

This implies $f\left(\frac{5}{2}\right) = f\left(\frac{1}{2}\right) + 3$. But then $f\left(\frac{5}{2}\right) = f\left(\frac{1}{2}\right)$. This is a contradiction.

Therefore, the solutions are $f(x) = 0$ for any $x \in \mathbb{Q}$, $f(x) = 2$ for any $x \in \mathbb{Q}$ and $f(x) = x$ for any $x \in \mathbb{Q}$. $\square$

Remarks. We shall discuss the same functional equation for $f : \mathbb{R} \to \mathbb{R}$ in problem 2.17.

Problem 2.14. Find all function(s) $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(x^2 + yf(z)) = xf(x) + zf(y) \tag{1}$$

for all $x, y, z \in \mathbb{R}$.

Analysis. If there is a term x^2 in the equation, sometimes it is useful to replace x by $-x$. By comparing with the original equation, we can obtain some relation between $f(x)$ and $f(-x)$.

It is not hard to see that both the zero function and the identity function are solutions. This suggests we may be able to prove that f satisfies the Cauchy equation and $f(xy) = f(x)f(y)$.

Solution. Putting $y = z = 0$ in (1), we obtain

$$f(x^2) = xf(x). \tag{2}$$

Putting $x = 0$ in (2), we have $f(0) = 0$. Replace x by $-x$ in (2). We get

$$f(x^2) = -xf(-x).$$

Comparing this with (2), we have $xf(x) = -xf(-x)$. For nonzero x, this yields

$$f(-x) = -f(x), \tag{3}$$

which also holds for $x = 0$. To use (3), we replace some of the variables by their negatives. For example, we can replace y by $-y$ in (1). This gives

$$f(x^2 - yf(z)) = xf(x) - zf(y).$$

Adding this to (1) and using (2), we have

$$f(x^2 + yf(z)) + f(x^2 - yf(z)) = 2xf(x) = 2f(x^2).$$

Note that z does not play any role here unless $f(z) = 0$ for all $z \in \mathbb{R}$. Indeed, the zero function is clearly a solution. If f is not the zero function, we can choose z such that $f(z) \neq 0$ and replace y by $\frac{y}{f(z)}$. Then we obtain

$$f(x^2 + y) + f(x^2 - y) = 2f(x^2) \tag{4}$$

for any $x, y \in \mathbb{R}$. This equation looks similar to the Cauchy equation. Therefore, it is natural to transform it to that.

Firstly, (4) can be rewritten as

$$f(x+y) + f(x-y) = 2f(x) \tag{5}$$

for $x \geqslant 0$. For $(-x) < 0$, by using (3) and (5), we also have

$$f((-x)+y) + f((-x)-y) = -f(x-y) - f(x+y) = -2f(x) = 2f(-x).$$

Thus, (5) holds for any $x, y \in \mathbb{R}$. As in problem 2.11, this can be transformed to the Cauchy equation

$$f(x) + f(y) = f(x+y).$$

As suggested, our next goal is to prove $f(xy) = f(x)f(y)$. In view of (1), we should work on y and z since they are more related to products. Thus, we put $x = 0$ to get

$$f(yf(z)) = zf(y). \tag{6}$$

Putting $y = 1$ in (6), we find that

$$f(f(z)) = f(1)z. \tag{7}$$

After obtaining a relation between $f(f(z))$ and z, there are two common ways to proceed. One way is to replace z by $f(z)$. Another way is to apply f to both sides of an equation.

For example, applying f to both sides of (6) and using (7), we deduce

$$f(1)yf(z) = f\big(f(yf(z))\big) = f(zf(y)).$$

Swapping y and z in (6), we see that $f(zf(y)) = yf(z)$. So the above equation can be rewritten as

$$f(1)yf(z) = yf(z).$$

As we have assumed f is not identically zero, we can choose y and z so that $yf(z) \neq 0$. This yields $f(1) = 1$. Thus, (7) becomes

$$f(f(z)) = z.$$

Now, we can also use the second idea. Indeed, replacing y by $f(y)$ in (6) and using $f(f(y)) = y$, we get

$$f(f(y)f(z)) = zf(f(y)) = yz.$$

Applying f to both sides, we finally obtain

$$f(y)f(z) = f\big(f(f(y)f(z))\big) = f(yz).$$

This is the condition that we need. By proposition 2.1, the only possibilities are $f(x) = 0$ for all $x \in \mathbb{R}$ and $f(x) = x$ for all $x \in \mathbb{R}$. Clearly, both are solutions. $\square$

Remarks. Apart from demonstrating how to use proposition 2.1 to work on a functional equation on $\mathbb{R}$, we have also gone through several important techniques in solving functional equations, like replacing x by $-x$, replacing x by $f(x)$ and applying f to both sides of the equation. In general, there are far more variations of these substitutions. To decide which substitutions to use, one should keep all the results we have in mind (like $f(-x) = -f(x)$ and $f(f(x)) = x$) and see how to use those results.

Problem 2.15. (IMO HK TST 2020 Test 2 Problem 4) Find all real-valued functions f defined on the set of real numbers such that

$$f(f(x)+y)+f(x+f(y))=2f(xf(y)) \tag{1}$$

for any real numbers x and y.

Analysis. We shall combine several techniques we have introduced to work on this selection test problem. The first idea that comes to our mind is to swap the variables x and y, since the equation looks similar to problem 2.7.

Solution. Clearly, all constant functions are solutions. In the following, we show that there is no solution if f is not a constant function.

Swapping x and y, we get $f(f(y)+x)+f(y+f(x))=2f(yf(x))$. Comparing with (1), we get

$$f(xf(y))=f(yf(x)). \tag{2}$$

Then we substitute some small values to get more information. Putting $x=0$ in (2), we find that $f(0)=f(yf(0))$. If $f(0)\neq 0$, then $yf(0)$ runs through all real values. This implies f is a constant function, which is a contradiction. So we have $f(0)=0$. To use this, we put $x=0$ in (1) to get

$$f(y)+f(f(y))=0. \tag{3}$$

In view of this relation between $f(f(y))$ and $f(y)$, we replace x by $f(x)$ in (1). Using (3), this gives

$$f(-f(x)+y)+f(f(x)+f(y))=2f(f(x)f(y)).$$

Swapping x and y, and comparing with this equation again, we get

$$f(-f(x)+y)=f(-f(y)+x). \tag{4}$$

Putting $y=f(x)$ in (4) and using (3), we have

$$f(f(x)+x)=0. \tag{5}$$

To use this result, a natural step is to put $x=y$ in (1). This gives

$$f(xf(x))=0$$

for any $x \in \mathbb{R}$. In particular, by putting $x = 1$, we get $f(f(1)) = 0$. Then we put $y = 1$ in (3) to deduce $f(1) = 0$.

Now, putting $x = 1$ in (2), we obtain $f(f(y)) = 0$. Together with (3), we find that $f(y) = 0$ for any $y \in \mathbb{R}$. This is a contradiction as we have assumed f is non-constant.

Therefore, the only solutions are the constant functions. $\square$

Sometimes we may try to relate the term $f(x)$ to some simple terms such as $f(kx)$, $f(x+k)$ and $f(x^k)$ where $k \in \mathbb{N}$. By setting up sufficiently many equations involving these simple terms, we may be able to solve for $f(x)$.

Problem 2.16. Find all function(s) $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(x)^2 = 2f(xy) + 2f(x(x-y)) - 4x + 5 \tag{1}$$

for all $x, y \in \mathbb{R}$.

Analysis. There is no special technique in this solution. In fact, we shall demonstrate how to apply various substitutions to find out the relations between different terms, and then solve for $f(x)$ directly. Though this may not be the best solution to this particular problem, the thinking steps are quite straightforward, and can be applied to many problems of this kind.

Solution. We first find the value of $f(0)$. Putting $x = 0$ in (1), we get

$$f(0)^2 = 4f(0) + 5.$$

This gives $f(0) = -1, 5$.

Case 1. $f(0) = -1$

Since we have found the value of $f(0)$, we try to put some values so that some terms are eliminated. We either put $y = 0$ or $y = x$ in (1) to get

$$f(x)^2 = 2f(x^2) - 4x + 3. \tag{2}$$

This gives a relation between $f(x)$ and $f(x^2)$. This can be easily modified to give a relation between $f(x)$ and $f(-x)$. Indeed, replacing x by $-x$ in (2), we have

$$f(-x)^2 = 2f(x^2) + 4x + 3. \tag{3}$$

The difference of (2) and (3) gives

$$f(-x)^2 - f(x)^2 = 8x. \tag{4}$$

Next, we can eliminate some terms by equating some expressions. For example, by considering $xy = x(x-y)$, we may put $x = 2y$ in (1). This gives

$$f(2y)^2 = 4f(2y^2) - 8y + 5. \tag{5}$$

Now, the square term on the left can be removed by using (2). Indeed, putting $x = 2y$ in (2) gives

$$f(2y)^2 = 2f(4y^2) - 8y + 3. \tag{6}$$

Comparing (5) and (6), we obtain $f(4y^2) = 2f(2y^2) + 1$. By the change of variable $x = \sqrt{2}y$, this yields

$$f(2x^2) = 2f(x^2) + 1. \tag{7}$$

On the other hand, since we have obtained relations involving $f(-x)$, one idea is to apply some substitutions that create the negative of some terms. For example, we replace x by $-x$ and y by x in (1) to obtain

$$f(-x)^2 = 2f(-x^2) + 2f(2x^2) + 4x + 5.$$

Taking the difference with (2), this gives

$$f(-x)^2 - f(x)^2 = 2f(-x^2) - 2f(x^2) + 2f(2x^2) + 8x + 2.$$

Using (4) and (7), we have

$$\begin{aligned} & f(-x)^2 - f(x)^2 = 2f(-x^2) - 2f(x^2) + 2f(2x^2) + 8x + 2 \\ \Rightarrow \quad & 8x = 2f(-x^2) - 2f(x^2) + 2(2f(x^2) + 1) + 8x + 2 \\ \Rightarrow \quad & f(-x^2) + f(x^2) = -2. \end{aligned}$$

This is the same as

$$f(-x) + f(x) = -2 \tag{8}$$

no matter x is positive or not. This gives another relation between $f(-x)$ and $f(x)$. Therefore, combining (4) and (8), we have

$$f(-x) - f(x) = \frac{f(-x)^2 - f(x)^2}{f(-x) + f(x)} = -4x.$$

Hence, using (8) again, we easily obtain

$$f(x) = \frac{[f(-x) + f(x)] - [f(-x) - f(x)]}{2} = 2x - 1$$

for all $x \in \mathbb{R}$. We check that

$$\begin{aligned} 2f(xy) + 2f(x(x-y)) - 4x + 5 &= 2(2xy - 1) + 2(2x(x-y) - 1) - 4x + 5 \\ &= 4x^2 - 4x + 1 = (2x-1)^2 = f(x)^2. \end{aligned}$$

Case 2. $f(0) = 5$

Similarly, putting $y = 0$ in (1), we get

$$f(x)^2 = 2f(x^2) - 4x + 15. \tag{9}$$

Replacing x by $-x$ in (9), we have

$$f(-x)^2 = 2f(x^2) + 4x + 15. \tag{10}$$

The difference of (9) and (10) gives

$$f(-x)^2 - f(x)^2 = 8x. \tag{11}$$

Putting $x = 2y$ in (1), we have

$$f(2y)^2 = 4f(2y^2) - 8y + 5. \tag{12}$$

Substitute $x = 2y$ in (9). This yields

$$f(2y)^2 = 2f(4y^2) - 8y + 15. \tag{13}$$

Comparing (12) and (13), we obtain $f(4y^2) = 2f(2y^2) - 5$. By the change of variable $x = \sqrt{2}y$, this yields

$$f(2x^2) = 2f(x^2) - 5. \tag{14}$$

Next, we replace x by $-x$ and y by x in (1) to obtain

$$f(-x)^2 = 2f(-x^2) + 2f(2x^2) + 4x + 5.$$

Taking the difference with (9), this gives

$$f(-x)^2 - f(x)^2 = 2f(-x^2) - 2f(x^2) + 2f(2x^2) + 8x - 10.$$

Using (11) and (14), we have

$$\begin{aligned}
& f(-x)^2 - f(x)^2 = 2f(-x^2) - 2f(x^2) + 2f(2x^2) + 8x - 10 \\
\Rightarrow \quad & 8x = 2f(-x^2) - 2f(x^2) + 2(2f(x^2) - 5) + 8x - 10 \\
\Rightarrow \quad & f(-x^2) + f(x^2) = 10.
\end{aligned}$$

This is the same as

$$f(-x) + f(x) = 10 \tag{15}$$

for all $x \in \mathbb{R}$. Combining (11) and (15), we have

$$f(-x) - f(x) = \frac{f(-x)^2 - f(x)^2}{f(-x) + f(x)} = \frac{4}{5}x.$$

Hence, using (15) again, we easily obtain

$$f(x) = \frac{[f(-x) + f(x)] - [f(-x) - f(x)]}{2} = -\frac{2}{5}x + 5$$

for all $x \in \mathbb{R}$. However, we check that this function does not satisfy the given equation. For example, we put $x = 5$ and $y = 0$ in (1). We find that $f(5)^2 = 3^2 = 9$ while

$$2f(0) + 2f(25) - 4(5) + 5 = 2(5) + 2(-5) - 4(5) + 5 = -15.$$

The two values do not agree with each other. So this solution is rejected.

Therefore, the only solution is $f(x) = 2x - 1$. $\square$

Remarks. For the case $f(0) = 5$, since it leads to no solution, there should be a better way to reject this case. However, since the substitutions in case 1 can be applied directly in case 2, we do not need to think of another new method to handle it. This solution also tells us the importance of checking the solution.

Problem 2.17. Find all function(s) $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(x+y) + f(x)f(y) = f(x) + f(y) + f(xy) \tag{1}$$

for all $x, y \in \mathbb{R}$.

Analysis. This is a generalization of problem 2.13. This time we shall introduce another technique. Sometimes it may be useful to introduce more variables to the equation so that we can have more ways to work on. This can also distort the symmetry of the equation and hence new equations can be obtained by swapping variables.

Solution. As in the solution of problem 2.13, the constant function $f(x) = 2$ is a solution. Other than that, we must have $f(0) = 0$. Also, we have $f(x) = 0$ for all $x \in \mathbb{Q}$ or $f(x) = x$ for all $x \in \mathbb{Q}$.

Replace y by $y + z$ in (1). This gives

$$f(x+y+z) = f(x) + f(y+z) + f(xy+xz) - f(x)f(y+z).$$

We can further break each term of the form $f(u+v)$ into $f(u), f(v), f(uv)$ using (1). Indeed, by

$$f(y+z) = f(y) + f(z) + f(yz) - f(y)f(z)$$

and

$$f(xy+xz) = f(xy) + f(xz) + f(x^2yz) - f(xy)f(xz),$$

the above equation becomes

$$\begin{aligned} f(x+y+z) = f(x) + f(y) + f(z) + f(xy) + f(yz) + f(xz) - f(x)f(y) - f(y)f(z) \\ - f(x)f(z) + f(x)f(y)f(z) + f(x^2yz) - f(xy)f(xz) - f(x)f(yz). \end{aligned}$$

Though it looks clumsy, we can now use the symmetry to simplify it. By swapping x and y and comparing with the same equation, we deduce

$$f(x^2yz) - f(xy)f(xz) - f(x)f(yz) = f(xy^2z) - f(xy)f(yz) - f(y)f(xz). \tag{2}$$

After this mysterious procedure, we obtain another functional equation with more variables. Then we can continue with the usual steps. We put $y = 1$ in (2) to cancel two of the terms. This gives

$$f(x^2z) - f(x)f(xz) = f(xz) - f(1)f(xz).$$

Note that the left-hand side can be simplified using (1). Indeed, we have

$$f(x(xz)) - f(x)f(xz) = f(x + xz) - f(x) - f(xz).$$

Thus, the above equation becomes

$$f(x + xz) = f(x) + \big(2 - f(1)\big)f(xz). \tag{3}$$

If $f(n) = 0$ for all $n \in \mathbb{Z}$, we put $z = -1$ in (3) so that $f(x) = -2f(-x)$. It follows that

$$f(x) = -2f(-x) = 4f(x),$$

and hence f is identically zero.

If $f(n) = n$ for all $n \in \mathbb{Z}$, we replace z by $\dfrac{y}{x}$ in (3) for nonzero x. This yields the Cauchy equation $f(x + y) = f(x) + f(y)$, which also holds for $x = 0$. Using (1), we obtain the product condition $f(xy) = f(x)f(y)$. By proposition 2.1 and the fact $f(1) = 1$, the remaining solution must be the identity function.

It is trivial to check that all 3 solutions $f(x) = 0$, $f(x) = 2$ and $f(x) = x$ satisfy the given equation. □

We now summarize all basic tricks introduced in this subsection.

- basic techniques
  - guessing the answers
  - change of variables (problems 2.1, 2.2, 2.3)
  - rejecting piecewise functions (problem 2.6)
  - making use of symmetry (problems 2.7, 2.8, 2.9)
- substitutions
  - putting $x = 0$, $x = \pm 1$, $x = y$, etc. (problems 2.4, 2.5)
  - equating two terms (problems 2.6, 2.16)
  - replacing x by $-x$ (problems 2.14, 2.16)
  - swapping x and y (problems 2.7, 2.15, 2.17)
  - replacing x by $f(x)$ (problems 2.9, 2.12, 2.14, 2.15)
  - applying f to both sides (problem 2.14)
- Cauchy equation and related techniques
  - induction (problems 2.10, 2.13)
  - extending results from $\mathbb{Z}$ to $\mathbb{Q}$ (problems 2.10, 2.13)
  - Cauchy equation (problems 2.10, 2.11, 2.12, 2.14, 2.17)

- special techniques
  - defining a new function (problem 2.11)
  - adding a new variable (problem 2.17)

Of course it does not mean that these are the only techniques and substitutions that one should use when working on functional equations. These are just some common methods that apply to many different problems. In general, to solve a particular problem, one has to be flexible and think of substitutions which can simplify the given equations or provide useful information.

2.2 Injectivity, Surjectivity and Bijectivity

In this subsection, we shall introduce several special kinds of functions which are commonly used.

Definition 2.1. A function $f : A \to B$ is called *injective*(單射) or *one-to-one* if $f(a_1) = f(a_2)$ implies $a_1 = a_2$ for any $a_1, a_2 \in A$.

In other words, the images of distinct elements are distinct. For example, the functions $f : \mathbb{R} \to \mathbb{R}$ and $g : \mathbb{R} \to \mathbb{R}$ defined by

$$f(x) = x \quad \text{and} \quad g(x) = \begin{cases} x+1 & \text{if } x \geqslant 0, \\ 2x & \text{if } x < 0 \end{cases}$$

are injective functions. However, the function $h : \mathbb{R} \to \mathbb{R}$ defined by $h(x) = x^2$ is not, since $h(1) = h(-1) = 1$. The concept of injectivity depends on the domain of the function. For example, if we restrict h to the domain $\mathbb{R}_0^+$, then h becomes injective.

For general problems in functional equation, $f(a_1) = f(a_2)$ does not necessarily imply $a_1 = a_2$. So if we are able to prove that f is injective, we can cancel the function signs on both sides. This is the most useful property of injective functions for our purpose.

Definition 2.2. A function $f : A \to B$ is called *surjective*(滿射) or *onto* if for any $b \in B$, there exists $a \in A$ such that $f(a) = b$.

In other words, the range is the same as the codomain of the function. For example, the functions $f : \mathbb{R} \to \mathbb{R}$ and $g : \mathbb{R} \to \mathbb{R}$ defined by

$$f(x) = x \quad \text{and} \quad g(x) = \begin{cases} 2x & \text{if } x \geqslant 0, \\ \dfrac{x}{2} & \text{if } x < 0 \end{cases}$$

are surjective functions. However, the function $h : \mathbb{R} \to \mathbb{R}$ defined by $h(x) = x^2$ is not, since $h(x) \neq -1$ for any $x \in \mathbb{R}$. Similar to the injectivity, the concept of surjectivity depends on the codomain of the function. For example, if the codomain of h is changed to $\mathbb{R}_0^+$, then h becomes surjective.

Once we have shown that a function is surjective, we can replace $f(x)$ by any particular value like $0, \pm 1$ or y, which may simplify some equations. To prove that f is surjective, we usually try to rearrange the terms to obtain the form $x = f(g(x))$ where x is arbitrary. This can sometimes be done in an obvious way.

It is also useful if we can prove that a function is surjective on some subset of the codomain. For example, if we can show that $f(x) = a$ is solvable for every $a \geqslant 0$, this may already give us some useful information to work on.

Lastly, we have one more definition.

Definition 2.3. A function $f : A \to B$ is called *bijective*(雙射) or *one-to-one correspondence* if it is both injective and surjective.

These special functions are important because most solutions to functional equations are of these types. For example, linear functions $f(x) = ax + b$ where $a \neq 0$ are bijective. On the other hand, the constant functions $f(x) = c$ are neither injective nor surjective. But this special case can usually be considered separately.

We first work on some simple examples to illustrate how to show that a function is injective or surjective.

Problem 2.18.

(a) Suppose $f : \mathbb{R} \to \mathbb{R}$ satisfies

$$f(x + f(y)) = x + f^3(f(y) + y) \tag{1}$$

for all $x, y \in \mathbb{R}$. Prove that f is bijective.

(b) Suppose $g : \mathbb{R} \to \mathbb{R}$ satisfies

$$g(g(x) + y) = 2x + g^3(x + y) + g(2y - x) \tag{2}$$

for all $x, y \in \mathbb{R}$. Prove that g is surjective.

(c) Suppose $h : \mathbb{R} \to \mathbb{R}$ satisfies

$$h(h(x) + y) = 2x + h(h(y) - x) \tag{3}$$

for all $x, y \in \mathbb{R}$. Prove that h is bijective.

Analysis. Part (a) has the form $f(f_1(x,y)) = x + f(f_2(x,y))$. When there is a single x outside the functions, it is usually easy to prove the injectivity or surjectivity. For the injectivity, we first assume $f(\alpha) = f(\beta)$. Then we substitute suitable expressions to generate the term $f(\alpha)$ in one equation, and substitute analogous things to generate the term $f(\beta)$ in another equation. By comparing the equations, we may obtain $\alpha = \beta$. Therefore, it is preferable to have expressions containing α and β outside the functions. For the surjectivity, we just need to substitute a certain y to make all but one function a constant. Then we can use the single x outside the functions to show that the expression covers all real values.

For (b), there are 3 terms on the two sides involving the function g. One way is to cancel two terms, such as equating $g(x) + y = 2y - x$. The remaining step is trivial.

Lastly, the surjectivity of (c) can be proved similarly. For the injectivity, we shall see how we can use the surjectivity to prove the result.

Solution.

(a) Firstly, consider $f(\alpha) = f(\beta)$. Putting $x = \alpha - f(y)$ and $x = \beta - f(y)$ in (1) respectively, we have

$$\begin{cases} f(\alpha) = \alpha - f(y) + f^3(f(y) + y), \\ f(\beta) = \beta - f(y) + f^3(f(y) + y). \end{cases}$$

Comparing these equations, we get $\alpha = \beta$. This means f is injective.

Next, we put $y = 0$ in (1). This gives $f(x + f(0)) = x + c$ where $c = f^4(0)$. By the change of variable $z = x + c$, we have $f(z - c + f(0)) = z$, meaning that f is surjective, as z can be any real number.

(b) We substitute y so that $g(x) + y = 2y - x$. In other words, we put $y = g(x) + x$ in (2). Then we have $g^3(g(x) + 2x) = -2x$, and hence $g\left(g_1\left(-\frac{x}{2}\right)\right) = x$ where $g_1(x) = g^2(g(x) + 2x)$. Since x is arbitrary, this shows g is surjective.

(c) For the surjectivity, we put $y = -h(x)$ in (3) so that $h(0) - 2x = h(h_1(x))$, where $h_1(x) = h(-h(x)) - x$. Replacing x by $\frac{h(0) - x}{2}$, we get $x = h\left(h_1\left(\frac{h(0) - x}{2}\right)\right)$. Since x is arbitrary, this shows h is surjective.

We now prove the injectivity. Suppose $h(\alpha) = h(\beta)$. Putting $x = \alpha$ and $x = \beta$ in (3) respectively, we have

$$\begin{cases} h(h(\alpha) + y) = 2\alpha + h(h(y) - \alpha), \\ h(h(\beta) + y) = 2\beta + h(h(y) - \beta). \end{cases}$$

Note that we do not immediately obtain $\alpha = \beta$ by comparing the two equations. The same happens if we consider the substitutions $y = \alpha$ and $y = \beta$ as well. The only information that we can deduce from the above equation is that

$$2\alpha + h(h(y) - \alpha) = 2\beta + h(h(y) - \beta) \tag{4}$$

for all $y \in \mathbb{R}$. As y is arbitrary, we still have some flexibility. For example, Noting that h is surjective, there exists y such that $h(y) = \alpha + \beta$. Putting this y in (4), we obtain

$$2\alpha + h(\beta) = 2\beta + h(\alpha).$$

As $h(\alpha) = h(\beta)$, this yields $\alpha = \beta$ as desired. □

Remarks. Proving that a function is injective can be difficult. Part (c) shows that the proof of the injectivity may require some tricky substitutions. But at the same time, this also provides a simple application of the surjectivity of a function. Without this, it is hard to proceed after obtaining (4).

Problem 2.19. Find all function(s) $f : \mathbb{R} \to \mathbb{R}$ such that

$$f\big(f(x)^2 + f(y)\big) = xf(x) + y \tag{1}$$

for all $x, y \in \mathbb{R}$.

Analysis. This functional equation looks similar to the one in problem 2.18(a). So there should be no problem in proving the bijectivity. We shall see how this property can be used to simplify the equation.

Solution. Suppose $f(\alpha) = f(\beta)$. Putting $y = \alpha$ and $y = \beta$ respectively in (1), we get

$$\begin{cases} f\big(f(x)^2 + f(\alpha)\big) &= xf(x) + \alpha, \\ f\big(f(x)^2 + f(\beta)\big) &= xf(x) + \beta. \end{cases}$$

Comparing the two equations, we obtain $\alpha = \beta$, so f is injective.

Next, replacing y by $y - xf(x)$ in (1), we see that any $y \in \mathbb{R}$ is the image of something. Therefore, f is also surjective.

Now, one way of using the surjectivity is to set a constant value for $f(x)$. For example, there exists x such that $f(x) = 0$. Putting this x in (1), the equation is simplified to

$$f(f(y)) = y. \tag{2}$$

Again, we recall those standard steps after obtaining a relation like this. We replace x by $f(x)$ in (1). This yields $f\big(f(f(x))^2 + f(y)\big) = f(x)f(f(x)) + y$. Then by using (2), this is simplified to

$$f(x^2 + f(y)) = xf(x) + y.$$

Comparing this with (1), we find that

$$f\big(f(x)^2 + f(y)\big) = f(x^2 + f(y)).$$

Here is one situation that we can use the injectivity. Indeed, this means we can remove f from both sides to obtain

$$f(x)^2 + f(y) = x^2 + f(y).$$

Clearly, this yields $f(x) = \pm x$ for each $x \in \mathbb{R}$.

Then we can use the standard trick of rejecting the piecewise functions. Suppose $f(a) = a$ and $f(b) = -b$ for some $a, b \neq 0$. Substituting $x = a$ and $y = b$ in (1), we have $f\big(f(a)^2 + f(b)\big) = af(a) + b$, which means

$$f(a^2 - b) = a^2 + b.$$

As $f(x) = \pm x$ for each x, we must have $a^2 - b = \pm(a^2 + b)$, i.e. $a = 0$ or $b = 0$. Both are not possible. Therefore, either $f(x) = x$ for all $x \in \mathbb{R}$ or $f(x) = -x$ for all $x \in \mathbb{R}$. It is easy to check that both are solutions. □

Problem 2.20. Find all function(s) $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(f(x) + yz) = x + f(y)f(z) \tag{1}$$

for all $x, y, z \in \mathbb{R}$.

Analysis. Again, it is easy to prove that f is bijective. This is another simple example to demonstrate the use of the bijectivity.

Solution. Suppose $f(\alpha) = f(\beta)$. Putting $x = \alpha$ and $x = \beta$ respectively in (1), we get

$$\begin{cases} f(f(\alpha) + yz) &= \alpha + f(y)f(z), \\ f(f(\beta) + yz) &= \beta + f(y)f(z). \end{cases}$$

Obviously, this yields $\alpha = \beta$ so that f is injective.

Next, replacing x by $x - f(y)f(z)$ in (1), we see that any $x \in \mathbb{R}$ is the image of something. Therefore, f is also surjective.

This time we try to use the injectivity to remove the function signs. This means we want to get an equation of the form $f(\cdots) = f(\cdots)$. For example, we can put $x = 0$ and $f(z) = 1$ in (1). Note that we cannot do so at the beginning since there may not be z such that $f(z) = 1$. But now the existence of z is guaranteed by the surjectivity of f. Thus, using this substitution, we obtain

$$f(f(0) + yz) = f(y)$$

for any $y \in \mathbb{R}$ (with z being fixed), and hence $f(0) + yz = y$ for all $y \in \mathbb{R}$. This is only true when $f(0) = 0$ and $z = 1$. Therefore, we have proved $f(0) = 0$ and $f(1) = 1$ in an indirect way.

The remaining steps are standard. For example, by putting $z = 0$ in (1), we obtain

$$f(f(x)) = x. \tag{2}$$

Then we again replace x by $f(x)$ and put $z = 1$ in (1). This gives the equation $f\big(f(f(x)) + y\big) = f(x) + f(y)$. By (2), this is essentially the Cauchy equation

$$f(x+y) = f(x) + f(y). \tag{3}$$

Next, putting $x = 0$ in (1), we have

$$f(yz) = f(y)f(z). \tag{4}$$

Equations (3) and (4) are the conditions in proposition 2.1. It follows that f can only be the zero function or the identity function. The former case is rejected since $f(1) = 1$. It suffices to check $f(x) = x$ satisfies the equation. Indeed,

$$f(f(x) + yz) = x + yz = x + f(y)f(z).$$

□

Problem 2.21. Find all function(s) $f : \mathbb{R} \to \mathbb{R}$ such that

$$f\big(2f(y + f(x))\big) = f(x+y) + f(x) + y \tag{1}$$

for all $x, y \in \mathbb{R}$.

Analysis. The problem can be easily solved if we have the injective condition. Indeed, by putting $y = -f(x)$ in (1) and using the injectivity, we immediately obtain $f(x) = x + c$ for some constant c. So the major part of the solution is to prove that f is injective.

Suppose $f(\alpha) = f(\beta)$. Following the standard procedure of substituting $x = \alpha$ and $x = \beta$ respectively, we cannot deduce $\alpha = \beta$ at once. However, if $\alpha > \beta$, we can show that f is periodic with period $\alpha - \beta$. This often happens when we try to prove the injectivity. We shall demonstrate how to use the periodic condition to derive some contradiction, or essentially prove that $\alpha - \beta$ must be 0.

Solution. Suppose $f(\alpha) = f(\beta)$. Substitute $x = \alpha$ and $x = \beta$ respectively in (1). We obtain

$$\begin{cases} f\big(2f(y + f(\alpha))\big) &= f(\alpha + y) + f(\alpha) + y, \\ f\big(2f(y + f(\beta))\big) &= f(\beta + y) + f(\beta) + y. \end{cases}$$

Comparing these equations, we get $f(\alpha + y) = f(\beta + y)$ for all $y \in \mathbb{R}$. If we replace y by $y - \beta$, this implies $f(y + d) = f(y)$ for any $y \in \mathbb{R}$ where $d = \alpha - \beta$. This basically means the function has period d (unless $d = 0$).

A simple way to use this periodicity is to add d to some variables. We replace y by $y + d$ in (1). This yields

$$f\big(2f(y + d + f(x))\big) = f(x + y + d) + f(x) + y + d.$$

As $f(z + d) = f(z)$ for any $z \in \mathbb{R}$, this implies

$$f\big(2f(y + f(x))\big) = f(x + y) + f(x) + y + d.$$

Comparing with (1), it is obvious that $d = 0$, and hence $\alpha = \beta$. Thus, f is injective.

Now, as suggested, we put $y = -f(x)$ in (1). We have $f(2f(0)) = f(x - f(x))$ and hence $2f(0) = x - f(x)$ by the injectivity. It follows that $f(x) = x + c$ for some constant c. We check that

$$\begin{aligned} f\big(2f(y + f(x))\big) = f\big(2f(y + x + c)\big) = f(2(x + y + 2c)) &= 2x + 2y + 5c, \\ f(x + y) + f(x) + y = (x + y + c) + (x + c) + y &= 2x + 2y + 2c. \end{aligned}$$

These are equal for any $x, y \in \mathbb{R}$ iff $c = 0$. Hence, the answer is $f(x) = x$ for all $x \in \mathbb{R}$. $\square$

Problem 2.22. (Kosovo MO 2019 Grade 11 Problem 4) Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that:

$$f(xy + f(x)) = xf(y) \tag{1}$$

for all $x, y \in \mathbb{R}$.

Analysis. Initially, we may not think of proving the injective or surjective condition since the zero function is clearly a solution. After doing some basic substitutions, we find that the case $f \equiv 0$ can be easily handled. But then there seems to be another solution that arises from $f(1) = 0$. Indeed, if one has tried all linear functions at the beginning, it is not hard to see that $f(x) = 1 - x$ is another solution, which is bijective. This suggests proving the bijective condition to simplify some equations.

Solution. Putting $x = 0$ in (1), we obtain $f(f(0)) = 0$. Putting $y = 0$ in (1), we obtain

$$f(f(x)) = xf(0). \tag{2}$$

Next, putting $x = f(0)$ and $y = 1$ in (1), and using $f(f(0)) = 0$, we find that $0 = f(0)f(1)$. This implies $f(0) = 0$ or $f(1) = 0$.

Case 1. $f(0) = 0$

Equation (2) becomes $f(f(x)) = 0$ for all $x \in \mathbb{R}$. Normally, it is not hard to deduce $f(x) = 0$ for all $x \in \mathbb{R}$ from this. For example, replacing x by $f(x)$ in (1), we get

$$f(yf(x)) = f(x)f(y). \tag{3}$$

On the other hand, if we apply f to both sides of (1), we get $f(xf(y)) = 0$. Swapping x and y, this becomes $f(yf(x)) = 0$. Thus, (3) becomes $f(x)f(y) = 0$. By putting $x = y$, this clearly yields $f(x) = 0$ for all $x \in \mathbb{R}$. Obviously, this is a solution.

Case 2. $f(0) \neq 0$ and $f(1) = 0$

Note that it is important to assume $f(0) \neq 0$ in this case, since otherwise we may still obtain the zero function as a solution. Now, putting $y = 1$ in (1), we obtain

$$f(x + f(x)) = 0. \tag{4}$$

If we can show that f is injective, then we are already done using this equation. In fact, it is also easy to prove the injective condition. Suppose $f(\alpha) = f(\beta)$. By putting $x = \alpha$ and $x = \beta$ in (2) respectively, we obtain

$$\alpha f(0) = f(f(\alpha)) = f(f(\beta)) = \beta f(0),$$

and hence $\alpha = \beta$ since $f(0) \neq 0$. Thus, f is injective. Now, by (4), we have

$$f(x + f(x)) = 0 = f(1),$$

and so $x + f(x) = 1$ for any $x \in \mathbb{R}$. This implies $f(x) = 1 - x$. We check that

$$f(xy + f(x)) = 1 - (xy + 1 - x) = x(1 - y) = xf(y).$$

Therefore, the solutions are $f(x) = 0$ for all $x \in \mathbb{R}$ and $f(x) + 1 - x$ for all $x \in \mathbb{R}$. □

Problem 2.23. Find all function(s) $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(x + y + f(y)) = f(f(x)) + 2y \tag{1}$$

for all $x, y \in \mathbb{R}$.

Analysis. If we can prove that f is injective, then the problem becomes easy since we can put $y = 0$ to get $f(\cdots) = f(\cdots)$, and then conclude $f(x) = x + c$ for some constant c. Suppose $f(\alpha) = f(\beta)$. Apart from putting $x = \alpha$ and $x = \beta$ respectively and comparing the two equations, sometimes it is also possible to use substitutions that involve α and β at the same time. Here gives one example of this.

Solution. Suppose $f(\alpha) = f(\beta)$. Putting $x = \alpha$ and $y = \beta$ in (1), we obtain

$$f(\alpha + \beta + f(\beta)) = f(f(\alpha)) + 2\beta. \tag{2}$$

On the other hand, putting $x = \beta$ and $y = \alpha$ in (1), we obtain

$$f(\beta + \alpha + f(\alpha)) = f(f(\beta)) + 2\alpha. \tag{3}$$

Comparing (2) and (3), we get $2\beta = 2\alpha$, hence $\beta = \alpha$. This shows f is injective.

Now, we put $y = 0$ in (1). This gives

$$f(x + f(0)) = f(f(x)).$$

By the injectivity of f, this yields $x + f(0) = f(x)$, i.e. $f(x) = x + c$ for some constant c. We check that

$$f(x + y + f(y)) = (x + y + (y + c)) + c = (x + 2c) + 2y = f(f(x)) + 2y.$$

Therefore, all such functions are solutions. □

Remarks. There are many different ways in proving that a function is injective or surjective. One should try different substitutions to see whether some terms can be cancelled, or make use of the periodicity or symmetry, etc.

Problem 2.24. (Dutch TST 2012 Test 2 Problem 5) Find all functions $f : \mathbb{R} \to \mathbb{R}$ satisfying

$$f(x + xy + f(y)) = \left(f(x) + \frac{1}{2}\right)\left(f(y) + \frac{1}{2}\right) \tag{1}$$

for all $x, y \in \mathbb{R}$.

Analysis. Sometimes we may not be able to prove the injective condition. There is a replacement of this, which is to prove that $f(\alpha) = f(\beta) = 0$ implies $\alpha = \beta$. In other words, for $f : A \to B$, if we define the *kernel*(核) of f to be

$$\{x \in A : f(x) = 0\},$$

then we want to show that the kernel of f consists of at most one element. Clearly, this holds if f is injective, so this is a weaker result. But in many situations, this result is as useful as the injective condition in general.

Solution. At first, one may try to substitute some variables to be 0 to reduce the number of terms. But this time it does not seem to provide useful information immediately. Rather than that, we consider the substitution $y = -1$ in (1), which makes the term $x + xy$ vanish. This yields

$$f(f(-1)) = \left(f(x) + \frac{1}{2}\right)\left(f(-1) + \frac{1}{2}\right). \tag{2}$$

If $f(-1) \neq -\frac{1}{2}$, this implies f is a constant function. Assume $f(x) = c$ for all $x \in \mathbb{R}$. Then the equation becomes $c = \left(c + \frac{1}{2}\right)^2$, i.e. $c^2 + \frac{1}{4} = 0$. This is impossible. Therefore, we must have $f(-1) = -\frac{1}{2}$. Then (2) becomes $f\left(-\frac{1}{2}\right) = 0$.

Next, the key idea is to prove the following.

Claim. If $f(t) = 0$, then $t = -\frac{1}{2}$.

Proof. Suppose $f(t) = 0$. Putting $y = -\frac{1}{2}$ in (1), we obtain

$$f\left(\frac{x}{2}\right) = \frac{1}{2}\left(f(x) + \frac{1}{2}\right).$$

By further putting $x = 2t$, we find that $0 = \frac{1}{2}\left(f(2t) + \frac{1}{2}\right)$. It follows that $f(2t) = -\frac{1}{2}$. In view of this, we substitute $y = 2t$ in (1). This yields

$$f\left(x + 2tx - \frac{1}{2}\right) = 0.$$

If $1+2t \neq 0$, then $x+2tx-\frac{1}{2}$ is a linear polynomial in x, and so it runs through all real numbers. This shows f is constant 0, which is clearly impossible. Therefore, we must have $1+2t=0$, i.e. $t=-\frac{1}{2}$ as desired. $\square$

Now, we are almost done. Indeed, putting $x=-1$ in (1), since $f(-1)=-\frac{1}{2}$, we have

$$f(-1-y+f(y))=0.$$

By the claim, this implies $-1-y+f(y)=-\frac{1}{2}$ for all $y \in \mathbb{R}$. Equivalently, $f(y)=y+\frac{1}{2}$ for all $y \in \mathbb{R}$. We check that

$$f(x+xy+f(y)) = \left(x+xy+\left(y+\frac{1}{2}\right)\right)+\frac{1}{2} = xy+x+y+1,$$
$$\left(f(x)+\frac{1}{2}\right)\left(f(y)+\frac{1}{2}\right) = (x+1)(y+1) = xy+x+y+1.$$

So it is the only solution. $\square$

Problem 2.25. Find all function(s) $f:\mathbb{R}\to\mathbb{R}$ such that

$$f(2x+y+f(y)) = 2f(f(x))+2y \tag{1}$$

for all $x, y \in \mathbb{R}$.

Analysis. This problem looks similar to problem 2.23. However, the same idea does not work and it does not seem easy to prove the injective condition. Therefore, we look for a relaxed condition and consider the kernel of f.

Solution. Putting $x=y=0$ in (1), we get $f(f(0))=2f(f(0))$. This implies $f(f(0))=0$. Then putting $x=0$ in (1), we get

$$f(y+f(y)) = 2y. \tag{2}$$

Firstly, we can use this to prove that the kernel of f is $\{0\}$.

Claim. If $f(t)=0$, then $t=0$.

Proof. Indeed, if $f(t)=0$, then by putting $y=t$ in (2), we find that $0=2t$, i.e. $t=0$. $\square$

Secondly, we can use (2) to prove that f is surjective since $2y$ runs through all real values. (No matter this is useful or not, we always think of proving such things if it is easy to do so.)

Thirdly, equation (2) is a result that helps us eliminate nested functions (as similar to the equation $f(f(y)) = 2y$). In view of this, we consider the special substitution as follows. We replace y by $f(y) + y$ in (1). Using (2), this becomes

$$f(2x + f(y) + y + 2y) = 2f(f(x)) + 2(f(y) + y).$$

Note that we can rewrite this as

$$f\big(2(x + y) + y + f(y)\big) = 2f(f(x)) + 2f(y) + 2y. \tag{3}$$

Thus, we replace x by $x + y$ in (1) to get

$$f\big(2(x + y) + y + f(y)\big) = 2f(f(x + y)) + 2y. \tag{4}$$

Comparing (3) and (4), we deduce

$$f(f(x)) + f(y) = f(f(x + y)). \tag{5}$$

The next idea is to equate two terms so that the remaining equation is $f(\cdots) = 0$. For example, given any y, we can always find $x \in \mathbb{R}$ such that $f(x + y) = y$ since f is surjective and $x + y$ runs through all real values. Putting this x in (5), we plainly obtain $f(f(x)) = 0$ (for this specific x). By the claim, this implies $f(x) = 0$ and hence $x = 0$. In other words, we must have $f(y) = y$ for any $y \in \mathbb{R}$. The checking is trivial. □

Problem 2.26. Find all function(s) $f : \mathbb{Z} \to \mathbb{Z}$ such that

$$f(m - n + f(n)) = f(m) + f(n) \tag{1}$$

for all $m, n \in \mathbb{Z}$.

Analysis. This problem is of another style since the function is from $\mathbb{Z}$ to $\mathbb{Z}$. The first natural idea is to equate some terms such as $m - n + f(n) = n$ to obtain a simple equation like $f(2n - f(n)) = 0$. If we can prove that f is injective or the kernel of f is $\{0\}$, then we are done. This leads to the consideration of the kernel of f.

By assuming $f(d) = 0$ for some $d \neq 0$, we can easily deduce f is periodic. Periodic functions defined on $\mathbb{Z}$ are easy to investigate. One simple property of such functions is that they must be bounded. We can usually derive some contradiction using this boundedness.

Solution. Suppose there exists a nonzero integer d such that $f(d) = 0$. By putting $n = d$ in (1), we obtain

$$f(m - d) = f(m)$$

for any $m \in \mathbb{Z}$. Since $d \neq 0$, this shows f is periodic. In particular, as f is defined on $\mathbb{Z}$, there are only finitely many possible images. In other words, $S = \{f(n) : n \in \mathbb{Z}\}$ is a finite set. So $\max S$ and $\min S$ are well-defined.

If $\max S = M > 0$, let k be an integer such that $f(k) = M$. Putting $m = n = k$ in (1), we find that $f(f(k)) = 2M > M$. This contradicts $\max S = M$. Similarly, we can prove that $\min S$ cannot be negative. As a result, we have

$$0 \leqslant \min S \leqslant \max S \leqslant 0,$$

so that $S = \{0\}$. In other words, $f(n) = 0$ for all $n \in \mathbb{Z}$, which is clearly a solution.

Now, we assume $f(n) \neq 0$ for any $n \neq 0$. We put $m = 2n - f(n)$ in (1) so that $m - n + f(n) = n$. Then the equation becomes

$$f(2n - f(n)) = 0.$$

By the assumption, this implies $2n - f(n) = 0$, i.e. $f(n) = 2n$ for any $n \in \mathbb{Z}$. We check that this is another solution. Indeed,

$$f(m - n + f(n)) = 2(m - n + 2n) = 2m + 2n = f(m) + f(n).$$

Therefore, the solutions are $f(n) = 0$ for any $n \in \mathbb{Z}$ and $f(n) = 2n$ for any $n \in \mathbb{Z}$. $\square$

Remarks. For functions defined on $\mathbb{N}$ or $\mathbb{Z}$, induction is still the most important tool. In addition, there are some more ideas that may apply to problems involving such functions due to the discreteness of $\mathbb{N}$ and $\mathbb{Z}$. Note that such functions are basically the same as sequences, so it shares many common properties with sequences. Some more problems will be given in section 3.

Problem 2.27. (IMO Shortlist 2009 A7) Find all functions f from the set of real numbers into the set of real numbers which satisfy for all real x, y the identity

$$f(xf(x+y)) = f(yf(x)) + x^2. \tag{1}$$

Analysis. This time we see how to combine all techniques used before to solve this hard problem. This includes the concepts of surjectivity, injectivity and kernel, and also use various substitutions.

Solution. The proof is decomposed into several parts as follows.

Claim 1. f is surjective

Proof. We put $x = 0$ in (1) to get $f(0) = f(yf(0))$. If $f(0) \neq 0$, then we can replace y by $\dfrac{y}{f(0)}$ to get $f(y) = f(0)$ for all $y \in \mathbb{R}$. This means f is a constant function. However, this is impossible in view of (1). So we must have $f(0) = 0$. To use this fact, we may put $y = 0$ and $y = -x$ respectively in (1). These give

$$f(xf(x)) = x^2 \tag{2}$$

and

$$f(-xf(x)) = -x^2.$$

The two identities together imply f is surjective as both nonnegative and negative numbers are images under f. $\square$

Claim 2. $f(1) = 1$ or $f(-1) = 1$

Proof. By claim 1, there exists $t \in \mathbb{R}$ such that $f(t) = 1$. Putting $x = t$ in (2), we obtain

$$t^2 = f(tf(t)) = f(t) = 1$$

so that $t = \pm 1$. In other words, we have $f(1) = 1$ or $f(-1) = 1$. □

One should notice that both $f(x) = \pm x$ are solutions, so both cases are possible. The two cases are quite similar, and can be considered as the same case up to symmetry. A standard way is to prove the following.

Claim 3. If f is a solution, then g defined by $g(x) = f(-x)$ for all $x \in \mathbb{R}$ is another solution.

Proof. We replace x and y by $-x$ and $-y$ in (1) to get

$$g(xg(x+y)) = f(-xf(-x-y)) = f(-yf(-x)) + (-x)^2 = g(yg(x)) + x^2.$$

This shows g is also a solution. □

By claim 2 and claim 3, it suffices to consider the case $f(1) = 1$.

Claim 4. $f(n) = n$ for all $n \in \mathbb{Z}$

Proof. Putting $x = 1$ in (1), we get

$$f(f(y+1)) = f(y) + 1. \tag{3}$$

Therefore, whenever $f(x) = x$, we have

$$f(x-1) = f(f(x)) - 1 = f(x) - 1 = x - 1.$$

Using the base case $f(1) = 1$, we deduce $f(n) = n$ for all integers $n \leqslant 1$ by backward induction.

Next, we assume x is a negative integer in (2). This yields $f(x^2) = x^2$. Thus, we have $f(n) = n$ for all perfect squares n. Recall that $f(x) = x$ implies $f(x-1) = x-1$. As every positive integer is smaller than some perfect square, we can use backward induction to conclude $f(n) = n$ for all $n \in \mathbb{Z}$. □

Claim 5. f is injective

Proof. We first prove that $f(t) = 0$ implies $t = 0$. Indeed, this follows readily by putting $x = t$ in (2).

Now, suppose $f(\alpha) = f(\beta)$. We put $x = \alpha$ and $y = \beta - \alpha$ in (1). Note that

$$\begin{aligned}
& f(\alpha f(\beta)) = f\big((\beta-\alpha)f(\alpha)\big) + \alpha^2 \\
\Rightarrow \quad & f(\alpha f(\alpha)) = f\big((\beta-\alpha)f(\alpha)\big) + \alpha^2 \\
\Rightarrow \quad & f((\beta-\alpha)f(\alpha)) = 0 \quad \text{(by (2))} \\
\Rightarrow \quad & (\beta-\alpha)f(\alpha) = 0 \quad \text{(by the above observation)}.
\end{aligned}$$

This implies $\alpha = \beta$ or $f(\alpha) = 0$. The latter case implies $f(\alpha) = f(\beta) = 0$, which also gives $\alpha = \beta = 0$. Thus, f is injective. $\square$

Now, there could be better ways to work out the remaining solution. Here, we try to demonstrate how to use the bijectivity to find out relations between common terms.

Firstly, replace x by $-x$ in (2) to get $f(-xf(-x)) = x^2$. Comparing this with (2), we get

$$f(-xf(-x)) = f(xf(x)).$$

By claim 5, this yields $-xf(-x) = xf(x)$, or simply

$$f(-x) = -f(x) \tag{4}$$

(note that this also holds when $x = 0$).

Secondly, we shall prove $f(2x) = 2f(x)$ (this can be modified to give $f(kx) = kf(x)$ for $k \in \mathbb{Z}$, but we do not need this stronger result). The only trick is to generate some terms involving $2x$. Note that $3x$ and $4x$ may also be involved. For example, putting $y = x$ in (1) and using (2), we deduce

$$f(xf(2x)) = f(xf(x)) + x^2 = 2x^2. \tag{5}$$

Next, we put $y = 3x$ in (1) to get

$$f(xf(4x)) = f(3xf(x)) + x^2. \tag{6}$$

Also, we put $y = -3x$ in (1). The relation $f(xf(-2x)) = f(-3xf(x)) + x^2$ can be rewritten as

$$-f(xf(2x)) = -f(3xf(x)) + x^2 \tag{7}$$

using (4). Summing up (5), (6) and (7), we obtain

$$f(xf(4x)) = 4x^2. \tag{8}$$

Now, replace x by $\frac{x}{2}$ in (8) and use (2). We get

$$f\left(\frac{x}{2}f(2x)\right) = x^2 = f(xf(x)).$$

By claim 5, this implies $\frac{x}{2}f(2x) = xf(x)$, and hence

$$f(2x) = 2f(x) \tag{9}$$

(which is also true for $x = 0$).

Lastly, we put $x=2$ in (1) to get $f(2f(y+2))=f(yf(2))+4$. We have

$$\begin{aligned}
& & f(2f(y+2)) &= f(yf(2))+4 & \\
&\Rightarrow & 2f(f(y+2)) &= f(2y)+4 & \text{(by (9) and claim 4)} \\
&\Rightarrow & 2f(f(y+2)) &= 2f(y)+4 & \text{(by (9))} \\
&\Rightarrow & f\big(f\big((y+1)+1\big)\big) &= f(y)+2 & \\
&\Rightarrow & f(y+1)+1 &= f(y)+2 & \text{(by (3))} \\
&\Rightarrow & f(y+1) &= f(y)+1 & \\
&\Rightarrow & f(y+1) &= f(f(y+1)) & \text{(by (3))} \\
&\Rightarrow & y+1 &= f(y+1) & \text{(by claim 5)} \\
&\Rightarrow & f(x) &= x. &
\end{aligned}$$

By claim 3, the possible solutions are $f(x)=x$ for all $x\in\mathbb{R}$ and $f(x)=-x$ for all $x\in\mathbb{R}$. The checking is trivial. $\square$

Remarks. The substitution steps in the latter part seem to be tedious and artificial. But the idea is simply to find the relationship between $f(x)$ and $f(-x)$, and express $f(kx)$ (sometimes $f(x+k)$) in terms of $f(x)$. All these should have been done in the rough work and we only need to pick the useful results to complete the proof.

We summarize all tricks related to the injectivity and the surjectivity of functions introduced in this subsection.

- injective functions
  - proving injectivity via the standard trick (problems 2.18, 2.19, 2.20, 2.22)
  - proving injectivity via special substitutions (problems 2.18, 2.21, 2.23, 2.27)
  - application: removing f from both sides
- kernel
  - proving $f(\alpha)=f(\beta)=0$ implies $\alpha=\beta$ (problems 2.24, 2.25, 2.26)
- periodic functions
  - using periodicity (problems 2.21, 2.26)
  - using boundedness of a periodic function defined on $\mathbb{Z}$ (problem 2.26)
- surjective functions
  - proving surjectivity (problems 2.18, 2.19, 2.20, 2.25, 2.27)
  - application: setting $f(x)$ to be a particular value such as 0 or 1

3 Inequalities and Analysis

This section is mainly about problems that involve both functions and inequalities. There are several focuses in this section. Firstly, we go through a bit of analysis. In particular, we shall see how the concept of limits, the denseness of $\mathbb{Q}$, and the monotone convergence theorem may help us solve some kinds of MO problems.

Secondly, we discuss more problems in functional equations for which the functions are defined on $\mathbb{R}^+$ or $\mathbb{N}$. There are more techniques which can be applied to such problems, some of which are related to inequalities.

Lastly, we go through another kind of problems called functional inequalities.

3.1 Analysis

Although analysis is considered as a branch of mathematics at university level, some basic understanding on this topic could be useful. But it is fine to skip this part if one finds it hard to follow. First we consider the convergence of sequences.

Definition 3.1. A *sequence*(數列) is a function defined on $\mathbb{N}$.

This is a formal way to define a sequence in mathematics. For example, $f : \mathbb{N} \to \mathbb{R}$ is a sequence of real numbers. Indeed, by listing out the images $f(1), f(2), f(3), \ldots$ one by one, we can get all the information of the sequence. So sometimes we just call this list a sequence. Under this definition, we can say a sequence is bounded, monotonic, increasing, periodic, etc. by using the same definitions for functions.

Instead of using the function notations, it is more common to use subscripts to label the terms. For example, we write $a_n = f(n)$ so that the sequence becomes $a_1, a_2, a_3, \ldots$. Some other common notations for a sequence include $\{a_n\}$, $\{a_n\}_{n\in\mathbb{N}}$, (a_n), etc. In some occasions, it may be more convenient to deal with a sequence $\{a_n\}_{n\geqslant 0} = \{a_0, a_1, a_2, \ldots\}$ that starts with the zeroth term.

Definition 3.2. Let $\{a_n\}$ be a sequence of real numbers. We say that $\{a_n\}$ *converges*(收斂) to some real number L if the following holds. For any $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that $|a_n - L| < \varepsilon$ for any positive integer $n \geqslant N$.

In that case, we say that $\{a_n\}$ is *convergent*(收斂) and say that L is the *limit*(極限) of $\{a_n\}$. We use the notation

$$\lim_{n\to\infty} a_n = L.$$

If a sequence does not converge to any real number, we say that it is *divergent*(發散).

Another simplified notation for $\lim_{n\to\infty} a_n = L$ is $a_n \to L$. This means if $a_n \to L$, then the tail part of the sequence is getting closer and closer to L. Therefore, the limit of a sequence and whether or not the limit exists are independent of the first few terms of the sequence. Here are some basic facts about convergent sequences. Suppose $\{a_n\}$ and $\{b_n\}$ are convergent sequences with limits A and B respectively.

- The limit A is uniquely determined.
- The sequence $\{a_n\}$ must be bounded.
- If $a_n \leqslant b_n$ for any $n \in \mathbb{N}$, then $A \leqslant B$.
- If $m \leqslant a_n \leqslant M$ for any $n \in \mathbb{N}$, then $m \leqslant A \leqslant M$.

Definition 3.3. Let $\{a_n\}$ be a sequence of real numbers. We say that $\{a_n\}$ *diverges to positive infinity*(趨向無窮大) if the following holds. For any $M \in \mathbb{R}$, there exists $N \in \mathbb{N}$ such that $a_n > M$ for any positive integer $n \geqslant N$. In that case, we write

$$\lim_{n\to\infty} a_n = \infty.$$

We can also use the notation $a_n \to \infty$. This means the tail part of the sequence is getting larger and larger. One can define the meaning of diverging to negative infinity in an analogous way.

For our purpose, we mainly need to know the intuitive meaning of the limit of a sequence, and know how to find the limit of some common sequences. Normally we do not need to use the tedious definition. Here are some basic examples.

- The limit of a constant sequence is the same constant.
- Define $a_n = c^n$ where c is a constant.
  - If $c > 1$, then $\{a_n\}$ diverges to positive infinity.
  - If $c = 1$, then $\{a_n\}$ converges to 1.
  - If $-1 < c < 1$, then $\{a_n\}$ converges to 0.
  - If $c \leqslant -1$, then $\{a_n\}$ diverges.
- Define $b_n = n^c$ where c is a constant.
  - If $c > 0$, then $\{b_n\}$ diverges to positive infinity.
  - If $c = 0$, then $\{b_n\}$ converges to 1.
  - If $c < 0$, then $\{b_n\}$ converges to 0.
- Define $c_n = \dfrac{(-1)^n}{n}$. Then $\{c_n\}$ converges to 0.
- Define $d_n = n^{\frac{1}{n}}$. Then $\{d_n\}$ converges to 1.

- Define $e_n = \left(1+\frac{1}{n}\right)^n$. Then $\{e_n\}$ converges to $e \approx 2.718$.

The limits of some more complicated sequences can be found based on these simple sequences and the following results. Let $\{a_n\}$ and $\{b_n\}$ be convergent sequences with limits A and B respectively, and let $\{c_n\}$ be a sequence diverging to positive infinity. Then we have the following.

- $\lim\limits_{n\to\infty}(a_n+b_n) = A+B$
- $\lim\limits_{n\to\infty} a_n b_n = AB$
- $\lim\limits_{n\to\infty} \frac{a_n}{b_n} = \frac{A}{B}$ if $b_n \neq 0$ for all $n \in \mathbb{N}$ and $B \neq 0$
- $\lim\limits_{n\to\infty}(a_n+c_n) = \infty$
- $\lim\limits_{n\to\infty} a_n c_n = \infty$ if $A > 0$, and $\lim\limits_{n\to\infty} a_n c_n = -\infty$ if $A < 0$
- $\lim\limits_{n\to\infty} \frac{a_n}{c_n} = 0$
- $\lim\limits_{n\to\infty} a_n^{\frac{1}{c_n}} = 1$ if $a_n > 0$ for all $n \in \mathbb{N}$ and $A > 0$

As an example, we have

$$\lim_{n\to\infty} \frac{2n^2+4\sqrt{n}}{3n^2-2^{\frac{1}{n}}} = \lim_{n\to\infty} \frac{2+4n^{-\frac{3}{2}}}{3-n^{-2}\cdot 2^{\frac{1}{n}}} = \frac{2+4\lim\limits_{n\to\infty} n^{-\frac{3}{2}}}{3-\left(\lim\limits_{n\to\infty} n^{-2}\right)\left(\lim\limits_{n\to\infty} 2^{\frac{1}{n}}\right)} = \frac{2+4\cdot 0}{3-0\cdot 1} = \frac{2}{3}.$$

The following theorem is useful in proving that some sequences are convergent.

Theorem 3.1. (Monotone convergence theorem(單調收斂定理)) Let $\{a_n\}$ be an increasing sequence of real numbers. Then the following hold.

(a) If $\{a_n\}$ is bounded above, then it is convergent.

(b) If $\{a_n\}$ is not bounded above, then it diverges to positive infinity.

Similar results hold for a decreasing sequence.

Problem 3.1. Let $a_1 = 0$ and let $a_{n+1} = \frac{a_n^2+1}{2}$ for any $n \in \mathbb{N}$. Prove that $\{a_n\}$ is convergent and find the limit of $\{a_n\}$.

Analysis. This is a standard problem that shows an application of the monotone convergence theorem. By computing a few terms $\{a_n\} = \left\{0, \frac{1}{2}, \frac{5}{8}, \frac{89}{128}, \ldots\right\}$, it is not hard to guess that $\{a_n\}$ is increasing. For problems involving recurrence relations, usually we can use induction to prove all desired properties of the sequence. Also, we shall provide a standard method to find the limit of a recurrence sequence.

Solution. The proof is decomposed into 3 parts.

Claim 1. $\{a_n\}$ is increasing

Proof. Indeed, we have $a_{n+1} - a_n = \dfrac{a_n^2+1}{2} - a_n = \dfrac{(a_n-1)^2}{2} \geqslant 0$ for any $n \in \mathbb{N}$. □

Claim 2. $\{a_n\}$ is bounded above

Proof. We want to find some constant M such that $a_n \leqslant M$ for any $n \in \mathbb{N}$. The idea is to use induction. Clearly, all terms are nonnegative. Therefore, whenever $a_k \leqslant M$, we have

$$a_{k+1} = \frac{a_k^2+1}{2} \leqslant \frac{M^2+1}{2}.$$

This is bounded above by M iff $M^2+1 \leqslant 2M$, i.e. $(M-1)^2 \leqslant 0$. Thus, we can choose $M = 1$. In that case, whenever $a_k \leqslant 1$, we have $a_{k+1} \leqslant 1$. The base case $a_1 \leqslant 1$ is also satisfied. Therefore, $\{a_n\}$ is bounded above by 1 by induction. □

By the monotone convergence theorem, $\{a_n\}$ is convergent.

Claim 3. the limit of $\{a_n\}$ is 1

Proof. Let $a_n \to L$. To find the limit L, we take limits on both sides of $a_{n+1} = \dfrac{a_n^2+1}{2}$. This gives

$$\lim_{n\to\infty} a_{n+1} = \lim_{n\to\infty} \frac{a_n^2+1}{2},$$

and hence $L = \dfrac{L^2+1}{2}$. We easily get the only solution $L = 1$. This shows $\{a_n\}$ converges to 1. □

Next, we consider the limit of a function $f : \mathbb{R} \to \mathbb{R}$.

Definition 3.4. Let $f : \mathbb{R} \to \mathbb{R}$ be a function, and let $a \in \mathbb{R}$. We say that the *limit*(極限) of f at a is $L \in \mathbb{R}$ if the following holds. For any $\varepsilon > 0$, there exists $\delta > 0$ such that if $0 < |x-a| < \delta$, then $|f(x) - L| < \varepsilon$.

In that case, we write

$$\lim_{x\to a} f(x) = L.$$

We can also say that f tends to L when x tends to a, or write $f(x) \to L$ when $x \to a$. Loosely speaking, when x is close to a but not equal to a, then $f(x)$ is close to L. Since x is always different from a, the value of $f(a)$ is irrelevant.

Analogous to the limit of sequences, we have the following basic facts. Suppose $\lim\limits_{x\to a} f(x) = L$ and $\lim\limits_{x\to a} g(x) = L'$.

- The limit L is uniquely determined.
- If $f(x) \leqslant g(x)$ for any $x \in \mathbb{R}$, then $L \leqslant L'$.
- If $m \leqslant f(x) \leqslant M$ for any $x \in \mathbb{R}$, then $m \leqslant L \leqslant M$.
- $\lim\limits_{x\to a} c = c$ for any $c \in \mathbb{R}$
- $\lim\limits_{x\to a} x = a$
- $\lim\limits_{x\to a} (f+g)(x) = L + L'$
- $\lim\limits_{x\to a} (fg)(x) = LL'$
- $\lim\limits_{x\to a} \left(\dfrac{f}{g}\right)(x) = \dfrac{L}{L'}$ if $L' \neq 0$

Definition 3.5. Let $f : \mathbb{R} \to \mathbb{R}$ be a function, and let $a \in \mathbb{R}$. We say that the *right-hand limit*(右極限) of f at a is $L \in \mathbb{R}$ if the following holds. For any $\varepsilon > 0$, there exists $\delta > 0$ such that if $a < x < a + \delta$, then $|f(x) - L| < \varepsilon$.

In that case, we write

$$\lim_{x\to a^+} f(x) = L.$$

One can define the *left-hand limit*(左極限) $\lim\limits_{x\to a^-} f(x)$ in an analogous way. These are called *one-sided limits*(單側極限). These mean we only consider the behaviour of $f(x)$ when x approaches a from one side. One basic result is that

$$\lim_{x\to a} f(x) = L \quad \Leftrightarrow \quad \lim_{x\to a^+} f(x) = \lim_{x\to a^-} f(x) = L.$$

Definition 3.6. Let $f : \mathbb{R} \to \mathbb{R}$ be a function. We say that f tends to $L \in \mathbb{R}$ when x tends to ∞ if the following holds. For any $\varepsilon > 0$, there exists $M > 0$ such that if $x > M$, then $|f(x) - L| < \varepsilon$.

In that case, we write

$$\lim_{x\to \infty} f(x) = L.$$

One can define $\lim_{x\to-\infty} f(x) = L$, $\lim_{x\to a} f(x) = \pm\infty$ and $\lim_{x\to\pm\infty} f(x) = \pm\infty$ in analogous ways. These are just more formal ways to express the intuitive understanding of limits.

Using the concept of limits, we can now define a continuous function rigorously.

Definition 3.7. Let $f : \mathbb{R} \to \mathbb{R}$ be a function, and let $a \in \mathbb{R}$. We say that f is *continuous*(連續) at a if

$$\lim_{x\to a} f(x) = f(a).$$

Otherwise, we say that f is *discontinuous*(不連續) at a.

We say that f is a *continuous function*(連續函數) if it is continuous at every point.

In general we cannot assume a function is continuous, and the continuity condition is seldom given in MO problems. Therefore, our only usage is to use this to compute the limits of some simple functions. Indeed, in view of the definition, that means we can compute the limit of $f(x)$ when $x \to a$ by substituting $x = a$ if f is continuous.

Most common functions like polynomials (e.g. $f(x) = x^2$), exponential functions (e.g. $f(x) = 2^x$) and logarithm functions (e.g. $\log x$) are continuous (on their domains). Also, it is known that if f and g are continuous, so are $f + g$, fg, $\dfrac{f}{g}$ and $g \circ f$.

Using these facts, we can easily compute some limits. For example, we have

$$\lim_{x\to 2} \frac{3x + \sqrt{2x}}{2^x - 1} = \frac{3\lim_{x\to 2} x + \sqrt{2\lim_{x\to 2} x}}{\lim_{x\to 2} 2^x - 1} = \frac{3(2) + \sqrt{2(2)}}{2^2 - 1} = \frac{8}{3}.$$

The concepts of limits of sequences and limits of functions seem to be closely related. Indeed, there is a close connection as given below.

Theorem 3.2. (Sequential criterion(海涅定理)) Let $f : \mathbb{R} \to \mathbb{R}$ be a function, and let $a \in \mathbb{R}$. Then $\lim_{x\to a} f(x) = L$ iff the following holds. For any sequence $\{x_n\}$ which converges to a and $x_n \neq a$ for all n, the sequence $\{f(x_n)\}$ converges to L.

As a corollary, f is continuous at a iff the following holds. For any sequence $\{x_n\}$ which converges to a, the sequence $\{f(x_n)\}$ converges to $f(a)$. Analogous results hold if we consider one-sided limits instead.

Proposition 3.1. Let $f : \mathbb{R} \to \mathbb{R}$ be a monotonic function. Then the one-sided limits $\lim_{x\to a^+} f(x)$ and $\lim_{x\to a^-} f(x)$ exist for any $a \in \mathbb{R}$.

This is an analogue of the monotone convergence theorem. For example, if f is increasing, then when x approaches a from the left (which means $x < a$), the value of $f(x)$ is increasing, and it is bounded above by $f(a)$. Intuitively, it looks obvious that the left-hand limit should exist. As a corollary, a surjective monotonic function $f : \mathbb{R} \to \mathbb{R}$ must be continuous (since both $\lim\limits_{x \to a^+} f(x)$ and $\lim\limits_{x \to a^-} f(x)$ must be equal to $f(a)$, or otherwise there is a gap between $f(a)$ and these limits so that f cannot be surjective).

A well-known property of a continuous function is the following.

Theorem 3.3. (Intermediate value theorem(介值定理)) Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous function, and let $a < b$ be real numbers. For any k such that $f(a) < k < f(b)$ or $f(b) < k < f(a)$, there exists $c \in (a, b)$ such that $f(c) = k$.

Lastly, we introduce one more concept.

Definition 3.8. A subset X of $\mathbb{R}$ is said to be *dense*(稠密) in $\mathbb{R}$ if the following holds. For any $x, y \in \mathbb{R}$ with $x < y$, there exists $r \in X$ such that $x < r < y$.

Theorem 3.4. The set $\mathbb{Q}$ is dense in $\mathbb{R}$.

Proof. Consider any $x, y \in \mathbb{R}$ with $x < y$. There exists $n \in \mathbb{N}$ such that $n(y - x) > 1$. Since the interval (nx, ny) has length larger than 1, there exists an integer m in this interval. This implies $nx < m < ny$, and hence $x < \frac{m}{n} < y$. This shows the existence of a rational number $\frac{m}{n}$ in the interval (x, y). □

Another way to see this is to consider the decimal expansions of x and y. Since $x < y$, there must be some 'gaps' between the two numbers, and so we should be able to increase x by a little bit to obtain a terminating decimal (which must be a rational number) that is still smaller than y. But this argument is less rigorous.

Similarly, we can prove that the set of all irrational numbers is dense in $\mathbb{R}$.

Finally, we have introduced all the necessary tools, and we can return to our discussion on function problems in MO.

Problem 3.2. (Proposition 2.1) Let $f : \mathbb{R} \to \mathbb{R}$ be a function satisfying the Cauchy equation

$$f(x + y) = f(x) + f(y) \tag{1}$$

for all $x, y \in \mathbb{R}$. Suppose f satisfies one of the following conditions.

(i) $f(xy) = f(x)f(y)$ for all $x, y \in \mathbb{R}$

(ii) $f(x) \geqslant 0$ for all $x \geqslant 0$

(iii) f is monotonic

(iv) f is continuous

Prove that $f(x) = cx$ for all $x \in \mathbb{R}$ where c is a real constant.

Analysis. Using our new concepts, now we can provide a proof to this proposition.

Solution. There are several implications to show.

Claim 1. (i) implies (ii)

Proof. If (i) holds, by putting $x = y$ in $f(xy) = f(x)f(y)$, we obtain $f(x^2) = f(x)^2 \geqslant 0$. This proves (ii). □

Claim 2. (ii) implies f is increasing

Proof. Suppose (ii) holds. For any $x, y \in \mathbb{R}$ with $y \geqslant x$, we replace y by $y - x$ in (1) to get

$$f(y) = f(x) + f(y - x) \geqslant f(x)$$

since $y - x \geqslant 0$. This shows f is an increasing function. □

Claim 3. if f is increasing, then $f(x) = cx$ for some constant c

Proof. Since f satisfies the Cauchy equation, we have $f(r) = cr$ for any $r \in \mathbb{Q}$ where c is a constant. Note that $c \geqslant 0$ since f is increasing. For any $x \in \mathbb{R}$ and any $\varepsilon > 0$, there exist rational numbers $q_1 \in (x - \varepsilon, x)$ and $q_2 \in (x, x + \varepsilon)$ since $\mathbb{Q}$ is dense in $\mathbb{R}$. Now, as f is increasing, we get

$$c(x - \varepsilon) \leqslant cq_1 = f(q_1) \leqslant f(x) \leqslant f(q_2) = cq_2 \leqslant c(x + \varepsilon).$$

By taking the limit $\varepsilon \to 0$, we obtain $cx \leqslant f(x) \leqslant cx$, and hence $f(x) = cx$. □

Similarly, we can show that $f(x) = cx$ if f is decreasing. Combining claim 1, claim 2 and claim 3, we have already proven that each of (i), (ii) and (iii) implies $f(x) = cx$.

Claim 4. if f is continuous, then $f(x) = cx$

Proof. For any $x \in \mathbb{R}$, since $\mathbb{Q}$ is dense in $\mathbb{R}$, there exists a sequence $\{q_n\}$ of rational numbers which converges to x. As f is continuous, we have

$$f(x) = \lim_{y \to x} f(y) = \lim_{n \to \infty} f(q_n) = \lim_{n \to \infty} cq_n = cx.$$

□

Problem 3.3. (Singapore MO Open 2015 Problem 3) Find all functions $f:\mathbb{R}\to\mathbb{R}$ such that

$$f(x)f(yf(x)-1)=x^2f(y)-f(x)\quad \forall\, x,y\in\mathbb{R}. \tag{1}$$

Analysis. Although there are some alternative methods to work on this problem, we try to use a more natural approach to start, i.e. to find the values of some small numbers, guess the formula and prove that it holds for all rational numbers. To move one step further from $\mathbb{Q}$ to $\mathbb{R}$, one common trick is to use the denseness of $\mathbb{Q}$. We illustrate how to use this idea in the following solution.

Solution. The zero function is clearly a solution. In the following, we assume f is not constant zero. Putting $x=0$ in (1), we get

$$f(0)f(yf(0)-1)=-f(0).$$

If $f(0)\neq 0$, this implies $f(yf(0)-1)=-1$ for any y. Note that $yf(0)-1$ takes all possible real values. Thus, f is the constant function -1. It is not hard to see that it does not satisfy (1). Therefore, we must have $f(0)=0$.

Putting $y=0$ in (1), we get $f(x)f(-1)=-f(x)$. As f is not constant zero, this implies $f(-1)=-1$. Putting $x=-1$ in (1), we have

$$-f(-y-1)=f(y)+1. \tag{2}$$

Replacing y by $-yf(x)$, this becomes $f(yf(x)-1)=-f(-yf(x))-1$. Substituting this in (1) and simplifying, we obtain

$$-f(x)f(-yf(x))=x^2f(y). \tag{3}$$

Note that we can deduce (1) using (2) and (3). Therefore, we can actually work on these simpler equations.

Equation (3) easily implies the kernel of f only contains 0 (if $f(x)=0$, then $x^2f(y)=0$ for all y, and hence $x=0$ as f is not constant zero). Thus, putting $y=x\neq 0$ in (3), we obtain

$$f(-xf(x))=-x^2. \tag{4}$$

Note that this also holds when $x=0$.

Also, putting $y=-1$ in (3), we find that

$$f(x)f(f(x))=x^2.$$

In view of this, we can replace x by $f(x)$ in (4) and obtain

$$f(-x^2)=-f(x)^2. \tag{5}$$

Putting $x=1$, this yields $f(1)^2=1$, and hence $f(1)=\pm 1$. The case $f(1)=-1$ is rejected since otherwise (2) implies $f(-2)=-(f(1)+1)=0$, contradicting the fact that the kernel of f only contains 0. So we must have $f(1)=1$. Now, putting $x=1$ in (3), we obtain

$$f(-y)=-f(y). \tag{6}$$

Thus, (2) and (3) can be rewritten as

$$f(y+1) = f(y) + 1, \tag{7}$$

$$f(x)f(yf(x)) = x^2 f(y). \tag{8}$$

It suffices to work on these two equations. In many problems, if we can reduce the given equation into similar equations like these, where one equation gives the additive structure and one equation gives the multiplicative structure, we are almost done. One can be even more sure about this if the given equation can be deduced using these equations. The remaining steps are rather standard.

Claim 1. $f(n) = n$ for any $n \in \mathbb{Z}$

Proof. This can be easily proven by induction using (7) and any base case such as $f(0) = 0$. □

Claim 2. $f(nx) = nf(x)$ for any $n \in \mathbb{Z}$ and $x \in \mathbb{R}$

Proof. Putting $x = n$ in (8) and using claim 1, we see that $nf(ny) = n^2 f(y)$. No matter $n = 0$ or not, we still have $f(ny) = nf(y)$. □

As a corollary, by putting $x = \frac{m}{n}$ in claim 2 for $m \in \mathbb{Z}$ and $n \in \mathbb{N}$, we see that $f(r) = r$ for all $r \in \mathbb{Q}$.

Claim 3. $f(x) = x$ for any $x \in \mathbb{R}$

Proof. To finish this step, usually we need some analytic conditions like those listed in proposition 2.1. For example, by (5) and (6), we easily see that $f(x) \geqslant 0$ if $x \geqslant 0$, and $f(x) \leqslant 0$ if $x \leqslant 0$.

The usual trick of using the denseness of $\mathbb{Q}$ is to consider some rational numbers close to x, and use their images (which are known) to deduce some relations. Indeed, consider two rational numbers $\frac{a}{b}$ and $\frac{c}{d}$ such that $\frac{a}{b} < x < \frac{c}{d}$ (with $a, c \in \mathbb{Z}$ and $b, d \in \mathbb{N}$). This implies $ad < bdx < bc$. We can use this to give a bound for $f(x)$. Using (7) repeatedly, we find that

$$f(bdx) = f(bdx - ad) + ad \geqslant ad$$

since $bdx - ad > 0$. Similarly, we have

$$f(bdx) = f(bdx - bc) + bc \leqslant bc.$$

Thus, we know that $ad \leqslant f(bdx) \leqslant bc$. By claim 2, this implies $ad \leqslant bdf(x) \leqslant bc$, and hence

$$\frac{a}{b} \leqslant f(x) \leqslant \frac{c}{d}.$$

Now, since $\mathbb{Q}$ is dense in $\mathbb{R}$, we can find rational numbers which are arbitrarily close to x. By taking limits, the above inequalities give $x \leqslant f(x) \leqslant x$, and hence $f(x) = x$. □

Lastly, it is easy to check that $f(x) = x$ is another solution. So both the zero function and the identity function are solutions. □

3.2 Functional Equations and the Positive Condition

In this subsection, we investigate functional equations of the form $f : \mathbb{R}^+ \to \mathbb{R}^+$ or $f : \mathbb{N} \to \mathbb{N}$. Such functional equations are usually harder because we cannot use substitutions like $x = 0$. We shall go through some useful tricks which mostly apply to such problems.

The first simple idea is to use the trivial inequality $f(x) > 0$ for any x. This positive condition is needed in many problems of this kind.

Problem 3.4. Find all function(s) $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that

$$f(x) + f(y) = 2f\left(\frac{x+y}{2}\right)$$

for all $x, y \in \mathbb{R}^+$.

Analysis. The equation is the same as that of problem 2.11. However, since we cannot substitute $x = 0$, we cannot transform this to the Cauchy equation.

Recalling the proof of the Cauchy equation, it should be possible to show that all rational numbers satisfy the same formula. Then we again use the denseness of $\mathbb{Q}$ to derive some results for $f(x)$ where $x \in \mathbb{R}^+$. To do so, we can use the trivial condition $f(x) > 0$ in view of the codoamin of f.

Solution. Note that the equation shows the image of an arithmetic sequence is an arithmetic sequence. This observation can help simplify some arguments. For example, by letting $f(2) - f(1) = d$ and $f(1) = c + d$, we easily obtain $f(n) = c + nd$ for all $n \in \mathbb{N}$ since $f(1), f(2), \ldots, f(n)$ is an arithmetic sequence with common difference d.

Next, for any $n \in \mathbb{N}$, we consider the arithmetic sequence $\frac{1}{n}, \frac{2}{n}, \ldots$. Since 1 and 2 are two terms of this arithmetic sequence, we easily deduce

$$f\left(\frac{m+1}{n}\right) - f\left(\frac{m}{n}\right) = \frac{1}{n}(f(2) - f(1)) = \frac{d}{n}$$

for any $m \in \mathbb{N}$. Also, since $f(1) = c + d$, it is not hard to find that $f\left(\frac{m}{n}\right) = c + \frac{m}{n} \cdot d$ for any $m \in \mathbb{N}$. Thus, $f(r) = c + rd$ for any $r \in \mathbb{Q}^+$. Note that $c, d > 0$ since $f(r) > 0$ for all $r \in \mathbb{Q}^+$.

Then we proceed to consider any $x \in \mathbb{R}^+$. The idea is to consider some rational number r which is sufficiently close to x. Since the difference $\varepsilon = |x - r|$ is small, the number $x - k\varepsilon$ is positive for some large $k \in \mathbb{N}$. We can express $f(x - k\varepsilon)$ in terms of $f(x)$ and $f(r) = c + rd$. And since $f(x - k\varepsilon) > 0$, we can obtain some bounds on $f(x)$. Usually, by taking limit $r \to x$, we are able to deduce $f(x)$ uniquely.

Here are the details. For any $n \in \mathbb{N}$, there exists $r \in \mathbb{Q}^+$ such that $x - \frac{x}{n} < r < x$. By considering the arithmetic sequence $x, x - \varepsilon, x - 2\varepsilon, \dots, x - (n-1)\varepsilon$ where $\varepsilon = x - r > 0$ (note that the last term is positive since $x - (n-1)\varepsilon > x - \frac{n-1}{n}x > 0$), we see that

$$f(x - (n-1)\varepsilon) = f(x) - (n-1)(f(x) - f(r)) = (n-1)(c + rd) - (n-2)f(x).$$

Since $f(x - (n-1)\varepsilon) > 0$, this implies $f(x) < \frac{(n-1)(c+rd)}{n-2}$. When n tends to infinity, since $r \to x$, this yields

$$f(x) \leqslant c + xd.$$

Similarly, there exists $r' \in \mathbb{Q}^+$ such that $x < r' < x + \frac{x}{n}$. Using similar arguments with $\varepsilon' = r' - x$, we find that

$$f(r' - (n-1)\varepsilon') = f(r') - (n-1)(f(r') - f(x)) = (n-1)f(x) - (n-2)(c + r'd),$$

and hence $f(x) > \frac{(n-2)(c+r'd)}{n-1}$. When n tends to infinity, this yields

$$f(x) \geqslant c + xd.$$

Thus, we must have $f(x) = c + xd$ for any $x \in \mathbb{R}^+$. We check that all such functions are solutions. Indeed,

$$f(x) + f(y) = c + xd + c + yd = 2\left(c + \frac{x+y}{2} \cdot d\right) = 2f\left(\frac{x+y}{2}\right).$$

□

Problem 3.5. Find all function(s) $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that

$$f(f(x) + y) + f(x + y) = 2x + 2f(y) \tag{1}$$

for all $x, y \in \mathbb{R}^+$.

Analysis. Using some standard tricks, it is not hard to deduce a relation between $f(f(x))$ and $f(x)$. Replacing x by $f(x)$ repeatedly, this gives a relation between $f^k(x)$ and $f(x)$. In ordinary functional equation problems, this relation is usually useless. However, by using the positive condition $f^k(x) > 0$, we can derive some bounds on $f(x)$, which could be useful.

Solution. Noting that (1) is not symmetric in x and y, we try to deduce something using the symmetry. Thus, we first replace y by $f(y)$ in (1) to obtain

$$f(f(x)+f(y))+f(x+f(y))=2x+2f(f(y)), \tag{2}$$

so that the first clumsy term is symmetric. Swapping x and y in (1), we have

$$f(f(y)+x)+f(x+y)=2y+2f(x). \tag{3}$$

Taking the difference of (2) and (3), we get

$$f(f(x)+f(y))-f(x+y)=2x+2f(f(y))-2y-2f(x).$$

Swapping x and y, and comparing with this equation, we obtain

$$2x+2f(f(y))-2y-2f(x)=2y+2f(f(x))-2x-2f(y).$$

This implies

$$f(f(x))+f(x)-2x=f(f(y))+f(y)-2y.$$

Therefore, $f(f(x))+f(x)-2x=c$ for some constant c. This can be rewritten as

$$f^2(x)=-f(x)+2x+c.$$

Replacing x by $f(x)$, we have

$$f^3(x)=-f^2(x)+2f(x)+c=-(-f(x)+2x+c)+2f(x)+c=3f(x)-2x.$$

Similarly, we define

$$f^k(x)=a_kf(x)+b_kx+d_kc$$

for each $k\in\mathbb{N}$. Replacing x by $f(x)$, we obtain

$$\begin{aligned}
f^{k+1}(x)&=a_kf^2(x)+b_kf(x)+d_kc\\
&=a_k(-f(x)+2x+c)+b_kf(x)+d_kc\\
&=(b_k-a_k)f(x)+2a_kx+(a_k+d_k)c.
\end{aligned}$$

Thus, we derive the recurrence relations

$$a_{k+1}=b_k-a_k, \tag{4}$$
$$b_{k+1}=2a_k, \tag{5}$$
$$d_{k+1}=a_k+d_k. \tag{6}$$

It is standard to solve these recurrence equations. Firstly, combining (4) and (5) to eliminate b_k, we obtain

$$a_{k+2}=b_{k+1}-a_{k+1}=2a_k-a_{k+1}.$$

The characteristic equation is $\lambda^2+\lambda-2=0$, and so $a_k=A(-2)^k+B$ for some constants A and B. Using $a_1=1$ and $a_2=-1$, we find that

$$a_k=-\frac{1}{3}(-2)^k+\frac{1}{3}$$

for any $k \in \mathbb{N}$. By (5), we have

$$b_k = 2a_{k-1} = 2\left(-\frac{1}{3}(-2)^{k-1} + \frac{1}{3}\right) = \frac{1}{3}(-2)^k + \frac{2}{3}.$$

Also, (6) can be rewritten as $d_k - d_{k-1} = a_{k-1}$, and so

$$d_k = \sum_{n=2}^{k}(d_n - d_{n-1}) + d_1 = \sum_{n=2}^{k}\left(-\frac{1}{3}(-2)^{n-1} + \frac{1}{3}\right) = \frac{1}{9}(-2)^k + \frac{k}{3} - \frac{1}{9}.$$

It follows that

$$f^k(x) = a_k f(x) + b_k x + d_k c = (-2)^k\left(-\frac{f(x)}{3} + \frac{x}{3} + \frac{c}{9}\right) + \frac{kc}{3} + \left(\frac{f(x)}{3} + \frac{2x}{3} - \frac{c}{9}\right).$$

For a fixed $x \in \mathbb{R}^+$, we denote this by $(-2)^k u + kv + w$ where u, v, w are fixed. Since $f^k(x) > 0$ for any $k \in \mathbb{N}$, it is not hard to guess u must be 0. Here is a rigorous proof.

- If $u > 0$, we consider $k = 2m - 1$ where $m \in \mathbb{N}$. As $f^k(x) > 0$, we have

  $$-2^{2m-1}u + (2m-1)v + w > 0.$$

  This is not true when $m \to \infty$.

- Similarly, if $u < 0$, then by considering $k = 2m$ where $m \in \mathbb{N}$, we have

  $$2^{2m}u + 2mv + w > 0.$$

  This is not true when $m \to \infty$.

Therefore, we must have $u = 0$. This implies $f(x) = x + \dfrac{c}{3}$. Now, we check that

$$f(f(x)+y) + f(x+y) = \left(x + y + \frac{2c}{3}\right) + \left(x + y + \frac{c}{3}\right) = 2x + 2y + c$$

and

$$2x + 2f(y) = 2x + 2\left(y + \frac{c}{3}\right) = 2x + 2y + \frac{2c}{3}$$

are equal iff $c = 0$. Thus, the only solution is $f(x) = x$ for all $x \in \mathbb{R}^+$. $\square$

Remarks. To use similar tricks, one has to be familiar with the growth of some simple functions. For example, exponential functions grow faster than any polynomials (this fact is used in the above solution). Polynomials of larger degrees grow faster. Also, polynomials of degree at least 1 grow faster than logarithm functions.

Next, we go through two shortlist problems involving $f : \mathbb{R}^+ \to \mathbb{R}^+$. There are some new ideas involved in each problem, so it is worth to take a look.

Problem 3.6. (IMO Shortlist 2005 A2) Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ which have the property:

$$f(x)f(y) = 2f(x + yf(x)) \tag{1}$$

for all positive real numbers x and y.

Analysis. It is not hard to guess the only solution is the constant function $f(x) = 2$. The first idea is to equate $x = x + yf(x)$ or $y = x + yf(x)$ to cancel two of the terms. The former is impossible, while the latter is possible if $y = \dfrac{x}{1 - f(x)} > 0$. Note that one should always check this positive condition when working on functions defined on $\mathbb{R}^+$, since otherwise we cannot substitute such a value.

By using this, we can already obtain a nontrivial lower bound for f. We can modify the argument to get better and better bounds. Eventually, we are able to obtain the bound $f(x) \geqslant 2$, which is the best possible in view of the solution. Using inequalities is the key idea in solving this problem.

Solution. If $f(x) < 1$ for some $x \in \mathbb{R}^+$, then we put $y = \dfrac{x}{1 - f(x)} > 0$ in (1) so that $y = x + yf(x)$. It follows that $f(x) = 2$, contradicting $f(x) < 1$. Therefore, we must have

$$f(x) \geqslant 1$$

for all $x \in \mathbb{R}^+$. We can use this to obtain a better bound. Indeed, putting $x = y$ in (1), we immediately obtain

$$f(x)^2 = 2f(x + xf(x)) \geqslant 2,$$

and hence $f(x) \geqslant \sqrt{2}$. In general, whenever we have $f(x) \geqslant c$ for some constant c, then by putting $x = y$ in (1), we obtain $f(x) \geqslant \sqrt{2c}$. Thus, we define $a_1 = \sqrt{2}$ and

$$a_{n+1} = \sqrt{2a_n} \tag{2}$$

for $n \geqslant 1$. Then we must have $f(x) \geqslant a_n$ for all $x \in \mathbb{R}^+$ and $n \in \mathbb{N}$.

Inductively, it is easy to prove that $\{a_n\}$ is increasing and bounded above by 2. Therefore, by the monotone convergence theorem, $\{a_n\}$ converges to some $L \in \mathbb{R}$. Indeed, by taking limits on both sides of (2), we have $L = \sqrt{2L}$, and so $L = 0, 2$. Clearly, $L \geqslant a_1 > 0$. Thus, $\{a_n\}$ converges to 2. In other words, we must have

$$f(x) \geqslant 2$$

for all $x \in \mathbb{R}^+$.

Another standard trick to proceed is to swap x and y in (1) since the equation is not symmetric. Comparing the result with (1), we obtain

$$f(x + yf(x)) = f(y + xf(y)).$$

If f is injective (which should not be the case), then $x + yf(x) = y + xf(y)$, i.e.

$$\frac{f(x)-1}{x} = \frac{f(y)-1}{y}.$$

This implies $\dfrac{f(x)-1}{x} = c$ for some constant c, and hence $f(x) = cx+1$. It is not hard to check that it does not satisfy (1).

Therefore, f is not injective, and there exist $\alpha > \beta$ such that $f(\alpha) = f(\beta)$. We try to make one side of (1) contain the term $f(\alpha)$ and the other side contain $f(\beta)$. Indeed, by putting $x = \beta$ and $y = \dfrac{\alpha-\beta}{f(\beta)} > 0$ in (1), we get

$$f(\beta)f\left(\frac{\alpha-\beta}{f(\beta)}\right) = 2f\left(\beta + \frac{\alpha-\beta}{f(\beta)}f(\beta)\right),$$

i.e.

$$f(\beta)f\left(\frac{\alpha-\beta}{f(\beta)}\right) = 2f(\alpha).$$

Since $f(\alpha) = f(\beta)$, this yields $f\left(\dfrac{\alpha-\beta}{f(\beta)}\right) = 2$. In other words, we have found some $d \in \mathbb{R}^+$ such that $f(d) = 2$. This can be a useful result. For example, if we can choose $x, y > 0$ such that $f(x + yf(x)) = 2$, then $f(x)f(y) = 2f(x + yf(x)) = 4$. As $f(x), f(y) \geqslant 2$, this forces $f(x) = f(y) = 2$. However, in order that at least one of x and y is arbitrary (and can be arbitrarily large), the term $x + yf(x) \geqslant x + 2y$ should not be bounded. So we need a sufficiently large d to use this argument.

Indeed, by putting $x = y = d$ in (1), we obtain

$$f(d)^2 = 2f(d + df(d)).$$

As $f(d) = 2$, this implies $f(3d) = 2$. Inductively, we have $f(3^k d) = 2$ for all $k \in \mathbb{N}$.

Now, for each fixed $x \in \mathbb{R}^+$, we put $y = \dfrac{3^k d - x}{f(x)} > 0$ in (1) where $k \in \mathbb{N}$ is sufficiently large. This gives

$$f(x)f\left(\frac{3^k d - x}{f(x)}\right) = 2f\left(x + \frac{3^k d - x}{f(x)} \cdot f(x)\right) = 2f(3^k d) = 4.$$

As each term on the left is at least 2, we must have $f(x) = f\left(\dfrac{3^k d - x}{f(x)}\right) = 2$. In particular, $f(x) = 2$ for all $x \in \mathbb{R}^+$. This is obviously a solution. □

Problem 3.7. (IMO Shortlist 2007 A4) Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that

$$f(x + f(y)) = f(x+y) + f(y) \tag{1}$$

for all $x, y \in \mathbb{R}^+$.

Analysis. Note that $f(x) = 2x$ should be the only solution. As we are more familiar with the identity function $g(x) = x$, sometimes it is better to convert the functional equation so that it has the solution $g(x) = x$. This idea may also be useful when solving general problems in functional equations, not necessarily for f defined on $\mathbb{R}^+$. However, for this kind of problems, the range of the new function needs not be $\mathbb{R}^+$ anymore.

Solution. Define $g : \mathbb{R}^+ \to \mathbb{R}$ such that $g(x) = f(x) - x$. Then

$$\begin{aligned} & f(x + f(y)) = f(x + y) + f(y) \\ \Leftrightarrow \quad & f(x + g(y) + y) = g(x + y) + x + y + g(y) + y \\ \Leftrightarrow \quad & g(x + g(y) + y) + x + g(y) + y = g(x + y) + x + y + g(y) + y \\ \Leftrightarrow \quad & g(x + y + g(y)) = g(x + y) + y. \end{aligned} \tag{2}$$

We now prove that g is injective. Suppose $g(\alpha) = g(\beta)$. By putting $y = \alpha$ and $y = \beta$ respectively in (2), we obtain

$$\begin{cases} g(x_1 + \alpha + g(\alpha)) = g(x_1 + \alpha) + \alpha, \\ g(x_2 + \beta + g(\beta)) = g(x_2 + \beta) + \beta. \end{cases}$$

In order to compare the two equations, we should not use the same x. Indeed, we choose $x_1 = x - \alpha$ and $x_2 = x - \beta$ where x is sufficiently large so that $x_1, x_2 > 0$. Then the two equations look similar, and indeed this forces $\alpha = \beta$. This means g is injective.

We shall use the trick introduced in problem 2.17 (just try every possible approach). Replacing y by $y + z$ in (2), we obtain

$$g(x + y + z + g(y + z)) = g(x + y + z) + y + z. \tag{3}$$

We can use (2) to rewrite the right-hand side in two steps. Actually, equation (2) allows us to combine two terms $g(u)$ and v into a single term whenever $u > v$. Firstly, we have

$$g((x + z) + y) + y = g\big((x + z) + y + g(y)\big).$$

Next, we have

$$\begin{aligned} g(x + y + z) + y + z &= g(x + y + z + g(y)) + z = g\big((x + y + g(y)) + z\big) + z \\ &= g\big(x + y + z + g(y) + g(z)\big). \end{aligned} \tag{4}$$

Comparing (3) and (4), and using the injectivity, we obtain the Cauchy equation

$$g(y + z) = g(y) + g(z).$$

In order to conclude $g(y) = cy$ for some constant c, we need one extra condition such as $g(x) > 0$ for all $x \in \mathbb{R}^+$. This means we have to prove $f(x) > x$. This is the place where we can use the fact $f(x) > 0$. Indeed, in view of equation (1), the terms $x + f(y)$ cannot be equal to $x + y$ or y or otherwise the remaining term has image 0.

More concretely, If $f(y) = y$ for some y, then (1) becomes $f(x+y) = f(x+y) + y$, i.e. $y = 0$. This is not possible. If $f(y) < y$ for some y, we put $x = y - f(y)$ in (1). The equation is $f(y) = f(2y - f(y)) + f(y)$, which again leads to the contradiction $f(2y - f(y)) = 0$. It follows that $g(x) = f(x) - x > 0$ for all $x \in \mathbb{R}^+$. By proposition 2.1, we know that $g(y) = cy$ for some $c \in \mathbb{R}^+$. Now,

$$g(x + y + g(y)) = c(x + y + cy)$$

and

$$g(x + y) + y = c(x + y) + y$$

are the same iff $c^2 y = y$. This holds for all $x, y \in \mathbb{R}^+$ iff $c = 1$ (as the image is positive), i.e. $f(x) = 2x$ for all $x \in \mathbb{R}^+$. □

Remarks. Note that g being injective is different from f being injective. Thus, if we find that it seems hard to show that a function is injective, we may think of transforming the function to another to establish this property. This alternative result could be identically useful.

In the above problems, we have used the following trick. Suppose after substituting x and y to be certain expressions, we can derive some contradiction. Then this implies the substitutions cannot be done, which means that one of the expressions is non-positive. This trick is often used in this kind of problems.

Problem 3.8. (Spain MO 2018 Problem 6) Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that

$$f(x + f(y)) = yf(xy + 1) \tag{1}$$

for every $x, y \in \mathbb{R}^+$.

Analysis. A common solution to functional equation problems in $f : \mathbb{R}^+ \to \mathbb{R}^+$ is $f(x) = \frac{1}{x}$, which is also the solution to this problem. In this solution, we use a less common approach, i.e. proof by contradiction. We shall prove that both $f(x) > \frac{1}{x}$ and $f(x) < \frac{1}{x}$ are impossible.

Solution. Firstly, we would like to see whether $x + f(y)$ and $xy + 1$ can be equal. Indeed, they are equal iff $(y-1)x = f(y) - 1$. Thus, if $y \neq 1$ and $\frac{f(y)-1}{y-1} > 0$, then we can put $x = \frac{f(y)-1}{y-1}$ in (1) so that $f(x + f(y)) = f(xy + 1)$. This gives $y = 1$, contradicting the assumption $y \neq 1$. Thus, for any $y \neq 1$, we must have $\frac{f(y)-1}{y-1} \leqslant 0$. In other words, if $y > 1$, then $f(y) \leqslant 1$, while if $y < 1$, then $f(y) \geqslant 1$.

Noting that $xf(x)$ should be a constant, we try to generate this term and use the above inequalities. Indeed, we replace x by $\frac{y-1}{y}$ in (1) to obtain

$$f\left(1-\frac{1}{y}+f(y)\right)=yf(y).$$

Note that this can only be done when $y>1$ since otherwise $x\notin\mathbb{R}^{+}$.

- If $f(y)>\frac{1}{y}$, then $1-\frac{1}{y}+f(y)>1$, and so $f\left(1-\frac{1}{y}+f(y)\right)\leqslant 1$. However, this contradicts $yf(y)>y\cdot\frac{1}{y}=1$.
- If $f(y)<\frac{1}{y}$, then $1-\frac{1}{y}+f(y)<1$, and so $f\left(1-\frac{1}{y}+f(y)\right)\geqslant 1$. However, this contradicts $yf(y)<y\cdot\frac{1}{y}=1$.

It follows that $f(y)=\frac{1}{y}$ for any $y>1$.

Next, for any $x\leqslant 1$, we replace x by $\frac{x}{2}$ and put $y=\frac{2}{x}$ in (1). Since $y>1$ and $xy+1>1$, this gives $f\left(\frac{x}{2}+\frac{x}{2}\right)=\frac{2}{x}f(2)$, and hence $f(x)=\frac{1}{x}$. So this holds for any $x\in\mathbb{R}^{+}$. The checking is trivial. □

Problem 3.9. Find all function(s) $f:\mathbb{R}^{+}\to\mathbb{R}^{+}$ such that

$$f(f(x)+xy)=x(1+f(y)) \tag{1}$$

for all $x,y\in\mathbb{R}^{+}$.

Analysis. In this problem, we see a solution using the monotonicity of a function. Indeed, the first key step is to show that f is increasing. Also, it is obvious that f is surjective. Thus, these imply the function f is continuous. It is rare that we can deduce such a property in some MO problems. But at the same time it can be a useful property.

Solution. We first prove the following main result.

Claim. f is increasing

Proof. Suppose on the contrary that there exist $a<b$ such that $f(a)>f(b)$. The key observation is that $y=\frac{f(a)-f(b)}{b-a}>0$. For this y, we have $f(a)+ay=f(b)+by$. Therefore, by putting $x=a$ and $x=b$ in (1) respectively, we obtain

$$\begin{cases} f(f(a)+ay) &= a(1+f(y)),\\ f(f(b)+by) &= b(1+f(y)).\end{cases}$$

Since $f(f(a)+ay) = f(f(b)+by)$, we have $a(1+f(y)) = b(1+f(y))$, and hence $a = b$, contradicting $a < b$. This proves the claim. $\square$

Next, replacing x by $\dfrac{x}{1+f(y)}$ in (1), we see that f is surjective. As f is a surjective increasing function, it must be continuous, which means we can now take limits. Note that the right-hand limit of f at 0 also exists, and it must be equal to 0 since f is surjective and increasing. Thus, we can consider the limits when y tends to 0 in (1). This gives

$$\lim_{y\to 0^+} f(f(x)+xy) = f\Big(\lim_{y\to 0^+} (f(x)+xy)\Big) = f(f(x))$$

and

$$\lim_{y\to 0^+} x(1+f(y)) = x\Big(1+\lim_{y\to 0^+} f(y)\Big) = x.$$

As they are equal, we have

$$f(f(x)) = x \tag{2}$$

for any $x \in \mathbb{R}^+$. Using this, we can easily show that $f(x) = x$.

- If $f(x) > x$, then $f(f(x)) \geqslant f(x)$ since f is increasing. Together with (2), it follows that $x = f(f(x)) \geqslant f(x)$, contradicting $f(x) > x$.
- If $f(x) < x$, then $f(f(x)) \leqslant f(x)$. But then $x = f(f(x)) \leqslant f(x)$, contradicting $f(x) < x$.

Therefore, we can only have $f(x) = x$ for all $x \in \mathbb{R}^+$. Clearly, this is a solution. $\square$

Remarks. Note that how we use various results in analysis to work on this problem. Firstly, once we have established $\lim_{x\to 0^+} f(x) = 0$, then we can basically substitute $y = 0$ in the functional equation by defining $f(0) = 0$. This crucial step is forbidden for $f : \mathbb{R}^+ \to \mathbb{R}^+$, but now we are able to do so. Secondly, if we can prove that f is monotonic and $f(f(x)) = x$, then we can conclude $f(x) = x$. This is one common usage of the monotonic condition.

Next, we move on to see some problems for which the functions are defined on $\mathbb{N}$. Here are some simple facts that can be easily proven, and they may be useful in solving such problems.

- Given a bounded function $f : \mathbb{N} \to \mathbb{N}$, there exists $n \in \mathbb{N}$ such that $f(m) = n$ has infinitely many solutions in m.
- Given $f : \mathbb{N} \to \mathbb{N}$, there exists $m \in \mathbb{N}$ such that $f(n) \geqslant f(m)$ for all $n \in \mathbb{N}$.
- Given $f : \mathbb{N} \to \mathbb{N}$, there exist infinitely many $m < n$ such that $f(m) \leqslant f(n)$.

Problem 3.10. (IMO Shortlist 2011 A4) Determine all pairs (f, g) of functions from the set of positive integers to itself that satisfy

$$f^{g(n)+1}(n) + g^{f(n)}(n) = f(n+1) - g(n+1) + 1 \tag{1}$$

for every positive integer n.

Analysis. Since the left-hand side involves a nested function with many layers, the usual techniques of solving functional equations can hardly be used in this problem. To handle such nested functions, some common approaches include establishing some bounds, or showing the functions have some periodicity.

In view of (1), the term $f(n+1)$ should be relatively large compared to the other terms. But then if f increases rapidly, then the term $f^{g(n)+1}(n)$ should be even much larger. So we can try to set up some inequalities on f.

Solution. Rearranging the terms in (1), and noting that all images are at least 1, we have

$$f(n+1) = f^{g(n)+1}(n) + g^{f(n)}(n) + g(n+1) - 1 \geqslant f^{g(n)+1}(n) + 1 + 1 - 1 > f^{g(n)+1}(n). \tag{2}$$

As $g(n) + 1 \geqslant 2$, this suggests applying more times of f does not increase the value by too much, while f cannot be a constant function. So we guess $f(n) = n$ is the only solution.

Recall that there must be a minimum value among $f(1), f(2), \ldots$ since any nonempty subset of $\mathbb{N}$ has a smallest element (this is called the well-ordering principle(良序原理)). Let m_1 be any positive integer such that $f(m_1)$ is the smallest possible. If $m_1 \geqslant 2$, we can put $n = m_1 - 1$ in (2) to obtain

$$f(m_1) > f\big(f^{g(m_1-1)}(m_1 - 1)\big).$$

This contradicts the minimality of $f(m_1)$. Thus, m_1 must be 1, i.e. $f(1)$ is the unique minimum value.

Similarly, we let $m_2 \geqslant 2$ be any positive integer such that $f(m_2)$ is the smallest among $f(2), f(3), \ldots$. Then by putting $n = m_2 - 1$ in (2), we have

$$f(m_2) > f\big(f^{g(m_2-1)}(m_2 - 1)\big).$$

From the choice of m_2, we must have $f^{g(m_2-1)}(m_2 - 1) = 1$. Since f only takes on positive integer values, and now we have shown that 1 belongs to the range of f, we must have $f(1) = 1$. Also, by $f^{g(m_2-1)}(m_2 - 1) = 1$, we can easily deduce $m_2 - 1 = 1$, which means $m_2 = 2$.

The same argument can be modified to show that $f(n) = n$ for all $n \in \mathbb{N}$. Indeed, suppose we have shown that $f(n) = n$ for $n = 1, 2, \ldots, k$, and $f(n) > f(k+1) > k$ for any $n > k+1$ (where the base case $f(1) = 1$ and $f(n) > f(2) > 1$ for any $n > 2$ is proven above). We let $m \geqslant k+2$ be any positive integer such that $f(m)$ is the smallest among $f(k+2), f(k+3), \ldots$. By putting $n = m - 1$ in (2), we have

$$f(m) > f\big(f^{g(m-1)}(m - 1)\big).$$

From the choice of m, the term $f^{g(m-1)}(m-1)$ can only be $1, 2, \dots, k+1$.

If $f^{g(m-1)}(m-1) = t$ for some $t = 1, 2, \dots, k$, then $m-1 = t$ by the inductive hypothesis. This leads to the contradiction

$$k + 2 \leqslant m = t + 1 \leqslant k + 1.$$

So we can only have $f^{g(m-1)}(m-1) = k+1$. Again, this has to be the minimum value among $f(k+1), f(k+2), \dots,$ and so we must have $f(k+1) = k+1$. By $f^{g(m-1)}(m-1) = k+1$, we easily obtain $m-1 = k+1$, and hence $m = k+2$. This implies

$$f(n) > f(m) = f(k+2) > k+1$$

for any $n > k+2$. By induction, we find that $f(n) = n$ for any $n \in \mathbb{N}$.

Now, (1) becomes

$$n + g^n(n) = n + 1 - g(n+1) + 1.$$

This implies $g^n(n) + g(n+1) = 2$. Since each term on the left is a positive integer, both of them are 1. In particular, $g(n+1) = 1$ for any $n \in \mathbb{N}$. Also, by putting $n = 1$, we get $g(1) = g(2) = 1$. Thus, $g(n) = 1$ for any $n \in \mathbb{N}$.

It is trivial to check that $f(n) = n$ and $g(n) = 1$ for all $n \in \mathbb{N}$ satisfy the equation. □

Problem 3.11. (IMO Shortlist 2009 A3/Problem 5) Determine all functions f from the set of positive integers to the set of positive integers such that, for all positive integers a and b, there exists a non-degenerate triangle with sides of lengths

$$a, \ f(b) \text{ and } f(b + f(a) - 1).$$

(A triangle is *non-degenerate* if its vertices are not collinear.)

Analysis. For the numbers to be the side lengths of a triangle, we have nothing to use except the triangle inequality. Since only integers are involved, we shall use $a + b \geqslant c + 1$ instead of $a + b > c$, as the former is stronger. This is another common trick we use when handling strict inequalities of integers.

To start with, we first consider some small values of a since we can say more when one of the side lengths has a small value.

Solution. Consider $a = 1$. Then 1, $f(b)$ and $f(b + f(1) - 1)$ are the side lengths of a triangle. For any triangle with side lengths $1, x, y$ where $x, y \in \mathbb{N}$, by the triangle inequality, we have

$$\begin{cases} 1 + x \geqslant y + 1, \\ 1 + y \geqslant x + 1 \end{cases}$$

This implies $x \geqslant y$ and $y \geqslant x$. Thus, we must have $x = y$. In our situation, we have

$$f(b) = f(b + f(1) - 1) \tag{1}$$

for all $b \in \mathbb{N}$.

We can use (1) to show that $f(1) = 1$. Suppose on the contrary that $d = f(1) - 1 > 0$. Then (1) becomes $f(b) = f(b+d)$ for all $b \in \mathbb{N}$. This means f is periodic. As f is defined on $\mathbb{N}$, this implies f is bounded. Let $f(n) \leqslant M$ for all $n \in \mathbb{N}$. By the triangle inequality, we have

$$a < f(b) + f(b + f(a) - 1) \leqslant 2M$$

for any $a \in \mathbb{N}$, which is clearly impossible. Therefore, we must have $f(1) = 1$.

Having the value of $f(1)$, it is natural to consider $b = 1$. Then a, 1 and $f(f(a))$ are the side lengths of a triangle. From above, this yields $f(f(a)) = a$ for any $a \in \mathbb{N}$. Thus, f is bijective.

The next natural step is to find $f(2)$. For example, we may consider $a = 2$ and $b = f(2)$. Then 2, $f(f(2)) = 2$ and $f(2f(2)-1)$ are the side lengths of a triangle. By the triangle inequality, we see that $f(2f(2)-1)$ can only be $1, 2, 3$.

- If $f(2f(2)-1) = 1$, then $f(2f(2)-1) = f(1)$, and hence, $2f(2)-1 = 1$ since f is injective. This yields $f(2) = 1 = f(1)$, which contradicts the injectivity.
- If $f(2f(2)-1) = 2$, then $2f(2)-1 = f\big(f(2f(2)-1)\big) = f(2)$. So $f(2) = 1$, which is again a contradiction.

So we must have $f(2f(2)-1) = 3$. This gives $f(3) = f\big(f(2f(2)-1)\big) = 2f(2)-1$. Although we cannot find out $f(2)$ explicitly, we can express all other $f(k)$'s in terms of $f(2)$ for $k \geqslant 2$. Indeed, one easily guesses the formula below by finding one more term $f(4)$. The proof is very much similar to the steps above.

Claim. $f(k) = (k-1)f(2) - (k-2)$ for any $k \geqslant 2$

Proof. We prove this by induction. The base cases $k = 2, 3$ have already been proven. Assume the claim holds for $k = 2, 3, \ldots, m$ where $m \geqslant 3$. When $k = m+1$, we consider $a = m$ and $b = f(2)$. Then m, $f(f(2)) = 2$ and $f\big(f(2) + f(m) - 1\big)$ are the side lengths of a triangle. By the triangle inequality, $f\big(f(2) + f(m) - 1\big)$ can only be $m-1, m, m+1$. Also, note that $f\big(f(2) + f(m) - 1\big) = f\big(mf(2) - (m-1)\big)$ by the inductive hypothesis.

- If $f\big(mf(2) - (m-1)\big) = m-1$, then

$$f(m-1) = f\Big(f\big(mf(2) - (m-1)\big)\Big) = mf(2) - (m-1).$$

  By the inductive hypothesis, this yields $(m-2)f(2) - (m-3) = mf(2) - (m-1)$, i.e. $f(2) = 1$, contradiction.

- If $f\big(mf(2) - (m-1)\big) = m$, then

$$f(m) = f\Big(f\big(mf(2) - (m-1)\big)\Big) = mf(2) - (m-1).$$

  By the inductive hypothesis, this yields $(m-1)f(2) - (m-2) = mf(2) - (m-1)$, i.e. $f(2) = 1$, contradiction.

So we must have $f\big(mf(2)-(m-1)\big)=m+1$. Then we have

$$f(m+1)=f\Big(f\big(mf(2)-(m-1)\big)\Big)=mf(2)-(m-1),$$

and hence the inductive step is completed. By induction, the claim is true. $\square$

Now, we may use the fact $f(f(2))=2$ to compute all terms. By putting $k=f(2)$ in the claim, we obtain

$$2=f(f(2))=(f(2)-1)f(2)-(f(2)-2),$$

and so $f(2)^2-2f(2)=0$. Since $f(2)>0$, the only solution is $f(2)=2$. Again by the claim, we find that

$$f(k)=2(k-1)-(k-2)=k$$

for any $k\geqslant 2$. This also holds when $k=1$. It remains to check that a, $f(b)=b$ and $f(b+f(a)-1)=a+b-1$ always satisfy the triangle inequality. Indeed, we have

$$\begin{aligned}a+b&>a+b-1,\\ a+(a+b-1)&>b,\\ b+(a+b-1)&>a.\end{aligned}$$

So $f(n)=n$ for any $n\in\mathbb{N}$. $\square$

3.3 Functional Inequalities

In a functional inequality problem, the given condition is an inequality relating some terms of a function f. Compared to functional equations, the conditions are generally weaker. Also, rather than finding out all possible functions, sometimes we may only be asked to show some properties of the functions since it may not be possible to find out all solutions. In this subsection, we are going to see several problems of this kind. Some tricks involved are similar to those used in functional equations involving $f:\mathbb{R}^+\to\mathbb{R}^+$.

Problem 3.12. Find all functions $f:\mathbb{R}^+\to\mathbb{R}^+$ such that

$$f(x+y)+f(x^x)\geqslant f(y^{xy})+f(xy)+f(f(x)) \tag{1}$$

for all $x,y\in\mathbb{R}^+$.

Analysis. Again, for functions whose codomain is $\mathbb{R}^+$, we have a trivial inequality $f(x)>0$. This can be used together with the given inequality to simplify the expression. For example, since there are two terms on the left and three terms on the right, if we can equate two pairs of terms on both sides, then we deduce a contradiction easily.

Solution. After some trial and error, we shall show the existence of $x, y \in \mathbb{R}^+$ such that $x^x = y^{xy}$ and $x + y = xy$. Indeed, $x^x = y^{xy}$ iff $x = y^y$. Thus, we put $x = y^y$ in (1) to obtain

$$f(y^y + y) \geqslant f(y^{y+1}) + f(f(y^y)). \tag{2}$$

Next, $y^y + y = y^{y+1}$ iff $y^y - y^{y-1} = 1$. Define $g(y) = y^y - y^{y-1}$. Note that $g(1) = 0$ and $g(2) = 2$. As g is continuous, by the intermediate value theorem, there exists $y \in (1, 2)$ such that $g(y) = 1$. We put this y in (2). This gives $0 \geqslant f(f(y^y))$. However, this is impossible since $f(f(y^y)) \in \mathbb{R}^+$. Thus, there is no function f satisfying (1). $\square$

Problem 3.13. (All-Russian MO 2000 Grade 11 Problem 1) Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(x + y) + f(y + z) + f(z + x) \geqslant 3f(x + 2y + 3z) \tag{1}$$

for all $x, y, z \in \mathbb{R}$.

Analysis. Due to the non-symmetry of this problem, the only solutions should be the constant functions. The only possibility to obtain equality from inequality is to deduce $a = b$ from $a \leqslant b$ and $b \leqslant a$. Therefore, our goal is to deduce $f(a) \leqslant f(b)$ and $f(b) \leqslant f(a)$ where at least one of a, b is arbitrary. As in the case of functional equations, the simplest way is to start with the substitution $x = 0$, etc.

Solution. Put $y = z = 0$ in (1). This gives $2f(x) + f(0) \geqslant 3f(x)$, and so

$$f(0) \geqslant f(x) \tag{2}$$

for any $x \in \mathbb{R}$. To get the opposite inequality, the right-hand side should be $f(0)$. Therefore, we replace x by $-2y - 3z$ in (1). This gives

$$f(-y - 3z) + f(y + z) + f(-2y - 2z) \geqslant 3f(0).$$

We can apply (2) to each term on the left. We have

$$3f(0) \geqslant f(-y - 3z) + f(y + z) + f(-2y - 2z) \geqslant 3f(0).$$

Therefore, equality holds. In particular, $f(y + z) = f(0)$ for any $y, z \in \mathbb{R}$. This shows f is a constant function.

Conversely, suppose $f(x) = c$ for all $x \in \mathbb{R}$ where c is a constant. Then both sides of (1) become $3c$ so that the inequality is satisfied. $\square$

Remarks. This basic idea is used in solving many problems in functional inequality, if we are asked to find all possible functions.

Problem 3.14. (Benelux MO 2013 Problem 2) Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(x + y) + y \leqslant f(f(f(x))) \tag{1}$$

holds for all $x, y \in \mathbb{R}$.

Analysis. The initial tricks are similar to those used to solve functional equations. For example, we equate the two terms $f(x+y)$ and $f^3(x)$ to remove the nested function $f^3(x)$. Note that whenever we have obtained an inequality $a \leqslant b$, we try to see if it is possible to deduce $b \leqslant a$ to conclude $a = b$. This is usually the only way that we can eventually work out the solution.

Solution. We put $y = f(f(x)) - x$ in (1) so that $x + y = f(f(x))$. This gives

$$f(f(x)) \leqslant x.$$

Replacing x by $f(x)$, this yields $f^3(x) \leqslant f(x)$. Thus, (1) implies

$$f(x+y) + y \leqslant f(x). \tag{2}$$

Since $x + y$ and x are basically independent, we can actually swap their roles to get the opposite relation. Indeed, we replace (x, y) by $(x + y, -y)$ in (2). Rearranging the terms, we obtain

$$f(x) \leqslant f(x+y) + y.$$

Comparing with (2), we have

$$f(x+y) + y = f(x)$$

for any $x, y \in \mathbb{R}$. Putting $x = 0$, we find that $f(y) + y = f(0)$. This shows $f(y) = c - y$ for some constant c. We check that all such functions are solutions. Indeed, we have

$$f(x+y) + y = c - (x+y) + y = c - x = f^3(x).$$

□

Problem 3.15. (Dutch TST 2018 Test 3 Problem 2) Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(x^2) - f(y^2) \leqslant (f(x) + y)(x - f(y)) \tag{1}$$

for all $x, y \in \mathbb{R}$.

Analysis. When handling inequalities, if we multiply both sides by -1, we need to change the inequality sign. If we can prove that f is an odd function, very often we can change an expression to its negative by replacing all variables by their negatives, while the inequality sign needs not be changed. Thus, we can compare the two results to obtain an equality. This is another useful trick in functional inequalities.

Solution. Putting $x = y = 0$ in (1), we obtain $0 \leqslant -f(0)^2$. This implies $f(0) = 0$. Putting $x = 0$ in (1) and multiplying both sides by -1, we obtain

$$f(y^2) \geqslant y f(y). \tag{2}$$

Putting $y = 0$ in (1), we get

$$f(x^2) \leqslant x f(x). \tag{3}$$

Comparing (2) and (3), we plainly obtain

$$f(x^2) = xf(x) \tag{4}$$

for any $x \in \mathbb{R}$. Replacing x by $-x$ and comparing with (4), we have $-xf(-x) = xf(x)$. This implies $f(-x) = -f(x)$ for $x \neq 0$, and hence f is an odd function (as $f(0) = 0$).

Now, using (4), the inequality (1) can be rewritten as

$$xf(x) - yf(y) \leqslant (f(x) + y)(x - f(y)).$$

After simplification, this becomes

$$f(x)f(y) \leqslant xy$$

for any $x, y \in \mathbb{R}$. Replacing y by $-y$ and noting that f is odd, this becomes $-f(x)f(y) \leqslant -xy$, or

$$f(x)f(y) \geqslant xy.$$

Clearly, the two inequalities combine to give

$$f(x)f(y) = xy. \tag{5}$$

In particular, $f(x)f(1) = x$ for any $x \in \mathbb{R}$. Also, by putting $x = y = 1$ in (5), we find that $f(1)^2 = 1$, and so $f(1) = \pm 1$. This shows $f(x) = x$ for all $x \in \mathbb{R}$ or $f(x) = -x$ for all $x \in \mathbb{R}$. In either case, (1) holds since the two sides are equal. So these are the only solutions. □

The concept of limits may be needed to solve some kinds of functional inequality problems. We shall see several applications below.

Problem 3.16. Let $f : \mathbb{R}^+ \to \mathbb{R}^+$ be a function such that

$$f(2x) \geqslant x + f(f(x)) \tag{1}$$

for all $x \in \mathbb{R}^+$. Prove that $f(x) \geqslant x$ for any $x \in \mathbb{R}^+$.

Analysis. The only idea is to seek for better and better lower bounds for $f(x)$. We start with the trivial bound $f(x) > 0$. Then we can use this to deduce a stronger bound $f(2x) \geqslant x$, which means $f(x) \geqslant \frac{x}{2}$.

By repeating the process indefinitely, we eventually obtain the desired bound. More formally, this means we have to take limits.

Solution. Using the trivial bound $f(f(x)) > 0$, by (1) we obtain $f(2x) > x$. Changing the variable, this becomes

$$f(x) > \frac{x}{2}$$

for all $x \in \mathbb{R}^+$. Next, using this bound, we deduce

$$f(2x) \geqslant x + f(f(x)) > x + \frac{f(x)}{2} > x + \frac{x}{4} = \frac{5}{4}x.$$

Replacing x by $\frac{x}{2}$, we obtain

$$f(x) > \frac{5}{8}x.$$

In general, if we have $f(x) > a_n x$ for all $x \in \mathbb{R}^+$ where $a_n \geqslant 0$ is a constant, then we can use (1) to deduce

$$f(2x) \geqslant x + f(f(x)) > x + a_n f(x) = x + a_n^2 x = (1 + a_n^2)x.$$

This implies

$$f(x) > \frac{1 + a_n^2}{2}x$$

for all $x \in \mathbb{R}^+$. Therefore, if we define $a_1 = 0$ and $a_{n+1} = \frac{1 + a_n^2}{2}$ for any $n \in \mathbb{N}$, then we have $f(x) > a_n x$ for all $x \in \mathbb{R}^+$ and all $n \in \mathbb{N}$ by induction. Using the result of problem 3.1, we know that $\{a_n\}$ converges to 1. Therefore, by taking limits of $f(x) > a_n x$, we obtain $f(x) \geqslant x$ for all $x \in \mathbb{R}^+$ as desired. □

Remarks. Note that when we take limits of a strict inequality, the result needs not be a strict inequality. For example, in this problem, even though $f(x) > a_n x$ and $\{a_n\}$ converges to 1, it is possible that $f(x) = x$ (which is clearly one of the solutions).

Problem 3.17. Suppose $f : \mathbb{R} \to \mathbb{R}$ is a function such that

$$f(x + y) \geqslant f(x) + y f(f(x)) \tag{1}$$

for all $x, y \in \mathbb{R}$. Prove that $f(0) \leqslant 0$.

Analysis. It is natural to use proof by contradiction. However, it may not be obvious how the condition $f(0) > 0$ can be used.

Rather than that, we first try to work out more properties of f before using the assumption $f(0) > 0$. Note that there is a term y on the right-hand side of (1). If $f(f(x)) > 0$ for some $x \in \mathbb{R}$. Then by considering $y \to \infty$, we can show that $f \to \infty$. It is more preferable to have this kind of understanding of limits since we can have a better picture on the behaviour of the function.

We first work out a plan as follows. In view of the above steps, we may first assume everything is nonnegative when x, y are large for simplicity. Then (1) implies both $f(x + y) \geqslant f(x)$ and $f(x + 1) \geqslant f(f(x))$. The former shows f is increasing in general. Using the increasing property, the latter inequality becomes $x + 1 \geqslant f(x)$. This shows $f(x)$ cannot be too large. But on the other hand, it is not hard to obtain a large lower bound for f using (1) since the right-hand side involves a nested function and a product of independent terms.

Solution. We first suppose $f(f(x)) > 0$ for some $x \in \mathbb{R}$. For this x, when y tends to ∞ in (1), the right-hand side of (1) tends to ∞. This shows

$$\lim_{x\to\infty} f(x) = \infty. \tag{2}$$

We first fix a large $a \in \mathbb{R}$ such that $f(a)$ and $f(f(a))$ are large, say $f(a), f(f(a)) > 1$. Such an a exists by (2). Putting $x = a$ in (1), we have

$$f(a+y) \geqslant f(a) + yf(f(a)).$$

We claim that this is larger than $a + y + 1$ for sufficiently large y. Indeed,

$$\begin{aligned} & f(a) + yf(f(a)) > a + y + 1 \\ \Leftrightarrow \quad & \big(f(f(a)) - 1\big)y > a + 1 - f(a) \\ \Leftrightarrow \quad & y > \frac{a+1-f(a)}{f(f(a))-1} \end{aligned}$$

since $f(f(a)) > 1$. Therefore, for any sufficiently large y, we have

$$f(a+y) > a + y + 1.$$

In other words,

$$f(x) > x + 1 \tag{3}$$

for all sufficiently large x.

Let b be a sufficiently large real number. We put $x = b+1$ and $y = f(b) - b - 1$ in (1). We may assume $y > 0$ by (3) and also $f(f(b+1)) > 0$ by (2). Thus, we have

$$f(f(b)) \geqslant f(b+1) + yf(f(b+1)) > f(b+1).$$

However, putting $x = b$ and $y = 1$ in (1) gives

$$f(b+1) \geqslant f(b) + f(f(b)) > f(f(b)).$$

The two inequalities contradict with each other.

Therefore, we must have $f(f(x)) \leqslant 0$ for all $x \in \mathbb{R}$. For any $y \leqslant 0$ in (1), we have

$$f(x+y) \geqslant f(x) + yf(f(x)) \geqslant f(x).$$

Therefore, f is decreasing.

Note that the above arguments have nothing to do with $f(0)$. Suppose on the contrary that $f(0) > 0$. Then we have $f(0) > 0 \geqslant f(f(x))$, so that $f(x) > 0$ for all $x \in \mathbb{R}$ by the decreasing property. However, this is a contradiction since $f(f(x)) \leqslant 0$ for all $x \in \mathbb{R}$. Hence, we must have $f(0) \leqslant 0$. $\square$

Problem 3.18. (IMO Shortlist 2009 A5) Let f be any function that maps the set of real numbers into the set of real numbers. Prove that there exist real numbers x and y such that $f(x - f(y)) > yf(x) + x$.

Analysis. There should be no way to find x and y explicitly since f is unknown. Again a natural way is to use proof by contradiction. The trick is very much like handling functional equations. But our aim is to derive a contradiction from the bounds.

Solution. Suppose on the contrary that

$$f(x - f(y)) \leqslant yf(x) + x \tag{1}$$

for all $x, y \in \mathbb{R}$. We put $y = 0$ in (1) to get $f(x - f(0)) \leqslant x$. This is somehow similar to the surjectivity of a function. For any $x \in \mathbb{R}$, instead of having $y \in \mathbb{R}$ with $f(y) = x$, we have $f(y) \leqslant x$ for some $y \in \mathbb{R}$. By changing the variable, this gives

$$f(x) \leqslant x + f(0). \tag{2}$$

Next, we put $x = f(1)$ and $y = 1$ in (1). Then we have

$$f(0) \leqslant f(f(1)) + f(1).$$

Applying (2) to the term $x = f(1)$, we find that

$$f(0) \leqslant f(f(1)) + f(1) \leqslant f(1) + f(0) + f(1).$$

Thus, we have $f(1) \geqslant 0$.

Having this more concrete result, it might be possible to obtain a contradiction by putting $x - f(y) = 1$ in (1). In that case, the left-hand side $f(1)$ is nonnegative. If we can choose a large y for which $f(x) = f(f(y) + 1)$ is very negative, then the right-hand side is negative and we are done. In view of (2), it suffices to find a large y for which $f(y)$ is small.

Here gives one possible argument. We put $x = 1$ in (1) to get

$$f(1 - f(y)) \leqslant yf(1) + 1. \tag{3}$$

If $f(1) = 0$, then we put $y = 1$ in (1) to obtain

$$f(x) \leqslant f(x) + x$$

for all $x \in \mathbb{R}$. This is not true when $x < 0$. So we must have $f(1) > 0$. Thus, putting $y = \dfrac{t-1}{f(1)}$ in (3) for some $t \in \mathbb{R}$, we get

$$f\left(1 - f\left(\frac{t-1}{f(1)}\right)\right) \leqslant t.$$

This is sufficient for our purpose. In fact, let $\alpha = 1 - f\left(\dfrac{t-1}{f(1)}\right)$ so that

$$f(\alpha) \leqslant t. \tag{4}$$

Using (2), we have

$$\alpha = 1 - f\left(\frac{t-1}{f(1)}\right) \geqslant 1 - \left(\frac{t-1}{f(1)} + f(0)\right) = -\frac{t}{f(1)} + c, \tag{5}$$

where the constant c is defined by $c = 1 + \dfrac{1}{f(1)} - f(0)$. When t tends to $-\infty$, the number α tends to ∞ since $f(1) > 0$. As suggested above, we put $x = f(\alpha) + 1$ and $y = \alpha$ in (1). If t is sufficiently negative, then we have

$$\begin{aligned}
f(1) &\leqslant \alpha f(f(\alpha)+1) + f(\alpha) + 1 \\
&\leqslant \alpha(f(\alpha) + 1 + f(0)) + f(\alpha) + 1 \quad \text{(by (2))} \\
&= (\alpha+1)f(\alpha) + \alpha(1 + f(0)) + 1 \\
&\leqslant (\alpha+1)t + \alpha(1+f(0)) + 1 \quad \text{(by (4))} \\
&= \alpha(t + 1 + f(0)) + t + 1 \\
&\leqslant \left(-\frac{t}{f(1)} + c\right)(t + 1 + f(0)) + t + 1 \quad \text{(by (5))} \\
&= -\frac{1}{f(1)}t^2 + c't + c''
\end{aligned}$$

for certain constants c', c''. As t tends to $-\infty$, the dominating term t^2 tends to ∞. Also, as $f(1) > 0$, this shows the last term tends to $-\infty$. This contradicts with $f(1) \geqslant 0$. So there must be $x, y \in \mathbb{R}$ satisfying

$$f(x - f(y)) > yf(x) + x.$$

$\square$

Lastly, we summarize the tricks that have been used in this section.

- techniques in analysis
  - using limits (problems 3.2, 3.3, 3.4, 3.5, 3.9, 3.16)
  - using increasing/decreasing property (problems 3.2, 3.9, 3.17)
  - using monotone convergence theorem (problems 3.6, 3.16)
  - using continuity (problems 3.2, 3.9)
  - using deneseness of $\mathbb{Q}$ (problems 3.2, 3.3, 3.4)
- functions $f : \mathbb{R}^+ \to \mathbb{R}^+$
  - using trivial bound $f(x) > 0$ (problems 3.4, 3.5, 3.12, 3.16)

- checking the positivity of an expression (problems 3.6, 3.7, 3.8, 3.9)

- functions $f : \mathbb{N} \to \mathbb{N}$
  - considering the minimum $f(m)$ (problem 3.10)
  - using non-strict inequalities for integers (problem 3.11)

- functional inequalities
  - proving $a \geqslant b$ and $b \geqslant a$ (problems 3.13, 3.14, 3.15)
  - considering sufficiently large/small numbers (problems 3.17, 3.18)

4 Functions and Number Theory

Sometimes we may need to use knowledge in number theory to solve functional equations. There are also some number theory problems where functions are involved. This requires us to integrate knowledge in both functions and number theory in order to solve such problems. In this section, we shall go through some of these problems.

Problem 4.1. (IMO Shortlist 2004 A4/Problem 2) Find all polynomials f with real coefficients such that for all reals a, b, c such that $ab+bc+ca = 0$ we have the following relations

$$f(a-b) + f(b-c) + f(c-a) = 2f(a+b+c). \tag{1}$$

Analysis. For general functional equation problems, we cannot assume a function to be a polynomial, though this is the case for most solutions to the problems. However, if this condition is given explicitly in the question, it is better for us to let $f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_0$ and work on it. The number theory ingredient will appear later.

Solution. We can only substitute values that satisfy $ab + bc + ca = 0$. The simplest way is to put $a = b = 0$ in (1). This gives

$$f(0) + f(-c) + f(c) = 2f(c)$$

for all $c \in \mathbb{R}$. If we further substitute $c = 0$, then we get $3f(0) = 2f(0)$, and hence $f(0) = 0$. This also gives $f(-c) = f(c)$. As f is a polynomial, these conditions have special meanings, i.e. f only consists of monomials of even powers, and the constant term vanishes.

Let $f(x) = a_n x^{2n} + a_{n-1}x^{2n-2} + \cdots + a_1 x^2$ where $a_n \neq 0$ except the special case $f \equiv 0$. Note that there is no use putting $a = 0$ anymore, since this forces $a = b = 0$ (up to permutation) and we get the same facts as above. That means we can only use nonzero values. The simplest way is to consider integral values first. Since $c = -\dfrac{ab}{a+b}$ (with $a + b \neq 0$), one possible substitution is $a = 3$, $b = 6$ and $c = -2$. Then (1) gives

$$f(-3) + f(8) + f(-5) = 2f(7).$$

As f is an even function, this means

$$f(3) + f(8) + f(5) = 2f(7).$$

Since the condition $ab + bc + ca = 0$ is homogeneous, whenever this relation holds, we also have $(ka)(kb) + (kb)(kc) + (kc)(ka) = 0$ for any k. This leads to a better result than the above one. This means we put $a = 3k$, $b = 6k$ and $c = -2k$ in (1) to obtain

$$f(3k) + f(8k) + f(5k) = 2f(7k)$$

for all $k \in \mathbb{R}$. Rewriting this using the form of f, this implies

$$\sum_{j=1}^{n} a_j((3k)^{2j} + (5k)^{2j} + (8k)^{2j} - 2(7k)^{2j}) = 0.$$

As this is true for all $k \in \mathbb{R}$, it is not hard to guess $s_j = a_j(3^{2j} + 5^{2j} + 8^{2j} - 2 \cdot 7^{2j})$ should be zero for all j. This can be done using a simple bound. Firstly, suppose on the contrary that $s_n \neq 0$ and let $M = \max\limits_{1 \leqslant j \leqslant n-1} \{|s_j|\}$. Then

$$k^{2n}|s_n| = \left|\sum_{j=1}^{n-1} k^{2j} s_j\right| \leqslant \sum_{j=1}^{n-1} k^{2j}|s_j| \leqslant M \sum_{j=1}^{n-1} k^{2j} = M\frac{k^{2n} - k^2}{k^2 - 1} \leqslant M\frac{k^{2n}}{\frac{1}{2}k^2} = 2Mk^{2n-2}$$

for sufficiently large k. However, this cannot be true for large k as $|s_n|$ and M are constants. So $s_n = 0$. In exactly the same way, we can show that $s_j = 0$ for any j by backward induction.

This remains to solve the Diophantine equation $3^{2t} + 5^{2t} + 8^{2t} = 2 \cdot 7^{2t}$. As the term 8^{2t} dominates other terms for large t, it is not hard to bound t.

For $t = 1$, $3^2 + 5^2 + 8^2 = 98 = 2 \cdot 7^2$.

For $t = 2$, $3^4 + 5^4 + 8^4 = 4802 = 2 \cdot 7^4$.

For $t = 3$, $8^6 = 262144 > 235298 = 2 \cdot 7^6$.

By simple induction, we can show that $8^{2t} > 2 \cdot 7^{2t}$ for $t \geqslant 3$. So the only possible polynomials are $f(x) = a_2x^4 + a_1x^2$ for some $a_1, a_2 \in \mathbb{R}$ (this includes the case $f \equiv 0$). In fact, these are all solutions and we check as follows.

Substituting $c = -\dfrac{ab}{a+b}$, we have to prove

$$f(a-b) + f\left(b + \frac{ab}{a+b}\right) + f\left(\frac{ab}{a+b} + a\right) = 2f\left(a + b - \frac{ab}{a+b}\right),$$

i.e.

$$f(a-b) + f\left(\frac{(2a+b)b}{a+b}\right) + f\left(\frac{(a+2b)a}{a+b}\right) = 2f\left(\frac{a^2+ab+b^2}{a+b}\right).$$

For fourth degree, we have

$$\begin{aligned} & (a-b)^4 + \left(\frac{(2a+b)b}{a+b}\right)^4 + \left(\frac{(a+2b)a}{a+b}\right)^4 = 2\left(\frac{a^2+ab+b^2}{a+b}\right)^4 \\ \Leftrightarrow \quad & (a^2-b^2)^4 + b^4(2a+b)^4 + a^4(a+2b)^4 = 2(a^2+ab+b^2)^4. \end{aligned}$$

The left-hand side is

$$\begin{aligned} & (a^8 - 4a^6b^2 + 6a^4b^4 - 4a^2b^6 + b^8) + (16a^4b^4 + 32a^3b^5 + 24a^2b^6 + 8ab^7 + b^8) \\ & + (a^8 + 8a^7b + 24a^6b^2 + 32a^5b^3 + 16a^4b^4) \end{aligned}$$

while the right-hand side is

$$2a^8 + 8a^7b + 20a^6b^2 + 32a^5b^3 + 38a^4b^4 + 32a^3b^5 + 20a^2b^6 + 8ab^7 + 2b^8.$$

For second degree, we have

$$\begin{aligned} & (a-b)^2 + \left(\frac{(2a+b)b}{a+b}\right)^2 + \left(\frac{(a+2b)a}{a+b}\right)^2 = 2\left(\frac{a^2+ab+b^2}{a+b}\right)^2 \\ \Leftrightarrow \quad & (a^2-b^2)^2 + b^2(2a+b)^2 + a^2(a+2b)^2 = 2(a^2+ab+b^2)^2. \end{aligned}$$

The left-hand side is

$$(a^4 - 2a^2b^2 + b^4) + (4a^2b^2 + 4ab^3 + b^4) + (a^4 + 4a^3b + 4a^2b^2)$$

while the right-hand side is

$$2a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + 2b^4.$$

Both are identities, and hence any $f(x) = a_2x^4 + a_1x^2$ are solutions. □

Remarks. When we solve Diophantine equations, establishing bounds for the variables through inequality is one of the useful methods. Therefore, all steps involved are basically algebra.

Next, we really go through some number theory problems involving functions.

Problem 4.2. (Canada MO 2017 Problem 2) Define a function $f(n)$ from the positive integers to the positive integers such that $f(f(n))$ is the number of positive integer divisors of n. Prove that if p is a prime, then $f(p)$ is prime.

Analysis. It is given that $f(f(n)) = d(n)$ for all $n \in \mathbb{N}$, where $d(n)$ is the number of positive divisors of n. The only trick we need is the usual trick of handling an equation involving $f(f(x))$, i.e. replacing n by $f(n)$ and applying f to both sides.

Solution. Consider any prime p. We have

$$f(f(p)) = d(p) = 2. \tag{1}$$

Applying f to both sides, we get $f^3(p) = f(2)$. Note that $f^3(p) = f^2(f(p)) = d(f(p))$, and so

$$d(f(p)) = f(2). \tag{2}$$

To prove that $f(p)$ is a prime, it suffices to prove $d(f(p)) = 2$. Thus, it remains to show $f(2) = 2$.

Putting $p = 2$ in (2), we have $d(f(2)) = f(2)$. This shows $f(2)$ has exactly $f(2)$ positive divisors. The only such positive integers are 1 and 2 (since we need all of $1, 2, \ldots, m$ to be positive divisors of m).

If $f(2) = 1$, then by putting $p = 2$ in (1), we get $f(1) = 2$. Also, by (2), we have $d(f(p)) = 1$, and so $f(p) = 1$ for any prime p.

Consider $n = 9$. We see that $f(f(9)) = d(9) = 3$. Thus, $f^3(9) = f(3) = 1$. This implies $d(f(9)) = 1$, and so $f(9) = 1$. But then

$$f(1) = f(f(9)) = 3,$$

contradicting $f(1) = 2$. Therefore, we must have $f(2) = 2$, and so $f(p)$ is a prime for any prime p. □

Problem 4.3. Find all function(s) $f : \mathbb{Z} \to \mathbb{N}$ such that

$$f(f(m)) + f(m+n) \mid 2m(2m+n) \tag{1}$$

for all $m, n \in \mathbb{Z}$.

Analysis. Note that the image under f is a positive integer. In particular, it is always at least 1. By choosing m, n for which $2m(2m+n)$ is small in absolute value, we can reduce to a few cases for $f(f(m))$ and $f(m+n)$. This idea helps us find some special values, and we can use these values to find general $f(k)$. The answer is a bit strange and we need a careful verification.

Solution. We put $m = 1$ and $n = -1$ in (1) to get

$$f(f(1)) + f(0) \mid 2.$$

As $f(f(1))$ and $f(0)$ are at least 1, this yields $f(f(1)) = f(0) = 1$. Similarly, by putting $m = -1$ and $n = 3$ in (1), we have

$$f(f(-1)) + f(2) \mid -2$$

and hence $f(f(-1)) = f(2) = 1$.

Next, for each $k \in \mathbb{Z}$, to generate the term $f(k)$, we may consider to substitute $m + n = k$ with fixed m. Since we know the values of $f(f(\pm 1))$, we can simply consider $m = \pm 1$. Indeed, we put $(m, n) = (1, k-1)$ and $(m, n) = (-1, k+1)$ in (1) respectively. These yield

$$1 + f(k) \mid 2(k+1) \quad \text{and} \quad 1 + f(k) \mid -2(k-1).$$

Summing these, we have $1 + f(k) \mid 4$. This implies $f(k) = 1, 3$.

Putting $m = 1$ and $n = 2k-1$ in (1), we get $1 + f(2k) \mid 2(2k+1)$. This shows $f(2k)$ cannot be 3, and hence $f(2k) = 1$ for any $k \in \mathbb{Z}$.

Then we put $m = 3$ and $n = 1$ in (1) to get $f(f(3)) + 1 \mid 42$. This implies $f(f(3)) = 1$. Note that we cannot have $f(3) = 3$, or otherwise $1 = f(f(3)) = f(3) = 3$. Hence, $f(3) = 1$ and $f(1) = f(f(3)) = 1$. The only possible solutions are

$$f(n) = \begin{cases} 1 & \text{if } n \text{ is even or } n = 1, 3, \\ 1 \text{ or } 3 & \text{if } n \text{ is odd and } n \neq 1, 3. \end{cases}$$

(Here, the choice of 1 or 3 is independent for each odd $n \neq 1, 3$.) We now show that all such functions are solutions. Firstly, $f(f(m)) = 1$ for all $m \in \mathbb{Z}$. As $2m(2m+n)$ is even, the condition is true if $f(m+n) = 1$. In the case $f(m+n) = 3$, m and n must have different parities, so that $4 \mid 2m(2m+n)$. Then the divisibility is still true. So all such functions are solutions. □

Remarks. The key idea in this solution is that a positive integer is at least 1. This is often useful in problems where the codomain of f is $\mathbb{N}$, just like the positive condition $f(x) > 0$ for problems in functional equations with $f : \mathbb{R}^+ \to \mathbb{R}^+$.

Problem 4.4. (IMO Shortlist 2004 N3) Find all functions $f : \mathbb{N} \to \mathbb{N}$ satisfying

$$\big(f(m)^2 + f(n)\big) \mid (m^2+n)^2 \tag{1}$$

for any two positive integers m and n.

Analysis. Apart from using the fact that $f(n) \geqslant 1$, we can also consider some substitutions involving prime numbers. For example, if m^2+n is a prime number, then there is only a few choices for the divisor of $(m^2+n)^2$. This is quite a common trick when working on such divisibility problems. The existence of infinitely many prime numbers is an important result.

Solution. Put $m = n = 1$ in (1). Then $f(1)^2 + f(1) \mid 4$. As $f(1)$ is a positive integer, it can only be 1. Then we put $m = 1$ and $n = p-1$ in (1) for some prime p. We get

$$1 + f(p-1) \mid p^2.$$

We can only have $f(p-1) = p-1$ or $f(p-1) = p^2-1$. In the latter case, by putting $m = p-1$ and $n = 1$ in (1), we get $(p^2-1)^2 + 1 \mid ((p-1)^2+1)^2$. Note that

$$\begin{aligned}
& (p^2-1)^2 + 1 \mid ((p-1)^2+1)^2 \\
\Rightarrow\quad & p^4 - 2p^2 + 2 \mid p^4 - 4p^3 + 8p^2 - 8p + 4 \\
\Rightarrow\quad & p^4 - 2p^2 + 2 \leqslant p^4 - 4p^3 + 8p^2 - 8p + 4 \\
\Rightarrow\quad & 4p^3 - 10p^2 + 8p - 2 \leqslant 0.
\end{aligned}$$

This cannot be true for large p (actually for any prime p). Hence, $f(k) = k$ for infinitely many integers k (of the form $k = p-1$). Suppose m is any such k in (1). This gives

$$k^2 + f(n) \mid (k^2+n)^2.$$

We try to rewrite this so that the right-hand side is independent of k. Indeed, as $k^2 + n \equiv -f(n) + n \pmod{k^2 + f(n)}$, this is the same as

$$k^2 + f(n) \mid (n - f(n))^2. \tag{2}$$

For each fixed $n \in \mathbb{N}$, the right-hand side of (2) is a constant, while the left-hand side can be arbitrarily large since there are infinitely many choices of k. Therefore, we must have $(n - f(n))^2 = 0$, which means $f(n) = n$. This is obviously a solution. □

Remarks. Very often, if we can show that $f(n)$ satisfies a formula for infinitely many $n \in \mathbb{N}$, we can conclude that it holds for all $n \in \mathbb{N}$ using a similar trick as given in this solution.

Problem 4.5. Find all surjective function(s) $f : \mathbb{Z} \to \mathbb{Z}$ such that

$$f(m+n) \mid f(mn) + mf(m) + 1 \tag{1}$$

for all $m, n \in \mathbb{Z}$.

Analysis. Apart from the techniques used in the above problems, here we need to make full use of the condition that f is surjective. This means we can replace $f(k)$ by any value we like. For example, it can be as large as possible, or we can set $f(k)$ to be a prime number, etc.

Also, just like the usual trick of solving functional equations, we can interchange m and n to get another divisibility with the same divisor $f(m+n)$. These can be combined to give extra results.

Solution. Firstly, we put $m = 0$ in (1) to get $f(n) \mid f(0) + 1$ for any $n \in \mathbb{Z}$. As $f(n)$ can be any integer by the surjectivity, we must have $f(0) = -1$.

Next, we put $(m, n) = (k, 1)$ and $(m, n) = (1, k)$ in (1) respectively. This gives

$$f(k+1) \mid f(k) + kf(k) + 1 \quad \text{and} \quad f(k+1) \mid f(k) + f(1) + 1.$$

Multiplying the second divisibility by $k+1$ and subtracting the first one, we have

$$f(k+1) \mid (k+1)f(1) + k.$$

By the change of variable $n = k+1$, this means

$$f(n) \mid n(f(1) + 1) - 1. \tag{2}$$

We claim that $f(1) + 1 = \pm 1$. If not, there exists a prime p which divides $f(1) + 1$. Then we choose n such that $f(n) = p$ in (2) to obtain the contradiction $p \mid -1$. Hence, we must have $f(1) + 1 = \pm 1$.

Case 1. $f(1) = 0$

Recall that $0 \mid n$ iff $n = 0$. We can use this to obtain some functional equations. Indeed, we put $m = 1-k$ and $n = k$ in (1) so that $0 \mid f((1-k)k) + (1-k)f(1-k) + 1$. This implies

$$f((1-k)k) + (1-k)f(1-k) + 1 = 0. \tag{3}$$

Similarly, we may as well put $m = k$ and $n = 1-k$ in (1) to get

$$f((1-k)k) + kf(k) + 1 = 0. \tag{4}$$

Comparing (3) and (4), we have

$$kf(k) = (1-k)f(1-k)$$

for all $k \in \mathbb{Z}$. This implies $1-k \mid kf(k)$, and hence $1-k \mid f(k)$ since $\gcd(1-k,k)=1$. Also, note that (2) gives $f(k) \mid k-1$ since $f(1)=0$. This shows $f(k)=\pm(k-1)$ for each k.

Now, for $k \neq 0,1$, we let $f((1-k)k)=\alpha((1-k)k-1)$ and $f(k)=\beta(k-1)$, where $\alpha,\beta=\pm 1$. Equation (4) gives $\alpha((1-k)k-1)+\beta k(k-1)+1=0$, i.e.

$$(-\alpha+\beta)k^2+(\alpha-\beta)k+(-\alpha+1)=0.$$

- If $\alpha=\beta$, we must have $\alpha=\beta=1$ so that $f(k)=k-1$.
- If $\alpha=1$ and $\beta=-1$, we get $-2k^2+2k=0$, which has no solution for $k\neq 0,1$.
- If $\alpha=-1$ and $\beta=1$, we get $2k^2-2k+2=0$, which has no integer solution.

It follows that $f(k)=k-1$ for any $k\neq 0,1$. Indeed, this also holds when $k=0,1$. We check that $f(k)=k-1$ for any $k\in\mathbb{Z}$ is a solution. This is because

$$f(m+n)=m+n-1$$

divides

$$f(mn)+mf(m)+1=mn-1+m(m-1)+1=m(m+n-1).$$

Also, it is obvious that f is surjective.

Case 2. $f(1)=-2$

The ideas are similar. Equation (2) becomes $f(k)\mid k+1$. If we choose k such that $f(k)=0$, then k can only be -1. This means $f(-1)=0$. Putting $(m,n)=(-1-k,k)$ and $(k,-1-k)$ in (1) respectively, we have

$$f((-1-k)k)+(-1-k)f(-1-k)+1=0$$

and

$$f((-1-k)k)+kf(k)+1=0. \tag{5}$$

Comparing these, we have

$$kf(k)=(-1-k)f(-1-k)$$

for all $k\in\mathbb{Z}$. This implies $k+1\mid kf(k)$, and hence $k+1\mid f(k)$ since $\gcd(k+1,k)=1$. Also, as $f(k)\mid k+1$, we have $f(k)=\pm(k+1)$ for each k.

Now, for $k\neq 0,-1$, we let $f((-1-k)k)=\alpha((-1-k)k+1)$ and $f(k)=\beta(k+1)$, where $\alpha,\beta=\pm1$. Equation (5) gives $\alpha((-1-k)k+1)+\beta k(k+1)+1=0$, i.e.

$$(-\alpha+\beta)k^2+(-\alpha+\beta)k+(\alpha+1)=0.$$

- If $\alpha=\beta$, we must have $\alpha=\beta=-1$ so that $f(k)=-(k+1)$.
- If $\alpha=1$ and $\beta=-1$, we get $-2k^2-2k+2=0$, which has no integer solution.

- If $\alpha = -1$ and $\beta = 1$, we get $2k^2 + 2k = 0$, which has no solution for $k \neq 0, -1$.

It follows that $f(k) = -k - 1$ for any $k \neq 0, -1$. Indeed, this also holds when $k = 0, -1$. We check that $f(k) = -k - 1$ for any $k \in \mathbb{Z}$ is a solution. This is because

$$f(m+n) = -m - n - 1$$

divides

$$f(mn) + mf(m) + 1 = -mn - 1 + m(-m-1) + 1 = m(-m-n-1).$$

Also, it is obvious that f is surjective.

Therefore, the solutions are $f(k) = k - 1$ for all $k \in \mathbb{Z}$ and $f(k) = -k - 1$ for all $k \in \mathbb{Z}$. $\square$

Remarks. If the surjective condition is removed, there are many other functions that satisfy the same divisibility. It is still possible to find out all such functions, but the original solution contains over 30 pages!

Problem 4.6. (IMO Shortlist 2011 N3) Let $n \geqslant 1$ be an odd integer. Determine all functions f from the set of integers to itself such that for all integers x and y the difference $f(x) - f(y)$ divides $x^n - y^n$.

Analysis. Note that adding a constant to the function does not affect the difference $f(x) - f(y)$. So we may assume $f(0) = 0$ for convenience. Again, we use the special fact that $0 \mid k$ iff $k = 0$. We can use this to prove that f is injective.

Similar to problem 4.4, we first look for the images of prime numbers. Also, for divisibility problems, the following result is useful. Whenever $a \mid b$ and $b \neq 0$, then $|a| \leqslant |b|$.

Solution. It is given that

$$f(x) - f(y) \mid x^n - y^n \tag{1}$$

for any $x, y \in \mathbb{Z}$. If $f(x)$ is a solution, then $g(x) = f(x) + c$ is also a solution for any $c \in \mathbb{Z}$. So we may assume $f(0) = 0$.

If $f(x) = f(y)$, then (1) becomes $0 \mid x^n - y^n$, and hence $x^n = y^n$. Since n is odd, this gives $x = y$, meaning that f is injective.

Next, we put $y = 0$ and $x = p$ in (1) where p is a prime number. Then we have $f(p) \mid p^n$ so that $f(p) = \pm p^{a_p}$ for some $0 \leqslant a_p \leqslant n$. As f is injective, at most two of the a_p's can be 0. For other primes p, we have $1 \leqslant a_p \leqslant n$.

For any $y \in \mathbb{Z}$, put $x = p$ in (1) for a sufficiently large prime p. We get

$$\pm p^{a_p} - f(y) \mid p^n - y^n.$$

Applying the Euclidean algorithm, we can reduce the size of the right-hand side. Indeed, let $n = qa_p + r$ where $0 \leqslant r \leqslant a_p - 1$. Then

$$p^n - y^n = (p^{a_p})^q p^r - y^n \equiv (\pm f(y))^q p^r - y^n \pmod{(\pm p^{a_p} - f(y))}.$$

If $(\pm f(y))^q p^r - y^n \neq 0$, then we have

$$|\pm p^{a_p} - f(y)| \leqslant |(\pm f(y))^q p^r - y^n| \leqslant |f(y)^q p^r| + |y^n|,$$

and hence

$$|p^{a_p}| \leqslant |f(y)^q p^r| + |y^n| + |f(y)|.$$

Since $r \leqslant a_p - 1$ and $q \leqslant n$, this cannot be true for large p. This shows

$$(\pm f(y))^q p^r - y^n = 0.$$

As this is true for all large p, and $f(y) \neq 0$ for $y \neq 0$ by the injectivity, we need $r = 0$ and $n = qa_p$. This yields $f(y) = \pm y^{a_p}$ (note that q is odd as $n = qa_p$).

Now, for distinct y_1 and y_2, we pick the same large prime p to verify $f(y_1) = \pm y_1^{a_p}$ and $f(y_2) = \pm y_2^{a_p}$. The choices of the signs and exponents are then the same since they only depend on p. Therefore, $f(y) = \pm y^d$ for any $y \in \mathbb{Z}$ where d is a positive divisor of n. In general, the function is $f(y) = \pm y^d + c$ for some $d \mid n$ and $c \in \mathbb{Z}$. We check that

$$f(x) - f(y) = \pm(x^d - y^d) \mid x^n - y^n.$$

□

Problem 4.7. (CMO 2020 Problem 6) Find all possible functions $f : \mathbb{N} \to \mathbb{N}$ such that for any $x, y \in \mathbb{N}$,

$$f(f(x) + y) \quad \text{divides} \quad x + f(y).$$

Analysis. Just like usual problems in functional equations, it could be useful in solving divisibility problems if we know that f is injective. Proving the injectivity is possible in some situations, while the surjectivity could hardly be proven in most cases before we have solved the problem.

Since the divisor involves only a single f, there should be some solutions such that $f(x)$ is small for all x (for example, the constant function $f \equiv 1$ is certainly a solution). In those cases f cannot be injective. Thus, we have to consider both situations. So it makes no harm by first considering the case when f is injective. In terms of a contest problem, sometimes this is how partial credits are obtained.

Solution. It is given that

$$f(f(x) + y) \mid x + f(y) \tag{1}$$

for any $x, y \in \mathbb{N}$. We consider the following two cases.

Case 1. f is injective

The idea is to choose small x and small $f(y)$ in (1), so that we do not have much choices for $f(f(x)+y)$. It just turns out the injectivity can be used in this idea. Again, recall that there must be $t \in \mathbb{N}$ such that $f(t) = m$ is the smallest possible. Putting $x = 1$ and $y = t$ in (1), we get

$$f(f(1)+t) \mid 1+m.$$

This implies $f(f(1)+t) \leqslant 1+m$. By the choice of m, we must have $f(f(1)+t) = m$ or $m+1$.

If $f(f(1)+t) = m$, then $f(f(1)+t) = f(t)$. But this contradicts f is injective since $f(1)+t > t$.

This gives $f(f(1)+t) = m+1$. Similarly, we can prove that

$$f(kf(1)+t) = m+k \tag{2}$$

by induction on k. Indeed, suppose (2) holds for $k = 0, 1, \ldots, \ell-1$. We consider the case $k = \ell$. Putting $x = 1$ and $y = (\ell-1)f(1)+t$ in (1), we get

$$f(\ell f(1)+t) \mid 1+f\big((\ell-1)f(1)+t\big) = m+\ell.$$

This implies $f(\ell f(1)+t) \leqslant m+\ell$. Since $\ell f(1)+t > kf(1)+t$ for $k = 0, 1, \ldots, \ell-1$, it is impossible that $f(\ell f(1)+t) = m, m+1, \ldots, m+\ell-1$ by the injectivity. Thus, we must have $f(\ell f(1)+t) = m+\ell$. This proves the inductive step, and so (2) holds for all $k \in \mathbb{N}$.

Now, since the images of $kf(1)+t$ for $k \in \mathbb{N}$ already cover all except possibly a finite number of positive integers, the terms $kf(1)+t$ can only miss a finite number of positive integers by the injectivity. Thus, we must have $f(1) = 1$. It follows that $m = 1$ since m is the smallest image. And so $t = 1$ by the injectivity. Then (2) implies $f(k+1) = k+1$ for all $k \in \mathbb{N}$. So $f(n) = n$ for all $n \in \mathbb{N}$, which is clearly a solution.

Case 2. f is not injective

Suppose $f(\alpha) = f(\beta)$ for some distinct $\alpha, \beta \in \mathbb{N}$. The natural step is to put $x = \alpha$ and $x = \beta$ in (1) respectively. This gives

$$f(f(\alpha)+y) \mid \alpha+f(y),$$
$$f(f(\beta)+y) \mid \beta+f(y).$$

Note that $f(f(\alpha)+y) = f(f(\beta)+y)$. Subtracting these, we obtain

$$f(f(\alpha)+y) \mid \alpha-\beta \tag{3}$$

for any $y \in \mathbb{N}$. In other words, $f(x) \mid \alpha-\beta$ for any $x > f(\alpha)$. In particular, this implies f is bounded. Since f only takes positive integer values, the boundedness condition is useful.

Let S be the set of all positive integers which are the images of infinitely many terms. Then S is a finite nonempty set since f is bounded. Let $M = \max S$. Since M is the maximum, this time it is better to choose x and y such that $f(f(x)+y) = M$. Indeed, we put $x = 1$ in (1) and choose a sufficiently large y such that $f(f(1)+y) = M$. Such a y exists by the definition of S. Then we have

$$M \mid 1 + f(y). \tag{4}$$

Since y is large, we may assume $f(y) \in S$. Thus, we have $M \leqslant 1 + f(y) \leqslant 1 + M$. This shows $1 + f(y) = M$ or $M + 1$.

- If $1 + f(y) = M$ for all y, then we have $f(y) = M - 1$ so that $M - 1 \in S$. Recall the property (3). This relation holds for any α, β, y such that $f(\alpha) = f(\beta)$. As $M-1, M \in S$, we can choose y such that $f(f(\alpha)+y)$ is $M-1$ and M respectively. This shows

  $$M - 1 \mid \alpha - \beta \quad \text{and} \quad M \mid \alpha - \beta.$$

  It is more desirable to have small $|\alpha - \beta|$. Indeed, since each $f(x)$ can only be $1, 2, \ldots, M$ for sufficiently large x, there must be $\alpha, \beta \in \mathbb{N}$ such that $f(\alpha) = f(\beta)$ and $1 \leqslant \alpha - \beta \leqslant M$ by the pigeonhole principle. For such a pair of α and β, in order that $M \mid \alpha - \beta$, we must have $\alpha - \beta = M$. Then we need $M - 1 \mid \alpha - \beta = M$, and so $M = 2$.

  Also, note that we cannot have $f(x) = f(x+1)$ for some $x \in \mathbb{N}$, or otherwise by putting $\alpha = x$ and $\beta = x + 1$ in (3), we obtain $f(x) \mid 1$ for all $x > f(\alpha)$, contradicting $2 \in S$. Thus, the sequence $\{f(n)\}$ must be eventually periodic, ending with the pattern $1, 2, 1, 2, \ldots$.

  We put $x = s$ and $x = s+2$ in (1) respectively such that $f(s) = f(s+2) = 1$. This gives

  $$\begin{aligned} f(1+y) &\mid s + f(y), \\ f(1+y) &\mid s + 2 + f(y). \end{aligned}$$

  Taking the difference of these, we see that $f(1+y) \mid 2$ for any $y \in \mathbb{N}$. Thus, $f(n) = 1, 2$ for any $n \geqslant 2$. So the periodicity begins from the first or the second term.

  If $f(2n) = 1$ and $f(2n+1) = 2$ for all $n \in \mathbb{N}$, by putting $x = y = 2$ in (1), we find that $2 = f(3) \mid 3$, contradiction.

  Therefore, we must have $f(2n) = 2$ and $f(2n+1) = 1$ for all $n \in \mathbb{N}$. Also, by putting $x = 3$ and $y = 1$ in (1), we need $2 = f(2) \mid 3 + f(1)$, so that $f(1)$ is odd (not necessarily be 1). We now show that all such functions are solutions.

  Firstly, the divisibility (1) must hold if $f(f(x)+y) = 1$. Also, since $f(x)+y > 1$, it suffices to consider the case $f(f(x)+y) = 2$, i.e. $f(x)+y$ is even.

- If x is even, then y is even, so $x + f(y)$ is even (hence divisible by 2).
- If x is odd, then y is odd, so $x + f(y)$ is even (hence divisible by 2).

Thus, (1) always holds.

- If $1 + f(y) = M + 1$ for some y, recall from (4) that $M \mid M + 1$. This shows $M = 1$, which means $f(n) = 1$ for all large n. Similarly, we put $x = r$ and $x = r + 1$ in (1) such that $f(r) = f(r+1) = 1$. This gives

$$f(1+y) \mid r + f(y),$$
$$f(1+y) \mid r + 1 + f(y).$$

Taking the difference of these, we see that $f(1+y) \mid 1$ for any $y \in \mathbb{N}$. Thus, $f(n) = 1$ for any $n \geqslant 2$. It is obvious that all such functions (with arbitrary $f(1)$) are solutions since $f(x) + y \geqslant 2$ for any $x, y \in \mathbb{N}$.

Thus, we have found three solutions in total: $f(n) = n$ for all $n \in \mathbb{N}$,

$$f(n) = \begin{cases} c & \text{if } n = 1, \\ 1 & \text{if } n > 1 \text{ is odd}, \\ 2 & \text{if } n \text{ is even} \end{cases}$$

where $c \in \mathbb{N}$ is an odd constant, and

$$f(n) = \begin{cases} d & \text{if } n = 1, \\ 1 & \text{if } n > 1 \end{cases}$$

where $d \in \mathbb{N}$ is any constant. □

Remarks. Although the solution is long, this is mainly due to the tedious work of handling small cases. But then the key ideas are those we have been using when solving functional equations, especially those used for $f : \mathbb{N} \to \mathbb{N}$.

Problem 4.8. (IMO Shortlist 2007 N5) Find all surjective functions $f : \mathbb{N} \to \mathbb{N}$ such that for every $m, n \in \mathbb{N}$ and every prime p, the number $f(m+n)$ is divisible by p if and only if $f(m) + f(n)$ is divisible by p.

Analysis. It is not hard to show that $f(k) = 1$ implies $k = 1$, by using the fact that 1 has no prime divisors. This result allows us to compare $f(n)$ and $f(n+1)$. In particular, it is obvious that they are relatively prime. Indeed, since $f(n) = n$ should be the only solution, we aim at proving $f(n+1) = f(n) + 1$ and hence the proof is almost done. Again, this can be done by showing that $f(n+1) - f(n)$ has no prime divisors. In other words, the integer 1 plays an important role in this kind of problem involving primes.

Solution. It is given that

$$p \mid f(m+n) \quad \Leftrightarrow \quad p \mid f(m)+f(n) \tag{1}$$

for any prime p and any $m, n \in \mathbb{N}$. Since f is surjective, there exists k such that $f(k)=1$. If $k>1$, then we can put $m=1$ and $n=k-1$ in (1). As $f(m+n)=f(k)=1$ has no prime divisors, $f(m)+f(n)$ also has no prime divisors. However, this cannot be true since $f(m)+f(n) \geqslant 2$. Therefore, we must have $k=1$, which means $f(1)=1$.

We put $m=1$ in (1). Then $p \mid f(n+1)$ iff $p \mid f(n)+1$. If $f(n+1)$ and $f(n)$ have a common prime divisor p, then p is also a common divisor of $f(n)+1$ and $f(n)$, which is impossible. Therefore, $\big(f(n+1), f(n)\big)=1$ for any $n \in \mathbb{N}$.

As suggested, we suppose $f(n+1)-f(n)$ has a prime divisor p. Certainly, this does not guarantee both $f(n+1)$ and $f(n)$ are divisible by p. But we can always find $m \in \mathbb{N}$ such that $p \mid f(m)+f(n)$ as f is surjective. This implies $p \mid f(m)+f(n+1)$ as well. Using (1), we see that p divides $f(m+n)$. Replacing n by $n+1$ in (1), we also obtain $p \mid f(m+n+1)$. Thus, both $f(m+n)$ and $f(m+n+1)$ are divisible by p, which contradicts to $\big(f(m+n), f(m+n+1)\big)=1$.

Therefore, we know that $f(n+1)-f(n)$ has no prime divisors. Therefore,

$$f(n+1)-f(n)=\pm 1$$

for each $n \in \mathbb{N}$. Naturally, we should reject the case -1. This can be done, for example, by showing that f is injective. As usual, suppose $f(\alpha)=f(\beta)$ and we substitute $m=\alpha$ and $m=\beta$ respectively in (1). Since $f(\alpha)+f(n)=f(\beta)+f(n)$, we find that $f(\alpha+n)$ and $f(\beta+n)$ have the same prime divisors for any $n \in \mathbb{N}$. If $\alpha \neq \beta$, we may assume $\alpha>\beta$. Then this means $f(n+d)$ and $f(n)$ have the same prime divisors for any $n>\beta$ where $d=\alpha-\beta$. This behaves the same as the periodic condition. In particular, the prime divisors of any $f(n)$ for $n>\beta$ must be the prime divisors of one of

$$f(\beta+1), f(\beta+2), \ldots, f(\beta+d)$$

(depending on $n \pmod d$). This shows only a finite number of prime divisors are possible, which contradicts the surjective condition. Therefore, $\alpha=\beta$ and hence f is injective.

Now, from $f(1)=1$, $f(n+1)-f(n)=\pm 1$ and the fact that f is injective, it is easy to use induction to show $f(n)=n$ for all $n \in \mathbb{N}$, which is obviously a solution. $\square$

Remarks. One may try to solve the following problem, which requires similar ideas as this solution.

(IMO Shortlist 2010 N5/Problem 3) Determine all functions $g: \mathbb{N} \to \mathbb{N}$ such that

$$(g(m)+n)(m+g(n))$$

is a perfect square for all $m, n \in \mathbb{N}$.

Again, we summarize the tricks used in the above number theory problems involving functions. Certainly this is not a complete list, as all tricks introduced before could be useful, and there are still many useful knowledge in number theory which may be necessary in solving some other problems.

- tricks in functions
  - handling nested function $f(f(n))$ (problem 4.2)
  - using trivial bound $f(n) \geqslant 1$ (problems 4.3, 4.4)
  - using surjectivity (normally hard to prove) (problems 4.5, 4.8)
  - using injectivity (problems 4.6, 4.7, 4.8)
  - considering smallest $f(n)$ (problem 4.7)
  - considering largest M which is the image of infinitely many terms for bounded $f : \mathbb{N} \to \mathbb{N}$ (problem 4.7)
- knowledge in number theory
  - $n \mid p^a$ implies $n = \pm p^b$ where $0 \leqslant b \leqslant a$ (problems 4.4, 4.6)
  - $0 \mid n$ iff $n = 0$ (problems 4.5, 4.6)
  - the only integer having infinitely many divisors is 0 (problem 4.4)
  - the only integers having no prime divisor is ± 1 (problems 4.5, 4.8)
  - existence of infinitely many primes (problems 4.4, 4.6)
  - if $a \mid b$ and $b \neq 0$, then $|a| \leqslant |b|$ (problems 4.6, 4.7)

Links

Theorems

- Denseness of $\mathbb{Q}$
- Intermediate value theorem
- Monotone convergence theorem
- Sequential criterion
- Well-ordering principle

Propositions

- 2.1: Cauchy equation on $\mathbb{R}$
- 3.1: Existence of the one-sided limits of monotonic functions

Problems

- 2.1: Change of variable 1
- 2.2: Change of variable 2
- 2.3: Change of variable and solving equations
- 2.4: Basic substitution 1
- 2.5: Basic substitution 2
- 2.6: Substitution and equating terms
- 2.7: Swapping variables
- 2.8: Substitution and symmetry
- 2.9: Nested function
- 2.10: Cauchy equation
- 2.11: Application of the Cauchy equation 1
- 2.12: Application of the Cauchy equation 2
- 2.13: Induction
- 2.14: Application of the Cauchy equation on $\mathbb{R}$
- 2.15: IMO HK TST 2020 Test 2 Problem 4
- 2.16: Various substitutions
- 2.17: Introducing more variable
- 2.18: General proof of injectivity and surjectivity
- 2.19: Injectivity and surjectivity 1

- 2.20: Injectivity and surjectivity 2
- 2.21: Injectivity and periodicity
- 2.22: Kosovo MO 2019 Grade 11 Problem 4
- 2.23: Injectivity
- 2.24: Dutch TST 2012 Test 2 Problem 5
- 2.25: Kernel
- 2.26: Periodicity and boundedness
- 2.27: IMO Shortlist 2009 A7
- 3.1: Application of the monotone convergence theorem
- 3.2: Proof of proposition 2.1
- 3.3: Singapore MO Open 2015 Problem 3
- 3.4: Denseness of $\mathbb{Q}$ and positivity
- 3.5: Positivity and limit
- 3.6: IMO Shortlist 2005 A2
- 3.7: IMO Shortlist 2007 A4
- 3.8: Spain MO 2018 Problem 6
- 3.9: Increasing and continuity
- 3.10: IMO Shortlist 2011 A4
- 3.11: IMO 2009 Problem 5
- 3.12: Trivial bound in functional inequality
- 3.13: All-Russian MO 2000 Grade 11 Problem 1
- 3.14: Benelux MO 2013 Problem 2
- 3.15: Dutch TST 2018 Test 3 Problem 2
- 3.16: Functional inequality and monotone convergence theorem
- 3.17: Functional inequality and proof by contradiction
- 3.18: IMO Shortlist 2009 A5
- 4.1: IMO 2004 Problem 2
- 4.2: Canada MO 2017 Problem 2
- 4.3: Divisibility and function
- 4.4: IMO Shortlist 2004 N3
- 4.5: Divisibility and surjective function
- 4.6: IMO Shortlist 2011 N3

- 4.7: CMO 2020 Problem 6
- 4.8: IMO Shortlist 2007 N5

Terminologies and Notations

- divergent
- domain
- element
- empty set $\varnothing$
- even function
- function $f : A \to B$
- identity function
- image $f(a)$
- injective
- inverse f^{-1}
- kernel
- left-hand limit $\lim_{x \to a^-} f(x)$
- limit (function) $\lim_{x \to a} f(x)$
- limit (sequence) $\lim_{n \to \infty} a_n$
- mapping $f : A \to B$
- monotonic decreasing
- monotonic function
- monotonic increasing
- nested function $f^k(x)$
- odd function
- one-sided limit
- one-to-one
- one-to-one correspondence
- onto
- period
- periodic function
- preimage
- proper subset $A \subsetneq B$
- range
- restriction $f|_C$
- right-hand limit $\lim_{x \to a^+} f(x)$

- sequence
- set $\{\ldots\}$
- strictly decreasing
- strictly increasing
- subset $A \subset B$, $A \subseteq B$
- surjective
- unbounded

References

- Functional Equations (by Titu Andreescu and Iurie Boreico)
- Introduction to Real Analysis (by Robert G. Bartle and Donald R. Sherbert)

「吾生也有涯，而知也無涯。以有涯隨無涯，殆已！」

《莊子・內篇・養生主—庖丁解牛》